



A Dynamic Optimization Framework for Real-Time parking lot-level recommendation and Parking Spot Assignment in Urban Networks

by

Satvika Pullisani

A Thesis

Submitted to the Department of Physics and Computer Science

Faculty of Science

in fulfilment of the requirements for

Master of Applied Computing

Wilfrid Laurier University

Waterloo, Ontario, Canada

2026

© Satvika Pullisani 2026

Examining Committee Membership

The thesis of **Satvika Pullisani**, *A Dynamic Optimization Framework for Real-Time Parking Recommendation and Spot-Level Assignment in Urban Networks*, is approved by:

Supervisor:

Dr. Usama Mir

Assistant Professor, Department of Computer Science

Wilfrid Laurier University, Brantford, Ontario, Canada

Co-Supervisor:

Dr. Dariush Ebrahimi

Assistant Professor, Department of Computer Science

Wilfrid Laurier University, Waterloo, Ontario, Canada

Internal Member:

Dr. Internal Member Name

Assistant Professor, Department of Computer Science

Wilfrid Laurier University, Brantford, Ontario, Canada

External Examiner:

Dr. External Examiner Name

Title, Department Name

University Name, City, Province/Country

Author's Declaration

I hereby declare that I am the sole author of this thesis. This is a true copy of the thesis, including any required final revisions, as accepted by my examiners.

I understand that my thesis may be made electronically available to the public.

Abstract

Parking facility search is a major challenge in sustainable and intelligent urban transportation systems (ITS). In dense urban road networks, the time spent looking for an available parking space adds to traffic congestion, travel delay, inefficient road usage, higher fuel consumption, and driver frustration. Smart Parking Systems (SPS) have been developed to support parking recommendation through sensing, communication, and intelligent decision-making. However, giving a reliable recommendation in real time is still difficult, because traffic conditions and parking availability can change while the driver is still travelling toward the destination. As a result, a recommendation that is suitable at the start of a route may not remain suitable later in the same route.

Another limitation is that many existing systems perform the recommendation only at the parking lot level. In these approaches, a parking lot is treated as a single available parking option, and the individual parking spots inside the lot are not considered. However, parking lots differ in availability, parking cost, driving time, and walking distance to the destination. Parking spots inside the same lot also differ in availability, spot-specific cost, internal driving time from the entrance, internal walking time to the pedestrian exit, and external walking time to the final destination. Therefore, a complete parking recommendation needs to cover both levels, which include both the parking lot and the individual parking spots within the parking lot.

This thesis addresses this problem by developing two related frameworks. The first framework focuses on parking lot-level recommendation and is referred to as the Dynamic Parking Lot Optimization Framework (DPLOF). DPLOF recomputes the driving route and the recommended parking lot from the driver's current location using the latest traffic information available at that point. The framework considers driving time, walking time, and parking cost, together with user-defined constraints such as maximum walking distance and maximum parking cost. The second framework focuses on parking spot-level recommendation and is referred to as the Dynamic Parking Spot Optimization Framework (DPSOF). DPSOF continues the decision after a lot has been selected and assigns an individual parking spot inside that lot. In this framework, each available spot is modeled as a separate parking option with its own availability,

spot-specific cost, internal driving time, internal walking time, external walking time, and proximity to the pedestrian exit.

For both frameworks, this thesis develops optimization-based models and evaluates different solution methods for small-scale, medium-scale, and large-scale urban road networks. For parking lot-level recommendation, the optimization model is first applied to small-scale networks to obtain optimal results and to check that the formulation works as intended. Since solving the optimization model repeatedly becomes computationally difficult for larger networks, a heuristic method is developed for the medium-scale and large-scale cases. This heuristic is then compared with meta-heuristic methods such as Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Ant Colony Optimization (ACO). For parking spot-level recommendation, an optimization model is also formulated, and the same three meta-heuristic methods are applied as scalable comparative methods, so that both problems are studied using a consistent set of solution approaches.

The proposed frameworks are evaluated using synthetic urban-like datasets and a real road network case study built in Simulation of Urban MObility (SUMO). The synthetic datasets enable controlled variation of network size and experimental parameters, while the SUMO-based network provides a real-world urban road network with directed road links and actual road connectivity. The results for the lot-level framework show that the recommendation can be updated from the current vehicle position when traffic conditions change during travel, and that the proposed heuristic gives solution quality close to the optimization model at a much lower computational cost. The spot-level framework then narrows the decision further by selecting a suitable parking spot inside the recommended parking lot.

Overall, this thesis presents two dynamic parking recommendation frameworks, one at the parking lot level and the other at the parking spot level. By considering both levels together instead of treating them separately, the proposed work aims to reduce parking facility search time and to support better use of the parking facilities that already exist in urban road networks.

Acknowledgments

I would like to express my sincere gratitude to my parents for their unconditional love, encouragement, sacrifices, and constant support throughout my academic journey. Their belief in me has been a continuous source of motivation, and this work would not have been possible without them.

I would also like to thank my grandparents for their love, encouragement, guidance, and blessings. Their support has always inspired me to work hard and pursue my goals with dedication.

I am sincerely grateful to my supervisor, Dr. Usama Mir, for his continuous guidance, valuable suggestions, constructive feedback, and encouragement throughout this research. His expertise and support have played an important role in the successful completion of this thesis.

I would also like to express my sincere appreciation to my co-supervisor, Dr. Dariush Ebrahimi, for his valuable insights, thoughtful comments, and continuous support. His guidance greatly contributed to improving the quality of this research.

I would like to thank the Department of Physics and Computer Science at Wilfrid Laurier University for providing an excellent academic environment, the necessary resources, and the opportunity to pursue my graduate studies and research.

Finally, I would like to thank my friends and classmates for their encouragement, support, and friendship throughout my graduate studies. I would especially like to thank my close friend, Harish, for his constant encouragement, motivation, and support during both the challenging and rewarding moments of this journey.

To everyone who has supported me, directly or indirectly, I extend my sincere thanks.

Thesis Outcomes

This thesis has resulted in two major research outcomes, both prepared as journal papers for submission to IEEE Transactions journals. The corresponding manuscript versions have been prepared as part of this work.

Dedication

I dedicate this thesis to my beloved parents in recognition of their unwavering support, love, and encouragement.

Contents

List of Figures	1
List of Tables	4
List of Algorithm	5
1 Introduction	6
2 Related Work	11
3 Graph Creation	20
3.1 Synthetic Graph Generation	20
3.2 Real-Life Road Network	21
3.3 Parking Lot Positioning	21
3.4 Parking Spot Generation and Positioning	22
3.5 Driver Source and Destination Positioning	22
3.6 Edge Weight and Travel-Time Assignment	23
3.7 Summary	23
4 Proposed Dynamic Parking Optimization Framework for Parking Lot Selection	24
4.1 Introduction	24
4.2 Related Work	26
4.3 System Model and Problem Description	28
4.4 Problem Formulation	31
4.4.1 Network Representation and Decision Variables	32
4.4.2 Objective Function	32
4.4.3 Routing Feasibility Constraints	33
4.4.4 Travel-Time Accumulation	33
4.4.5 Parking Selection and Cost Constraints	33
4.4.6 Maximum Walking-Distance Constraint	34
4.5 Heuristic and Meta-Heuristic Methods	35
4.5.1 Proposed Heuristic Method	35
4.5.2 Meta-Heuristic Baseline Design	37
4.5.3 Genetic Algorithm	38
4.5.4 Particle Swarm Optimization	38

4.5.5	Ant Colony Optimization	39
4.6	Results and Discussion	40
4.6.1	Simulation Setup, Performance Metrics, and Parameters	40
4.6.2	Small-Scale Evaluation	41
4.6.3	Medium- and Large-Scale Networks: Scalability and Parameter Analysis	49
4.6.4	Realistic Urban Network Evaluation Using SUMO	56
4.7	Conclusion	63
5	Parking Spot-level recommendation Framework	65
5.1	Introduction	65
5.2	Related Work	68
5.3	System Model and Problem Description	70
5.4	Problem Formulation	75
5.4.1	Network Representation and Decision Variables	76
5.4.2	Objective Function	76
5.4.3	Parking-Spot Selection Constraint	77
5.4.4	Walking and Parking-Cost Constraints	77
5.4.5	Dynamic Availability and Reservation Update	78
5.4.6	Interpretation of the Selected Solution	78
5.5	Meta-Heuristic Baselines	79
5.5.1	Genetic Algorithm	79
5.5.2	Particle Swarm Optimization	80
5.5.3	Ant Colony Optimization	81
5.6	Performance Evaluation	82
5.6.1	Simulation Setup, Performance Metrics, and Parameters	82
5.6.2	Small-Scale Evaluation	84
5.6.3	Medium- and Large-Scale Networks: Scalability and Parameter Analysis	93
5.6.4	Realistic Urban Network Evaluation Using SUMO	101
5.7	Conclusion and Future Work	109
6	Conclusion	112
6.1	Limitations of the Work	114
	References	116
	Glossary	122
	Abbreviations	124
	List of Symbols	125

List of Figures

4.1	System model of the proposed dynamic parking lot optimization framework.	30
4.2	Small-scale runtime scalability of the optimization model, the proposed heuristic, and the meta-heuristic methods.	42
4.3	Performance comparison between the optimization model and the proposed heuristic as a function of network size.	43
4.4	Performance comparison between the optimization model and the proposed heuristic as a function of edge density.	44
4.5	Performance comparison between the optimization model and the proposed heuristic as a function of the maximum walking distance.	45
4.6	Performance comparison between the optimization model and the proposed heuristic as a function of the traffic level.	46
4.7	Performance comparison between the optimization model and the proposed heuristic as a function of the maximum parking budget.	47
4.8	Performance comparison between the optimization model and the proposed heuristic as a function of the parking supply.	48
4.9	Performance comparison between the optimization model and the proposed heuristic as a function of the parking-cost range.	49
4.10	Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of network size.	50
4.11	Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of edge density.	51
4.12	Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of the maximum walking distance.	52
4.13	Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of the traffic level.	53
4.14	Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of the maximum parking budget.	54
4.15	Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of the parking supply.	55
4.16	Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of the parking-cost range.	56
4.17	Baseline performance comparison between the proposed heuristic and the meta-heuristic methods on the fixed 2055-node Waterloo–Kitchener road network.	57

4.18	Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of edge density on the fixed real-road network.	58
4.19	Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of the maximum walking distance on the fixed real-road network.	59
4.20	Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of the traffic level on the fixed real-road network.	60
4.21	Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of the maximum parking budget on the fixed real-road network.	61
4.22	Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of the parking supply on the fixed real-road network.	62
4.23	Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of the parking-cost range on the fixed real-road network.	63
5.1	System model of the Dynamic Parking Spot Optimization Framework with (A) candidate parking facilities and (B) internal structure of $P1$. .	73
5.2	Small-scale runtime scalability of the optimization model and the meta-heuristic methods.	85
5.3	Performance comparison between the optimization model and the meta-heuristic methods as a function of network size.	86
5.4	Performance comparison between the optimization model and the meta-heuristic methods as a function of edge density.	87
5.5	Performance comparison between the optimization model and the meta-heuristic methods as a function of the maximum walking distance. . . .	88
5.6	Performance comparison between the optimization model and the meta-heuristic methods as a function of the traffic level.	89
5.7	Performance comparison between the optimization model and the meta-heuristic methods as a function of the maximum parking budget.	90
5.8	Performance comparison between the optimization model and the meta-heuristic methods as a function of the parking-spot availability.	91
5.9	Performance comparison between the optimization model and the meta-heuristic methods as a function of the number of parking spots per lot.	92
5.10	Performance comparison between the optimization model and the meta-heuristic methods as a function of the parking-cost range.	93
5.11	Performance comparison between the meta-heuristic methods as a function of network size.	94
5.12	Performance comparison between the meta-heuristic methods as a function of edge density.	95
5.13	Performance comparison between the meta-heuristic methods as a function of the maximum walking distance.	96

5.14	Performance comparison between the meta-heuristic methods as a function of the traffic level.	97
5.15	Performance comparison between the meta-heuristic methods as a function of the maximum parking budget.	98
5.16	Performance comparison between the meta-heuristic methods as a function of the parking-spot availability.	99
5.17	Performance comparison between the meta-heuristic methods as a function of the number of parking spots per lot.	100
5.18	Performance comparison between the meta-heuristic methods as a function of the parking-cost range.	101
5.19	Performance comparison between the meta-heuristic methods on the fixed 2055-node SUMO road network.	102
5.20	Performance comparison between the meta-heuristic methods as a function of edge density.	103
5.21	Performance comparison between the meta-heuristic methods as a function of the maximum walking distance.	104
5.22	Performance comparison between the meta-heuristic methods as a function of the traffic level.	105
5.23	Performance comparison between the meta-heuristic methods as a function of the maximum parking budget.	106
5.24	Performance comparison between the meta-heuristic methods as a function of the parking-spot availability.	107
5.25	Performance comparison between the meta-heuristic methods as a function of the number of parking spots per lot.	108
5.26	Performance comparison between the meta-heuristic methods as a function of the parking-cost range.	109

List of Tables

2.1	Summary of related work on smart parking recommendation, allocation, and parking spot assignment	15
4.1	Notation used in the DPLOF system model and problem formulation .	31
4.2	Controlled parameter settings used in the parameter-sensitivity analysis.	41
5.1	Notation used in the DPSOF system model and problem formulation .	75
5.2	Controlled parameter settings used in the spot-level parameter-sensitivity analysis.	84

List of Algorithm

1	Proposed Heuristic Method	36
2	Genetic Algorithm Baseline	38
3	Particle Swarm Optimization Baseline	39
4	Ant Colony Optimization Baseline	40
5	Genetic Algorithm Baseline	80
6	Particle Swarm Optimization Baseline	81
7	Ant Colony Optimization Baseline	82

Chapter 1

Introduction

Smart Parking Systems (SPS) are becoming an essential component of intelligent urban transportation networks, particularly in dense urban areas where parking search remains a significant challenge. The rapid growth of urban populations and vehicle ownership has increased parking demand, and this makes it more difficult for drivers to find a suitable parking facility near their destinations. Many drivers spend a considerable amount of time searching for an available parking lot and then for an available spot inside it, which contributes to traffic congestion, travel delay, unnecessary vehicle circulation, higher fuel consumption, increased emissions, and reduced road-network efficiency [1] [21]. Therefore, improving parking management has become an important goal for supporting urban mobility, reducing the traffic caused by parking facility search, and improving the overall performance of SPS [2].

To address these challenges, SPS have been developed to support parking monitoring, occupancy estimation, availability detection, reservation, pricing, and driver guidance through sensing, communication, and intelligent decision-making technologies [3] [22] [23]. In comparison with traditional parking facility search methods, these systems help drivers identify parking lots and spots more efficiently and reduce the unnecessary effort spent on searching [4]. Existing parking recommendation and allocation approaches commonly consider factors such as travel time, walking distance, parking cost, parking availability, reservation status, and user preferences. Based on these factors, several existing recommendation frameworks have been explored to improve parking recommendations, including multi-objective parking allocation [24], genetic-algorithm-based parking-lot selection [25], data-driven parking optimization [26], ant-colony-based dynamic parking recommendation [27], memetic optimization for street parking allocation [28], shared parking reservation [29], and parking allocation under peak demand conditions [30].

Although these advances have improved parking recommendations, providing reliable recommendations in a realistic urban road network remains a challenging problem. Traffic conditions and parking availability can change while a driver is traveling toward a destination, and this makes a recommendation

generated at the beginning of a route less suitable later in the same route. Parking facility recommendation therefore depends not only on the availability of a parking lot, but also on the driver’s current location, the current traffic condition, the driving time to the parking lot, the internal driving time to the spot, the walking time from the parking spot to the destination, the parking cost of the spot, and user-defined constraints such as maximum walking distance and maximum parking cost. Reservation policies [5], competition among parking lots [6], and pricing mechanisms [7] further influence parking access, demand distribution, utilization, and driver selection behavior. Consequently, an effective parking recommendation needs to consider both route-related information and parking-system attributes while recommending a parking facility.

Research on SPS has addressed a wide range of parking-management problems, including parking allocation, pricing, assignment, monitoring, reservation, demand estimation, and occupancy inference. Prior studies have improved parking-space utilization [8] [34], studied adaptive pricing under changing demand [9] [37], optimized travel cost among nearby parking lots [10], balanced parking demand across multiple parking facilities [11], and estimated parking occupancy under incomplete sensing infrastructure [12]. Other studies have examined transaction-based occupancy inference [13], reservation before vehicle arrival [14], parking-space-view-based service improvement [15] [48], uncertain parking availability [16], and curb-parking demand influenced by nearby buildings, businesses, and activity areas [17]. Although these studies have contributed significantly to the development of SPS, relatively little attention has been given to parking recommendation methods that continuously update both the routing and the parking decision while a driver is traveling through the road network.

Another limitation of many existing parking recommendation systems is that the recommendation is performed mainly at the parking-lot level. In these approaches, each parking lot is treated as a single parking option, and the individual parking spots inside the lot are generally not considered during the recommendation process. However, parking spots within the same lot may differ in availability, parking cost, internal driving time from the entrance, walking time to the pedestrian exit, and walking distance to the final destination. Reservation-based valet parking studies have shown that reservation improves parking reliability, but internal movement within the lot remains largely unexplored [31]. Survey studies on smart-city parking and IoT-driven parking management also show that existing systems give priority to sensing, pricing, reservation, and parking-lot recommendation rather than to parking-spot-level routing and assignment [32] [33]. As a result, even after a lot has been recommended, the driver may still need to spend additional time searching inside that lot before finding a suitable available spot.

To address these limitations, this thesis treats parking recommendation as a two-level recommendation process consisting of parking lot-level recommen-

dation followed by parking spot-level recommendation. The first level focuses on dynamic parking lot-level recommendation through the proposed Dynamic Parking Lot Optimization Framework (DPLOF). DPLOF updates the traffic information and recomputes both the driving route and the parking-lot recommendation from the driver’s current location as the vehicle moves through the road network. At every node location, the available parking lots are evaluated using an objective function that considers driving time, walking time, parking cost, and user-defined constraints such as maximum walking distance and maximum parking cost. Since the recommendation is updated repeatedly during travel, the framework adapts to changing traffic conditions without relying on future traffic prediction or future parking-price estimation.

The second level extends the recommendation from the parking-lot level to the parking-spot level through the proposed Dynamic Parking Spot Optimization Framework (DPSOF). Instead of treating the parking lot as the final recommendation, DPSOF further evaluates the available parking spots inside the selected lot. For every parking spot, the assignment cost includes external driving time, internal driving time, internal walking time, external walking time, and spot-specific parking cost. This allows the recommendation to be made at the parking-spot level while remaining fully compatible with the overall graph-based parking recommendation framework.

The parking lot-level recommendation problem is first formulated as an optimization model for small-scale road networks in order to obtain optimal solutions. As the network size increases, solving the optimization model repeatedly becomes computationally expensive, because the route and the parking recommendation must be updated continuously as traffic conditions change. To address this challenge, a heuristic solution approach is developed for medium-scale and large-scale road networks. The proposed heuristic is evaluated against three commonly used meta-heuristic methods, namely Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Ant Colony Optimization (ACO). GA has been commonly applied to transportation and routing problems [18]. PSO has been used for path planning and search-based optimization [19], whereas ACO has been applied to congestion-aware route optimization and related transportation problems [20].

For parking spot-level recommendation, both an optimization-based recommendation model and the same meta-heuristic methods are considered. The optimization model identifies a suitable parking spot from the set of currently available spots, while GA, PSO, and ACO are used as comparative solution methods based on the proposed spot-level cost structure [49].

The proposed frameworks are then evaluated using synthetic urban-like datasets and a real-road-network case study developed in Simulation of Urban MObility (SUMO). At the lot level, the datasets are varied in network size, edge

density, parking supply, traffic level, walking-distance threshold, parking-cost threshold, and parking-cost range. At the spot level, the datasets are varied in network size, edge density, number of parking spots, traffic level, walking-distance threshold, parking-cost threshold, and spot-specific cost range. Performance is measured using travel time, walking distance, parking cost, and runtime. The SUMO-based network provides a realistic urban road structure for evaluating the proposed frameworks. The main contributions of this thesis are summarized as follows:

- A dynamic parking recommendation framework is developed at both the parking-lot level and the parking-spot level to support real-time parking decisions under changing traffic conditions.
- At the parking-lot level, a Dynamic Parking Lot Optimization Framework (DPLOF) is proposed to minimize driving time, walking time, and parking cost under user-defined constraints by updating the traffic information and recomputing the route and the parking-lot recommendation at each intersection without relying on future conditions.
- At the parking-spot level, a Dynamic Parking Spot Optimization Framework (DPSOF) is proposed by extending parking recommendation from parking-lot-level selection to individual parking-spot-level recommendation.
- Each parking spot is modeled as a separate option with its own identifier, availability status, spot-specific cost, internal driving time to the spot, and walking time from the spot to the destination, while internal parking-lot movement is represented through weighted links connecting parking-lot entrances, parking spots, and pedestrian exits.
- Optimization-based models are formulated for both framework levels, and heuristic or meta-heuristic solution approaches are considered for medium-scale and large-scale instances where repeated optimization becomes computationally expensive.
- The parking-lot-level framework is evaluated using travel time, walking distance, parking cost, and runtime under variations in network size, edge density, parking supply, traffic level, walking-distance threshold, parking-cost threshold, and parking-cost range.
- The parking-spot-level framework is evaluated using external driving time, internal driving time, internal walking time, external walking time, spot-specific parking cost, and runtime under variations in network size, edge density, parking-spot availability, number of parking spots per lot, traffic level, walking-distance threshold, parking-cost threshold, and parking-cost range.
- SUMO-based real-road-network case studies are used to assess both frameworks under realistic urban traffic conditions.

The remainder of this thesis is organized as follows:

- Chapter 2 surveys the relevant literature on smart parking systems, parking recommendation, parking allocation, pricing, reservation, and optimization-based parking recommendation. It also identifies the research gaps and the distinguishing features of the proposed methodologies.
- Chapter 3 presents the graph creation methodology used throughout this thesis. It describes the generation of synthetic urban-like road networks, the SUMO-based real-road-network case study, parking lot and parking spot generation, source and destination placement, and the assignment of travel times, walking distances, and parking costs. This chapter establishes the experimental environment used to evaluate the proposed frameworks under both controlled and realistic urban scenarios.
- Chapter 4 addresses the first research problem of this thesis, which focuses on real-time parking lot-level recommendation. An optimum is formulated for small-scale road networks, while a heuristic solution approach together with GA, PSO, and ACO are considered for medium-scale and large-scale networks. The framework updates the traffic information and recomputes the route and the parking-lot recommendation while considering driving time, walking time, parking cost, and user-defined constraints.
- Chapter 5 investigates the second research problem of this thesis, which focuses on parking spot-level recommendation. An optimization-based recommendation model is first developed for small-scale instances, after which GA, PSO, and ACO are applied to larger problem instances. The framework extends parking recommendation from parking-lot selection to individual parking-spot recommendation by considering external driving time, internal driving time, internal walking time, external walking time, and spot-specific parking cost.
- Chapter 6 concludes the thesis by summarizing the findings and contributions of this research, discussing its limitations, and outlining future research directions for dynamic parking lot-level recommendation and parking spot-level recommendation.

Chapter 2

Related Work

Smart Parking Systems (SPS) have been studied as a way of reducing the time drivers spend searching for parking and improving the use of limited parking facilities in urban areas. Several survey papers give an overview of this broader research area. Aljohani et al. [21] reviewed parking solutions implemented in smart cities, including parking guidance, sensing, reservation, pricing, and management approaches. Alam et al. [22] explored Internet of Things (IoT)-driven parking management systems and evaluated their sensing technologies, implementation limitations, and future research directions. Agrawal et al. [23] reviewed smart parking techniques intended to improve the use of urban space, while Almalki and Ragab [24] discussed the challenges associated with smart parking in sustainable urban environments. These surveys describe how sensing, communication, automation, and intelligent decision-making have improved parking services in general. However, they remain descriptive in nature. None of them formulates a route-dependent recommendation process, and none of them separates the selection of a parking lot from the selection of a specific spot within that lot. This thesis addresses that gap by formulating parking-lot recommendation and parking-spot assignment as two connected optimization problems, DPLOF and DPSOF, instead of treating parking management as a single undivided activity.

A major direction in SPS research concerns parking occupancy prediction and availability estimation. Li et al. [1] developed a near-real-time method for predicting parking occupancy across different parking locations, and their later work used spatial and temporal relationships among nearby parking areas to improve the prediction [2]. Zhao et al. [3] treated parking meters as virtual sensors for estimating citywide on-street parking availability, while Song et al. [4] considered parking vacancy prediction when the sensor information is incomplete. These methods improve the quality and reliability of parking availability information. However, their contribution stops at estimating whether a parking lot is occupied. None of them recomputes the driver's route or re-evaluates a parking-lot recommendation as the driver moves and the traffic condition changes. DPLOF addresses this gap by repeatedly recomputing both the route and the parking-lot recommendation from the driver's current location using the latest available traffic information.

Parking monitoring and occupancy inference have also been examined using several data sources. Kim et al. [12] used unmanned aerial vehicles to monitor outdoor parking areas. Assemi et al. [13] inferred on-street parking occupancy from payment transaction records, which provides an alternative when complete

sensing infrastructure is unavailable. Nie et al. [17] estimated public curb-parking demand using the distribution of nearby commercial areas, offices, businesses, and public facilities. Abo-Zahhad and Abo-Zahhad [36] combined IoT and machine-learning techniques for smart parking management, while He et al. [46] considered occupancy-rate prediction together with customer preferences. In these studies, the resulting information is used as an input to a separate management process rather than as part of a driver-facing recommendation loop. None of them re-evaluates the driving route or the parking decision once new information becomes available. In the proposed frameworks, recommendations are recomputed at every intersection using the latest availability and traffic information.

Another important research stream focuses on parking pricing, reservation, and competition. Boateng et al. [5] proposed a reservation and dynamic-pricing framework for automated valet parking and charging using multi-agent reinforcement learning. Saharan et al. [6,33] developed DyPARK, which combines dynamic pricing and allocation for smart on-street parking. Hu et al. [7] examined autonomous valet parking management under demand uncertainty, while Hassija et al. [9,47] proposed a parking-slot allocation framework based on virtual voting and adaptive pricing. These studies show that parking demand can be influenced through pricing, reservation, and competition mechanisms. However, their emphasis is on the operation of the parking system rather than on the route-dependent decision made by a driver. None of them evaluates current traffic information, walking time, and fixed parking cost together in the way that a driver approaching a destination has to. In the frameworks, parking cost is instead treated as one of three route-dependent factors, together with driving time and walking time, that are re-evaluated at every decision point.

Reservation-based parking services have also been used to reduce uncertainty before a vehicle arrives at a parking facility. Hu et al. [14,49] proposed CRPark, which combines competition and reservation services in a smart parking system. Wang et al. [15,32] developed PRPSV to improve parking efficiency and reservation services using a parking-space view. Zhang et al. [48] studied reservation-enhanced autonomous valet parking under practical operating constraints, while Lian [29] proposed a shared parking-spot reservation and allocation model that gives priority to commuting demand. These approaches improve parking reliability and reduce uncertainty before arrival. However, reservation is treated as a separate service placed on top of parking-lot selection. The internal travel from the parking-lot entrance to a specific reserved spot, and from that spot to a pedestrian exit, is not represented in any of them. This is the internal structure that DPSOF adds, since each available spot is modeled as a graph node connected through weighted internal driving and walking links rather than as part of an undivided parking lot.

Parking allocation has received considerable attention because the available parking facilities have to be distributed among competing drivers. Tran Thi Kim et al. [11] formulated parking assignment as a problem of minimizing parking expenses while balancing demand among several parking lots. Tang et al. [10] introduced the SCP-tree method for identifying multiple nearby parking spots

while reducing group travel cost. Hassija et al. [9, 47] used virtual voting and adaptive pricing to support decentralized slot allocation. Abdelmagid et al. [25] proposed a dynamic modeling framework for real-time parking allocation, while Amidu et al. [27] developed a primal-dual algorithm for online parking allocation. Zheng et al. [28] examined parking-space allocation under peak demand, and Hu et al. [30] used multi-objective intelligent optimization for parking-space allocation. Although these studies improve parking utilization, fairness, and demand distribution across multiple parking lots, they do not consider a dynamic recommendation process in which parking decisions are repeatedly recomputed using the vehicle’s current location together with updated traffic and parking conditions. The frameworks are built around this repeated recomputation, since the route-and-parking-lot decision is re-evaluated at every intersection the driver reaches instead of being fixed at the start of the route.

Real-time and automated parking management systems have also been proposed. Kulkarni et al. [26] developed a smart automated and real-time parking management system, while Bhimavarapu et al. [43] presented an end-to-end system that combines real-time parking-space detection and path finding. Xie et al. [41] introduced a self-optimization method for multi-style smart parking using experience, knowledge, and data. Baikani et al. [45] proposed NeuroPark Guide as an artificial-intelligence-based parking guidance system. These systems show that automation and adaptive decision-making are practical. However, they treat detection, guidance, or system management as separate concerns rather than combining route-dependent driving time, walking time, parking cost, and individual parking-spot characteristics within one objective function. The frameworks instead evaluate all of these factors together within a single weighted objective, first at the parking-lot level and then at the parking-spot level.

The supply of parking facilities under uncertainty has also been investigated. Wang et al. [16] formulated the dynamic supply of parking permits as a stochastic control problem. Ahmed et al. [8] proposed a paging-based method for improving the utilization of smart parking systems. Nakazato et al. [31] investigated parking-lot allocation using rematching and dynamic parking-fee design. These approaches show that parking supply, rematching, and pricing can be adjusted to changing demand. However, the resulting decisions remain at the system level or the parking-lot level. None of them connects the selected parking lot to an individual available parking spot through internal vehicle and pedestrian links, which is the extension provided by DPSOF.

Beyond parking-lot recommendation, several studies have considered path planning and guidance to individual parking spaces. Cheng and Kang [35] applied a Genetic Algorithm to the path-planning problem for automatic navigation to parking spaces inside parking lots. Dong et al. [38] developed a chaotic Particle Swarm Optimization method for parking-space guidance, while Wang et al. [39] used parallel PSO for parking guidance in complex parking environments. Cheng and Nian [34] examined parking-lot optimization using a Genetic Algorithm, and Adeyiga et al. [40] compared GA and PSO in an intelligent parking system. These studies confirm that search-based methods can guide a vehicle toward a

parking space. However, in each of these studies, the parking spot is selected only after the parking lot has already been identified, without jointly considering both recommendation levels. The parking spot is not modeled as a separate graph node with its own availability, spot-specific parking cost, internal driving time, internal walking time, external walking time, and pedestrian-exit connection. DPSOF represents this missing decision layer by modeling every currently available spot as a separate graph node with its own cost and traversal attributes.

Other adaptive methods have been developed for dynamic parking guidance and street-parking optimization. Hassoune et al. [42] proposed a dynamic parking-guidance architecture based on Ant Colony Optimization and multi-agent systems. Akandwanaho and Govender [44] used a memetic algorithm for dynamic street parking-space optimization. Dimlo et al. [37] developed a GA-RBF parking-management model intended to reduce fuel consumption, travel time, and parking cost. These methods support adaptive and scalable parking guidance. However, their objectives and network representations remain at the lot level or the street-segment level. None of them evaluates currently available spots through both external road-network travel and internal parking-facility movement, which is the problem that DPSOF is formulated to solve.

From a solution-method perspective, exact optimization models are useful for obtaining optimal solutions and for validating parking formulations on small instances. Multi-objective parking allocation [24, 30], real-time parking allocation [25], online allocation [27], peak-demand allocation [28], and demand balancing across multiple parking lots [11] show how mathematical and algorithmic formulations can represent parking decisions. However, as the number of nodes, parking facilities, routes, and parking spots increases, the number of possible decisions grows rapidly, and none of these formulations is evaluated for the repeated, current-location-based recomputation that a moving vehicle requires. For this reason, exact optimization is used only for small-scale validation in this thesis, while a scalable heuristic and meta-heuristic baselines are developed for medium-scale and large-scale networks.

Heuristic and meta-heuristic methods are commonly used when exact optimization becomes computationally expensive. Ben Helal et al. [18] developed a hybrid GA for a green vehicle-routing problem, which shows the suitability of GA for transportation and routing optimization. Hu and Ma [19] applied PSO to task assignment and path planning, while Wang et al. [20] used an improved ACO method for congestion-aware transportation-route optimization. Within parking research, GA has been used for parking-lot selection [34], path planning to parking spaces [35], and fuel-time-cost optimization [37]. PSO has been applied to parking-space guidance [38, 39], and ACO has been used for dynamic parking guidance [42]. These studies confirm that GA, PSO, and ACO are suitable baselines for scalable parking optimization. However, these studies do not jointly address the repeated recommendation of a parking lot under changing traffic conditions and the recommendation of an individual parking spot within the selected parking lot.

Despite these advances, the existing research directions are often studied

separately. Parking-lot recommendation methods generally treat each lot as a single parking option, while parking-space studies usually begin only after the lot has already been selected, and limited attention has been given to a connected framework that first updates the parking-lot recommendation from the driver’s current location and then extends the decision to an individual available parking spot. The parking spots are also not fully represented, since internal driving time from the entrance to the spot, internal walking time from the spot to a pedestrian exit, external walking time from the exit to the final destination, and spot-specific parking cost are rarely considered together. In addition, few studies compare exact optimization, heuristic, and meta-heuristic methods under the same graph structure, parameters, constraints, and cost components.

These limitations motivate the two frameworks developed in this thesis. The Dynamic Parking Lot Optimization Framework (DPLOF) repeatedly updates the route and the parking-lot recommendation. The Dynamic Parking Spot Optimization Framework (DPSOF) then extends this process by representing currently available parking spots as separate decision options, so that parking recommendation is addressed at both the parking-lot level and the parking-spot level. Table 2.1 summarizes the main studies discussed in this chapter, together with their problem focus, solution approach, and limitations with respect to the proposed work.

Table 2.1: Summary of related work on smart parking recommendation, allocation, and parking spot assignment

Author / Reference	Problem Studied	Approach / Solution	Key Limitations
Aljohani et al. [21]	Parking solutions for smart cities	Survey of sensing, guidance, reservation, pricing, and parking management	Survey-level analysis; no integrated dynamic lot and spot recommendation
Alam et al. [22]	IoT-driven parking management	Survey of IoT architectures, sensing approaches, limitations, and future work	Focuses on system architecture rather than route and parking optimization
Agrawal et al. [23]	Smart parking techniques and urban-space use	Review of parking technologies and innovations	Does not formulate real-time parking recommendation or spot assignment
Almalki and Ragab [24]	Smart parking in sustainable environments	Review of challenges and future perspectives	Broad survey; no detailed routing or parking-spot model

Continued on next page

Table 2.1 continued from previous page

Author / Reference	Problem Studied	Approach / Solution	Key Limitations
Li et al. [1]	Near-real-time parking occupancy prediction	Integrated occupancy-prediction method across parking lots	Prediction-focused; no route or parking-lot recomputation
Li et al. [2]	Spatial-temporal parking occupancy prediction	Dynamic graph transformer using relationships among parking areas	Improves prediction but does not optimize parking decisions
Zhao et al. [3]	On-street parking availability prediction	Parking meters used as virtual sensors	No route optimization or spot-level assignment
Song et al. [4]	Parking vacancy prediction with missing sensors	Collaborative prediction under incomplete sensor data	Focuses on data recovery and prediction rather than recommendation
Kim et al. [12]	Outdoor parking monitoring	Unmanned aerial vehicle-based occupancy monitoring	Monitoring only; no parking recommendation framework
Assemi et al. [13]	On-street parking occupancy inference	Inference using payment transaction data	No route, cost, or walking-distance optimization
Nie et al. [17]	Public curb-parking demand estimation	Demand estimation using nearby activity and point-of-interest data	Demand estimation only; no driver-level recommendation
Boateng et al. [5]	Automated valet parking reservation and pricing	Multi-agent reinforcement learning framework	Focused on pricing and reservation rather than repeated route recomputation
Saharan et al. [6, 33]	Dynamic on-street parking pricing and allocation	DyPARK dynamic pricing and allocation scheme	Primarily system-level management; limited internal parking-lot representation
Hu et al. [7]	Valet parking under demand uncertainty	Dynamic pricing and autonomous valet parking management	No connected lot-level and spot-level routing model
Ahmed et al. [8]	Parking-space utilization	Paging-based reservation and allocation technique	Does not account for dynamic route changes or internal travel
Hassija et al. [9, 47]	Parking-slot allocation and adaptive pricing	Virtual voting and adaptive pricing framework	Allocation-focused; no current-location-based route recomputation

Continued on next page

Table 2.1 continued from previous page

Author / Reference	Problem Studied	Approach / Solution	Key Limitations
Tang et al. [10]	Nearby parking-spot search	SCP-tree for minimizing group travel cost	Does not model dynamic traffic and internal lot movement together
Tran Thi Kim et al. [11]	Parking assignment across multiple lots	Optimization for expense minimization and demand balancing	No repeated recommendation from the moving vehicle’s current location
Hu et al. [14, 49]	Competition and reservation services	CRPark smart parking framework	Reservation-focused; internal spot traversal not fully modeled
Wang et al. [15, 32]	Parking efficiency and reservation	PRPSV using parking-space view	Space-view management without full external and internal route cost
Wang et al. [16]	Parking supply under uncertainty	Stochastic control of parking permits	System-level supply decision; no individual route and spot assignment
Abdelmagid et al. [25]	Real-time parking allocation	Dynamic modeling framework	No detailed internal driving and pedestrian-exit representation
Kulkarni et al. [26]	Automated real-time parking management	Integrated smart parking management system	System-oriented; limited optimization of route and spot-level costs
Amidu et al. [27]	Online parking allocation	Primal-dual online allocation algorithm	Does not integrate repeated routing and walking constraints
Zheng et al. [28]	Allocation under peak demand	Intelligent parking-lot allocation model	Focuses on demand conditions rather than dynamic routing
Lian [29]	Shared parking reservation and allocation	Reservation model prioritizing commuting demand	No internal parking-facility traversal model
Hu et al. [30]	Multi-objective parking-space allocation	Intelligent optimization algorithm	No current-location-based dynamic lot recommendation
Nakazato et al. [31]	Parking-lot allocation and fee design	Rematching and dynamic parking fees	Lot-level allocation; spot-specific travel components not modeled

Continued on next page

Table 2.1 continued from previous page

Author / Reference	Problem Studied	Approach / Solution	Key Limitations
Cheng and Nian [34]	Parking-lot optimization	Genetic Algorithm-based analysis	Limited treatment of real-time traffic and connected lot/spot decisions
Cheng and Kang [35]	Navigation to parking spaces	Genetic Algorithm for parking-space path planning	Applied after facility selection; no parking-lot recommendation stage
Dimlo et al. [37]	Fuel, time, and cost reduction	GA-RBF parking-management model	Does not represent every available spot with internal travel links
Dong et al. [38]	Parking-space guidance	Chaotic Particle Swarm Optimization	Guidance-focused; no unified external and internal cost model
Wang et al. [39]	parking recommendation	Parallel Particle Swarm Optimization	No dynamic parking-lot recommendation from current vehicle location
Adeyiga et al. [40]	Comparison of GA and PSO	Comparative intelligent parking analysis	Method comparison without the proposed two-stage graph model
Xie et al. [41]	Adaptive smart parking	Self-optimization using experience, knowledge, and data	General optimization; no explicit internal spot assignment structure
Hassoune et al. [42]	Dynamic parking recommendation	Ant Colony Optimization and multi-agent architecture	No detailed spot-specific cost and pedestrian-exit modeling
Bhimavarapu et al. [43]	Real-time space detection and path finding	End-to-end detection and navigation system	Does not jointly optimize lot selection and spot assignment
Akandwanaho and Govender [44]	Dynamic street parking optimization	Memetic Algorithm	Street-parking focus; no parking-facility internal movement
Baikani et al. [45]	AI-based parking recommendation	NeuroPark Guide	General guidance; no exact and metaheuristic comparison under one formulation
He et al. [46]	Occupancy prediction and customer preferences	Prediction-based smart parking management	No dynamic route recomputation or internal spot links

Continued on next page

Table 2.1 continued from previous page

Author / Reference	Problem Studied	Approach / Solution	Key Limitations
Zhang et al. [48]	Reservation-enhanced valet parking	Autonomous valet parking with reservation constraints	Internal movement considered partially; no connected urban lot recommendation
Ben Helal et al. [18]	Green vehicle routing	Hybrid Genetic Algorithm	General routing problem; not designed for parking recommendation
Hu and Ma [19]	Task assignment and path planning	Dynamic-cluster Particle Swarm Optimization	Not specific to parking-lot and parking-spot decisions
Wang et al. [20]	Congestion-aware route optimization	Improved Ant Colony Optimization	Transportation routing focus; no parking cost or internal walking model

Chapter 3

Graph Creation

To evaluate the performance of the proposed parking recommendation frameworks, a set of different road networks is used. This set includes graphs that are synthetically generated urban-like road networks and a real-road-network case study developed in Simulation of Urban MObility (SUMO). The synthetic graphs are used to examine the behavior of the frameworks under controlled network sizes and parameter variations, while the SUMO-based graph is used to evaluate the frameworks on a more realistic road network.

The evaluation begins with the construction of the synthetic road networks and is followed by the use of the SUMO-based real-road network for fixed-network analysis. To support both parking lot-level recommendation and parking spot-level recommendation, parking lots, parking spots, source and destination nodes, edge distances, travel times, walking distances, and parking costs are all defined as part of the graph-creation process. This chapter presents the graph generation methodology and describes the main components used to build the experimental environment for evaluating the proposed frameworks.

3.1 Synthetic Graph Generation

The synthetic graphs are generated as directed urban-like road networks at different network sizes. They are used as controlled test cases for evaluating the proposed parking recommendation frameworks. Each graph is generated using node locations and edge links that resemble the structure of a realistic urban road network. A graph consists of nodes and directed edges, where the nodes represent intersections or locations and the edges represent road connections between two nodes.

Initially, the nodes are randomly generated and assigned coordinates within the network area so that the resulting road network does not follow a regular grid pattern. After the nodes are generated, directed edges are assigned as road connections between them. Some edges connect nearby nodes, while the remaining edges connect nodes that are farther apart. This creates multiple possible routes for a driver to travel from a source node to a destination node, which is similar to travel in an urban road network.

The synthetic datasets are generated as small-scale, medium-scale, and large-scale road networks. As the network size increases, additional nodes and edges are added to the graph. The network area is also scaled so that the graph

does not become too crowded. This helps to preserve a comparable road structure across the small-scale, medium-scale, and large-scale cases. As a result, each graph becomes larger and more complex than the previous one, while the generated graphs remain structurally consistent across the different network sizes.

3.2 Real-Life Road Network

Alongside the synthetic graphs, a real-road-network graph developed using SUMO is used to evaluate the proposed frameworks under realistic road-network conditions. The road-network data were obtained from OpenStreetMap through Overpass Turbo and were then imported into SUMO for simulation and pre-processing. Unlike the synthetic networks, which are generated artificially for controlled experiments, the SUMO-based graph preserves the actual road layout and connectivity of the selected area. This makes it more suitable for examining how the model behaves under realistic road conditions.

The SUMO road network is then converted into a graph representation in which intersections and other relevant road points are treated as nodes, and road segments are treated as directed edges. This produces a directed road graph that can be used within the same route-and-parking framework as the synthetic networks. The SUMO-based graph used in this thesis contains 2055 nodes and serves as the fixed real-road network for the experimental analysis presented later. Since it is derived from an actual road map, it provides a more realistic setting than the synthetic cases.

3.3 Parking Lot Positioning

Parking lots are represented as candidate parking nodes in the driving network. Their placement is important because the problem studied in this thesis is not only a routing problem. The driver first travels by vehicle to a selected parking lot and then walks from that parking lot to the final destination.

In the baseline setting, approximately 5% of the graph nodes are selected as candidate parking locations. These locations are selected near the destination region so that the parking decision remains meaningful. This follows the practical assumption that drivers usually look for parking close to their final destination.

For the parameter-sensitivity analysis, the number of candidate parking locations is varied by changing the parking-supply percentage, while the other parameters are kept fixed. This allows the effect of parking availability on route selection, walking distance, parking cost, and recommendation quality to be examined separately.

3.4 Parking Spot Generation and Positioning

After the parking lot nodes are selected, parking spots are generated for each candidate parking lot. This step supports the parking spot-level recommendation framework. While the parking lot-level recommendation framework selects a suitable parking lot, the spot-level recommendation framework selects a suitable individual spot inside the selected lot.

Each parking lot is assigned a variable number of parking spots rather than a fixed count. In the baseline setting, the number of spots per lot is generated within the range of 20 to 40, and approximately 10% of the generated spots are marked as available. Each parking lot is guaranteed to contain at least one available spot, which avoids cases in which a lot is included in the candidate set but contains no feasible option.

Each parking spot is assigned its own attributes, such as availability, spot-specific parking cost, internal driving time, internal walking time, external walking time, and proximity to the pedestrian exit. These attributes allow the model to compare different parking spots within the same parking lot. As a result, the recommendation process considers both the selected parking lot and the assigned parking spot.

Parking costs are assigned within controlled ranges. In the baseline setting, the parking cost varies from 8.0 to 20.5. Additional experiments use wider cost ranges to represent stronger variation in parking prices. This helps to examine how parking-cost variation affects the final recommendation.

For each parking candidate, the walking distance from the parking location to the destination is calculated using the spatial layout of the graph. A parking location is considered feasible only if this walking distance is within the maximum walking threshold $W_{\max}$. Therefore, the graph may contain several candidate parking locations, but only the feasible ones are treated as valid options for the driver.

3.5 Driver Source and Destination Positioning

The source and destination are selected so that each experiment produces a meaningful parking recommendation scenario. In the synthetic graphs, the source node is selected from the driving network and is placed far enough from the destination region. This avoids very short routes and ensures that both the driving decision and the walking decision affect the final result.

The destination is placed near the outer region of the network instead of being treated as an ordinary internal node. This supports the two-stage travel structure used in this thesis. In the first stage, the driver travels by vehicle from the source to a selected parking lot. In the second stage, the driver walks from the parking lot to the final destination.

For each scenario, the same source and destination are used for all compared methods. This ensures that the optimization model, the heuristic method,

and the meta-heuristic methods are evaluated under the same travel conditions. Therefore, any difference in performance is caused by the decision method rather than by a change in the source and destination.

3.6 Edge Weight and Travel-Time Assignment

Each directed road edge is assigned a distance and a travel time. The distance is calculated from the spatial positions of the two connected nodes. The travel time is then calculated from this distance and the driving speed assigned to that road segment.

In the synthetic networks, different road types are used to represent local roads, connector roads, and arterial roads. This introduces variation in the travel-time structure and makes the generated graphs more realistic than uniform random graphs. In the baseline setting, low-traffic driving speeds are sampled from 50 to 70 km/h. The other traffic scenarios use different speed ranges to represent very low, medium, and high traffic conditions.

When the traffic condition is changed, the travel times are recomputed while the road topology remains fixed. This allows the effect of traffic variation to be studied without changing the structure of the graph.

Walking travel is handled separately from vehicle travel. A fixed walking speed of 5.0 km/h is used to convert walking distance into walking time. This calculation is used in both parking lot-level recommendation and parking spot-level recommendation.

For the edge-density analysis, additional bidirectional driving links are added between eligible node pairs, while the other parameters remain fixed. The resulting number of directed edges is used to represent the edge-density setting in the experimental results.

3.7 Summary

This chapter described the graph creation process used in this thesis. Synthetic urban-like directed graphs are generated at different scales to support controlled evaluation, and a SUMO-based real-road-network case study is used to test the frameworks on a more realistic road structure.

Parking lots are modeled as candidate nodes near the destination region, and parking spots are generated inside the selected parking lots to support spot-level recommendation. The source and destination are placed so that the two-stage travel process of driving to a parking lot and then walking to the final destination is preserved. Finally, distances, travel times, walking times, and parking costs are assigned to support the evaluation of both parking lot-level recommendation and parking spot-level recommendation. These graph components provide the basis for the experimental analysis presented in the following chapters.

Chapter 4

Proposed Dynamic Parking Optimization Framework for Parking Lot Selection

Parking lot search contributes to congestion, travel delay, and inefficient use of road infrastructure, especially in dense urban areas. Existing parking recommendation approaches may become less effective when traffic conditions change while the driver is still driving toward the recommended parking lot. To address this limitation, this chapter proposes a Dynamic Parking Lot Optimization Framework (DPLOF) that updates the traffic information and recalculates the route and the parking-lot recommendation from the vehicle's current location. The objective is to minimize the total travel time, which includes both driving time and walking time, together with the parking cost, while satisfying user preferences such as a maximum walking distance and a maximum parking cost. An optimization model is proposed to validate the formulation on small-scale networks, while a heuristic approach is developed for medium-scale and large-scale networks where solving the optimization model repeatedly becomes computationally impractical. The heuristic is then evaluated against meta-heuristic baselines, namely Genetic Algorithm, Particle Swarm Optimization, and Ant Colony Optimization, using several performance metrics under controlled variations in network and parking conditions. The evaluation is carried out using synthetic urban-like datasets and a Simulation of Urban MObility (SUMO) based real-road-network case study for practical validation. The results suggest that DPLOF is suitable for real-time parking lot-level recommendation in urban environments.

4.1 Introduction

The rapid growth of population and vehicle ownership in dense urban areas has made parking a persistent challenge in smart transportation systems. Drivers often spend a considerable amount of time searching for a parking lot near their destination, which increases congestion, travel delay, fuel consumption, emissions, and driver frustration [1]. Therefore, efficient parking management is important for smart transportation systems, since it directly influences congestion mitigation, travel-time reduction, and overall road-network performance [2]. For this reason, Smart Parking Systems (SPS) have been developed to improve the moni-

toring, estimation, and prediction of parking-space occupancy [3]. In comparison with traditional search methods, these systems help drivers identify parking options more efficiently [4].

However, when a driver is already moving through a road network, selecting a suitable parking lot requires more than identifying parking availability alone. The decision has to minimize the total travel time, which includes the driving time from the current location to the parking lot and the walking time from the selected lot to the destination, together with the fixed parking cost, while also considering the current traffic condition and the user preferences. These factors show that parking-lot recommendation is affected by both route-related conditions and parking-system characteristics. Prior studies further show that reservation policies [5], competition among parking facilities [6], and pricing mechanisms [7] can influence parking access, demand distribution, utilization, and selection behavior. These studies form the basis for a broader body of SPS research that addresses different aspects of parking management.

Existing SPS research has addressed several important functions, including allocation, pricing, assignment, monitoring, reservation, and demand estimation. These studies show the importance of space utilization [8], adaptive pricing under changing demand [9], travel cost among nearby parking lots [10], demand balancing across parking lots [11], and occupancy estimation under incomplete sensing infrastructure [12]. Other works have considered transaction-based occupancy inference [13], reservation before arrival [14], parking-space-view-based service improvement [15], uncertain parking availability [16], and curb parking demand influenced by nearby buildings, businesses, and activity areas [17]. Despite these advances, limited attention has been given to parking-lot recommendation methods that update both the route and the parking decision while the vehicle is already on route.

To address this gap, this chapter proposes a Dynamic Parking Lot Optimization Framework (DPLOF) for real-time parking lot-level recommendation under changing traffic conditions. DPLOF performs repeated route-and-parking-lot recomputation during travel using continuously updated real-time traffic information. At each intersection, the system evaluates the available parking lots through an objective function that minimizes the total travel time, including both driving time and walking time, together with the fixed parking cost. Based on this evaluation, the route and the parking-lot recommendation are recomputed from the current vehicle location until the vehicle reaches the selected lot. This process avoids any dependence on future traffic prediction or future parking-price estimation. DPLOF follows a two-level route-and-parking-lot decision process in which the driver first travels from the current vehicle location to the selected parking lot and then walks from that lot to the destination. The recommendation also satisfies user preferences such as maximum walking distance and maximum parking cost. In this work, at least one parking lot is assumed to be available, and the parking cost is treated as a fixed value for each lot.

To solve the proposed parking lot-level recommendation problem, this chapter first presents an optimization model for small-scale networks. However,

since repeated route-and-parking-lot recomputation becomes computationally expensive as the network size increases, a heuristic solution approach is developed for medium-scale and large-scale networks in order to support real-time recommendation under changing traffic conditions. The proposed heuristic is evaluated against three meta-heuristic baselines, namely Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Ant Colony Optimization (ACO). GA is widely used for complex routing and transportation optimization problems [18], PSO has been applied to path planning and search-based optimization problems [19], and ACO has been used for congestion-aware route optimization and related transportation problems [20].

The proposed approach is evaluated using controlled synthetic urban-like datasets across different performance metrics and parameter variations. The evaluation considers driving time, walking time, parking cost, objective value and runtime under controlled variations in network size, edge density, parking supply, walking-distance threshold, traffic level, parking-cost threshold, and parking-cost range. A SUMO-based real-road-network instance is also used to validate the framework under a realistic urban topology. The main contributions of this chapter are as follows:

- A dynamic parking lot optimization framework is proposed to minimize driving time, walking time, and parking cost under user-defined constraints by updating the traffic information and recomputing the route and the parking-lot recommendation at each intersection without relying on future conditions.
- An optimization model is proposed for small-scale validation, and a heuristic solution approach is developed for medium-scale and large-scale networks where solving the optimization model repeatedly becomes computationally expensive.
- The proposed heuristic is evaluated against GA, PSO, and ACO using travel time, walking distance, parking cost, and runtime under controlled parameter variations, including network size, edge density, parking supply, traffic level, walking-distance threshold, parking-cost threshold, and parking-cost range.
- A SUMO-based real-road-network instance is used to validate the framework under a realistic urban topology.

The remainder of this chapter is organized as follows. Section 4.2 reviews related work on urban parking systems, parking optimization, and scalable search-based methods. Section 4.3 presents the system model. Section 4.4 details the problem formulation. Section 4.5 describes the proposed heuristic and the meta-heuristic baselines. Section 4.6 presents the experimental setup, the synthetic evaluation, the performance analysis, and the SUMO-based validation. Section 4.7 concludes the chapter and outlines future perspectives.

4.2 Related Work

SPS have emerged as an important solution for reducing the time drivers spend searching for parking in urban areas. Traditional parking systems provide lim-

ited support and often require drivers to search manually for an available parking lot or space, which increases congestion, fuel consumption, emissions, and travel delay. With the development of Intelligent Transportation Systems (ITS), prior research has improved parking-related services through occupancy prediction, pricing strategies, reservation services, allocation models, and monitoring methods. However, these studies often address individual components of the parking problem rather than the complete parking lot-level recommendation process for a driver who is already moving through an urban road network.

A major direction in the literature focuses on parking occupancy prediction. Accurate prediction of vacant parking spaces can improve parking guidance and reduce unnecessary search time. Li et al. [1] proposed a near-real-time method for predicting parking occupancy across different parking lots, and their later work improved the prediction accuracy by capturing the relationships among nearby parking areas and the occupancy changes over time [2]. Zhao et al. [3] used parking meters as virtual sensors to estimate on-street parking availability at the urban scale, and Song et al. [4] addressed missing sensor data through collaborative vacancy prediction. These studies improve parking availability estimation, but they mainly focus on prediction and sensing rather than on repeated route-and-parking-lot recomputation using current traffic information while the vehicle is already on route.

Another research stream focuses on parking pricing, competition, and reservation management. Boateng et al. [5] proposed a reservation and pricing framework for automated valet parking and charging using multi-agent reinforcement learning. Saharan et al. [6] developed a dynamic pricing and allocation model for smart on-street parking, Hu et al. [7] studied parking management under demand uncertainty, and Ahmed et al. [8] proposed a paging-based reservation strategy to improve parking-slot utilization. These studies show that pricing and reservation mechanisms can improve system-level parking management. However, their primary focus is on the operation of the parking system rather than on driver-centered parking lot-level recommendation that jointly evaluates route-dependent driving time, walking time, fixed parking cost, and user preferences.

Parking allocation and assignment have also been widely studied. Hassija et al. [9] proposed a decentralized parking-slot allocation framework using virtual voting and adaptive pricing to improve fairness and security. Tang et al. [10] introduced a spatial index tree, called the SCP-tree, to identify multiple nearby parking spots while reducing group travel cost, and Tran Thi Kim et al. [11] formulated parking assignment as a demand-balancing problem across multiple parking lots. These works show the importance of fair allocation, reduced travel cost, and balanced use of parking facilities. However, they generally do not model parking lot-level recommendation as a repeated two-level route-and-parking decision in which the driving route is updated from the current vehicle location under changing traffic conditions.

Several studies have focused on parking monitoring, demand estimation, occupancy inference, and reservation support. Kim et al. [12] explored the use of unmanned aerial vehicles for parking occupancy monitoring. Assemi et al. [13] in-

ferred parking occupancy from payment transaction data when complete sensing infrastructure is unavailable. Hu et al. [14] examined competition and reservation services in smart parking systems. Wang et al. [15] proposed a reservation optimization method based on a parking-space view, Wang et al. [16] studied the dynamic supply of parking spaces under uncertainty, and Nie et al. [17] estimated public street-parking demand using nearby activity locations such as commercial areas, offices, and public facilities. These studies improve parking data collection, occupancy estimation, demand understanding, and reservation support. However, they mainly strengthen individual SPS functions rather than combining current traffic information, route-dependent driving time, walking-distance feasibility, fixed parking cost, and user preferences within a single real-time parking lot-level recommendation framework.

Since parking lot-level recommendation in urban networks involves a large number of possible route and parking combinations, heuristic and meta-heuristic methods are often used in transportation optimization. Ben Helal et al. [18] showed the effectiveness of a hybrid GA for vehicle routing. Hu et al. [19] used PSO for path planning in complex environments, and Wang et al. [20] developed an improved ACO approach for congestion-aware routing. These studies show that GA, PSO, and ACO are relevant baselines for large-scale routing and search-based optimization problems. However, they are not designed for the proposed parking lot-level recommendation setting, where the solution has to combine current-location-based route recomputation, parking-lot selection, walking-distance feasibility, and fixed parking cost under the same objective structure.

Overall, existing research has improved parking prediction, pricing, allocation, monitoring, reservation, and transportation optimization. However, these directions are often studied separately and do not fully address real-time parking lot-level recommendation for a vehicle that is already moving through the road network. Limited attention has been given to methods that repeatedly recompute both the route and the parking-lot recommendation from the current vehicle location using updated traffic information, while considering walking time, walking-distance constraints, fixed parking cost, and user preferences without relying on future traffic prediction or future parking-price estimation. Motivated by this gap, the proposed DPLOF formulates parking lot-level recommendation as a two-level route-and-parking-lot optimization problem, uses an optimization model for small-scale validation, develops a scalable heuristic for medium-scale and large-scale networks, and evaluates the heuristic against GA, PSO, and ACO using synthetic urban-like datasets and a SUMO-based real-road-network instance.

4.3 System Model and Problem Description

This section describes the system model of the proposed DPLOF. The framework consists of a driver, a centralized recommendation server, real-time traffic information, parking-lot information, and an urban road network. These components work together to recommend a suitable parking lot under changing traffic condi-

tions. In this work, at least one parking lot is assumed to be available, and the parking cost is treated as a fixed value for each parking lot.

The urban environment is modeled as a directed graph $G = (N, E)$, where N denotes the set of road-network nodes and E denotes the set of directed edges. The nodes represent road intersections, the driver's source node S , the current vehicle location A , the destination node D , and the parking lots $p_i \in P$, where P is the set of parking lots. Each directed edge $(i, j) \in E$ represents a road segment and is associated with a fixed travel distance d_{ij} and a travel time t_{ij} . The distance d_{ij} remains fixed, while t_{ij} changes according to the current traffic condition. Therefore, the road-network structure remains fixed during the recommendation, but the edge travel times are updated as the traffic condition changes.

DPLOF follows a two-level route-and-parking-lot decision process. At the first level, the driver travels from the current vehicle location A to a selected parking lot p_i . At the second level, the driver walks from the selected parking lot to the destination D . Let $t_{p_i}^{drive}$ denote the estimated driving time from the current vehicle location A to parking lot p_i under the current traffic condition. The walking distance from p_i to D is denoted by $d_{p_i}^{walk}$. Since the walking speed is assumed to be constant, the corresponding walking time $t_{p_i}^{walk}$ can be derived from $d_{p_i}^{walk}$ when required.

As the vehicle moves through the road network, the centralized recommendation server updates the traffic state and recomputes the route and the parking-lot recommendation from the current vehicle location. When the vehicle reaches a new intersection, the system retrieves the latest traffic information, evaluates the parking lots using the objective function, and updates the recommended route and parking lot. This repeated recomputation continues until the vehicle reaches the selected parking lot. The model uses current traffic information only and does not forecast future traffic conditions or future parking-price changes.

Fig. ?? illustrates the system workflow. The parking lots p_1 and p_2 represent the parking-lot options near the destination. The quantities $t_{p_1}^{drive}$ and $t_{p_2}^{drive}$ denote the driving times from the current vehicle location to the respective parking lots, $t_{p_1}^{walk}$ and $t_{p_2}^{walk}$ denote the walking times from those parking lots to the destination, and C_{p_1} and C_{p_2} denote the fixed parking costs. As the driver progresses through the road network, the system updates the traffic information and re-evaluates the route and the parking-lot recommendation using the total travel time, the fixed parking cost, and the user preferences.

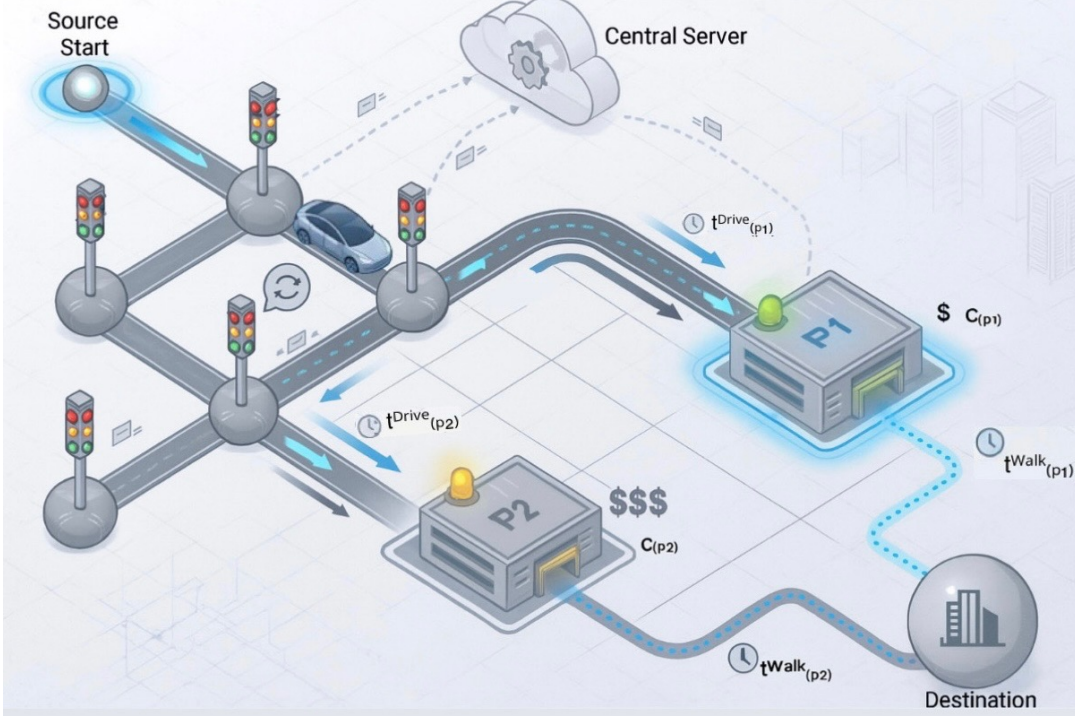


Figure 4.1: System model of the proposed dynamic parking lot optimization framework.

Each parking lot p_i is characterized by two main attributes, which are the walking distance $d_{p_i}^{walk}$ and the parking cost C_{p_i} . Parking availability is assumed in this work, and each generated instance includes at least one parking lot that satisfies the user preferences. A parking lot is excluded from consideration if its walking distance exceeds the maximum acceptable walking distance W_{max} or if its parking cost exceeds the maximum acceptable parking cost C_{max} . Therefore, the feasible parking-lot set is determined by the user-defined walking-distance and parking-cost constraints.

At each decision step, the system evaluates the tuple $(t_{p_i}^{drive}, t_{p_i}^{walk}, C_{p_i}, d_{p_i}^{walk})$ for every $p_i \in P$. The parking lots that satisfy the walking-distance and parking-cost constraints are retained as feasible options. These feasible parking lots are then evaluated by the optimization model, the proposed heuristic, and the meta-heuristic baselines under the same system model. The recommendation process is repeated as the driver moves from one intersection to another until the selected parking lot is reached.

Several assumptions are made in the system model. The urban road-network structure and the edge distances remain fixed during the recommendation interval, while the edge travel times may change according to the current traffic condition. The parking cost and the walking distance from each parking lot to the destination remain fixed. Parking availability is assumed, and at least one feasible parking lot is included in each generated instance. The model uses current traffic information only, and future traffic conditions and future parking-price changes are not forecasted. The notations used throughout the system

model and the problem formulation are summarized in Table 4.1.

Table 4.1: Notation used in the DPLOF system model and problem formulation

Parameters	
$G = (N, E)$	Directed urban road network
N	Set of road-network nodes
E	Set of directed road-network edges
S	Driver source node
D	Driver destination node
A	Current vehicle location during route progression
P	Set of parking lots
p_i	Parking lot i , where $p_i \in P$
$(i, j) \in E$	Directed edge from node i to node j
t_{ij}	Travel time on edge (i, j) under current traffic
d_{ij}	Travel distance on edge (i, j)
$t_{p_i}^{drive}$	Driving time from A to parking lot p_i
$t_{p_i}^{walk}$	Walking time from parking lot p_i to destination D
C_{p_i}	Fixed parking cost at parking lot p_i
$d_{p_i}^{walk}$	Walking distance from parking lot p_i to D
$W_{\max}$	Maximum acceptable walking distance
$C_{\max}$	Maximum acceptable parking cost
α	Weight balancing total travel time and parking cost
Decision Variables	
x_{ij}	1 if edge (i, j) is selected; 0 otherwise
$x_{L[p], D}$	1 if the walking link from parking lot p to D is selected; 0 otherwise
T_i	Accumulated travel time upon reaching node i
H_p	Heuristic score associated with parking lot p

4.4 Problem Formulation

This section formulates parking lot-level recommendation in DPLOF as a two-level route-and-parking-lot optimization problem over the network defined in the system model. The objective is to determine a route in which the driver first travels by vehicle from the current location to a selected parking lot and then walks from that lot to the destination, while jointly minimizing the total travel time and the parking cost under the user-defined constraints. At the initial decision step, the starting node corresponds to the source node S . During route

progression, the same formulation is re-evaluated by replacing S with the driver's current node A whenever the vehicle reaches a new intersection. In this structure, the destination is not reached directly through the driving network. Instead, the driving route terminates at a selected parking lot, and the final connection to the destination is completed through a parking-to-destination walking edge.

4.4.1 Network Representation and Decision Variables

Based on the network representation in the system model, decision variables are introduced for route and parking-lot selection. Using $G = (N, E)$, binary route-selection variables are defined over E , which contains both driving edges and parking-to-destination walking edges. For each $(i, j) \in E$, the binary decision variable in Equation (4.1) is defined as follows:

$$x_{ij} = \begin{cases} 1, & \text{if edge } (i, j) \text{ is included in the selected route,} \\ 0, & \text{otherwise.} \end{cases} \quad (4.1)$$

Let T_i denote the accumulated travel time upon reaching node $i \in N$. Since the route includes both the driving path to a parking lot and the final walking edge to the destination, the accumulated value at node D , denoted by T_D , represents the total travel time from the starting node to the destination.

As described in the system model, let P denote the set of parking lots considered during the recommendation, and let $L[p]$ denote the graph node corresponding to p . The selected parking lot is identified through the walking edge $(L[p], D)$. The parameter $C[p]$ denotes the fixed parking cost associated with p , and the variable $Cost$ denotes the parking cost of the selected lot. In the implementation, an auxiliary parking-selection variable is linked to $(L[p], D)$. However, the formulation is presented directly using the route variable associated with the final walking edge.

4.4.2 Objective Function

The objective function in Equation (4.2) minimizes a weighted combination of the total travel time and the parking cost:

$$\min \alpha T_D + (1 - \alpha) Cost, \quad (4.2)$$

where T_D denotes the total travel time, including both driving and walking times, $Cost$ is the parking cost of the selected parking lot, and $\alpha \in [0, 1]$ is the objective-weight parameter that controls the trade-off between travel time and parking cost. Equation (4.2) therefore balances travel efficiency and parking cost while reflecting the user preferences.

4.4.3 Routing Feasibility Constraints

The selected edges have to form a valid and continuous route from the starting node to a parking lot and then from that parking lot to the destination. To enforce this structure, the flow-conservation constraint in Equation (4.3) is imposed over the network:

$$\sum_{j:(i,j) \in E} x_{ij} - \sum_{j:(j,i) \in E} x_{ji} = \begin{cases} 1, & \text{if } i = S, \\ -1, & \text{if } i = D, \\ 0, & \text{otherwise.} \end{cases} \quad (4.3)$$

The net outgoing flow at S is set to 1, which shows that the route begins at the source or current node, while the net incoming flow at D is set to 1, which shows that the route terminates at the destination. All intermediate nodes satisfy balanced flow. Since the destination is reachable only through a parking-to-destination walking edge, Equation (4.3) ensures that the route first reaches a parking lot through the driving network and then arrives at the destination through the final walking connection.

To maintain route consistency, Equation (4.4) prevents the route from using both directions of the same bidirectional road segment:

$$x_{ij} + x_{ji} \leq 1, \quad \forall (i, j) \in E \text{ such that } (j, i) \in E. \quad (4.4)$$

4.4.4 Travel-Time Accumulation

The accumulation of travel time along the route is modeled in Equation (4.5):

$$x_{hi} = 1 \Rightarrow T_i = T_h + t_{hi}, \quad \forall (h, i) \in E, i \neq S. \quad (4.5)$$

Here, t_{hi} represents the travel time on edge (h, i) under the current traffic condition. For a selected edge, the accumulated travel time at node i has to equal the accumulated travel time at node h plus the travel time associated with edge (h, i) . Since the edge set includes both driving edges and the final parking-to-destination walking edge, the same propagation rule applies to both levels of the route.

The starting-node time is initialized using Equation (4.6):

$$T_S = 0. \quad (4.6)$$

As a result, T_D represents the total travel time of the complete route.

4.4.5 Parking Selection and Cost Constraints

The selected parking lot is determined through the final parking-to-destination walking edge. Since the complete route reaches the destination through exactly one such edge, the parking-selection constraint in Equation (4.7) ensures that a single parking lot is selected:

$$\sum_{p \in P: (L[p], D) \in E} x_{L[p], D} = 1. \quad (4.7)$$

The parking-cost variable is linked to the selected parking lot using Equation (4.8):

$$Cost = \sum_{p \in P: (L[p], D) \in E} C[p] x_{L[p], D}. \quad (4.8)$$

This ensures that $Cost$ is equal to the fixed parking cost of the selected parking lot. The user's maximum parking-cost preference is then enforced using Equation (4.9):

$$Cost \leq C_{\max}. \quad (4.9)$$

Together, Equations (4.7) to (4.9) ensure that exactly one parking lot is selected and that its cost satisfies the user's parking-cost constraint. In this work, at least one parking lot is assumed to be available at each recommendation step.

4.4.6 Maximum Walking-Distance Constraint

To satisfy the user's walking preference, the walking distance from the selected parking lot to the destination has to remain within the maximum acceptable walking distance, as enforced by Equation (4.10):

$$d_p^{walk} x_{L[p], D} \leq W_{\max}, \quad \forall p \in P \text{ such that } (L[p], D) \in E. \quad (4.10)$$

Here, d_p^{walk} denotes the walking distance from p to the destination through the parking-to-destination walking edge $(L[p], D)$, and $W_{\max}$ denotes the maximum acceptable walking distance. Equation (4.10) allows a parking lot to be selected only when its walking distance satisfies the user's walking-distance constraint.

Under the proposed formulation, a solution consists of a driving route, a selected parking lot, and a final walking edge from that lot to the destination. This structure ensures that the destination is reached through the selected parking lot rather than directly through the driving network. The formulation therefore combines route selection, travel-time accumulation, parking-cost control, and walking-distance feasibility within a single optimization model.

Since the recommendation is recomputed repeatedly as the vehicle progresses through the intersections of the road network, the computational complexity increases significantly with the network size. For this reason, the optimization model is mainly used for small-scale validation, while scalable heuristic and meta-heuristic approaches are developed for medium-scale and large-scale networks.

4.5 Heuristic and Meta-Heuristic Methods

This section presents the proposed heuristic and the meta-heuristic baselines used in DPLOF. The proposed heuristic is designed as the main scalable solution method for medium-scale and large-scale networks, where solving the optimization model repeatedly becomes computationally expensive. GA, PSO, and ACO are included as comparative baselines. All methods follow the same two-level route-and-parking structure, in which the driver travels by vehicle from the source or current node to a selected parking lot and then walks from that lot to the destination.

4.5.1 Proposed Heuristic Method

The proposed heuristic provides a scalable approximation of the optimization objective. Instead of solving the full optimization model at every recommendation step, it computes the shortest-path driving times from the current vehicle location to all candidate parking lots, filters out the infeasible options, removes the dominated candidates, and ranks the remaining parking lots using a weighted travel-time and parking-cost score. This design allows the recommendation to be recomputed efficiently when the vehicle reaches a new intersection or when the traffic information is updated.

Starting from the current node A , the heuristic first applies Dijkstra’s algorithm using the current edge travel times. For each candidate parking lot, it obtains the driving time from A , retrieves the walking distance from that lot to the destination D , and converts this walking distance into walking time using the fixed walking speed v_{walk} . A parking lot is retained only if it is available and satisfies both W_{max} and C_{max} . The remaining candidates are then reduced using dominance filtering. Finally, each retained parking lot is scored using the same weighted structure as the optimization objective, and the lot with the minimum score is selected. The procedure is summarized in Algorithm 1.

Algorithm 1 Proposed Heuristic Method

```

1: Data: Road network  $G$ , current node  $A$ , destination  $D$ , candidate parking lots  $P$ , edge travel
   times, walking distances, parking costs, availability,  $W_{\max}$ ,  $C_{\max}$ ,  $\alpha$ , walking speed  $v_{\text{walk}}$ 
2: Result: Recommended parking lot  $p^*$ 
3:  $F \leftarrow \emptyset$ ,  $p^* \leftarrow \emptyset$ ,  $H^* \leftarrow \infty$ 
4:  $\text{Dist} \leftarrow \text{Dijkstra}(G, A)$ 
5: for each  $p \in P$  do
6:    $t_p^{\text{drive}} \leftarrow \text{Dist}[p]$ 
7:    $d_p^{\text{walk}} \leftarrow \text{DirectWalkDistance}(p, D)$ 
8:    $t_p^{\text{walk}} \leftarrow d_p^{\text{walk}} / v_{\text{walk}}$ 
9:   if  $a_p > 0$  and  $d_p^{\text{walk}} \leq W_{\max}$  and  $C_p \leq C_{\max}$  then
10:     $F \leftarrow F \cup \{p\}$ 
11:   end if
12: end for
13:  $F \leftarrow \text{DominancePrune}(F)$ 
14: for each  $p \in F$  do
15:    $H_p \leftarrow \alpha (t_p^{\text{drive}} + t_p^{\text{walk}}) + (1 - \alpha)C_p$ 
16:   if  $H_p < H^*$  then
17:      $H^* \leftarrow H_p$ ,  $p^* \leftarrow p$ 
18:   end if
19: end for
20:
21: return  $p^*$ 

```

The minimum driving time from A to each candidate parking lot p_i is computed using Dijkstra's algorithm over the current edge travel times:

$$t_{p_i}^{\text{drive}} = \text{dist}_{\text{Dijkstra}}(A, p_i). \quad (4.11)$$

The walking distance from p_i to D is obtained from the direct parking-to-destination walking connection:

$$d_{p_i}^{\text{walk}} = \text{DirectWalkDistance}(p_i, D). \quad (4.12)$$

The corresponding walking time is derived using the fixed walking speed:

$$t_{p_i}^{\text{walk}} = \frac{d_{p_i}^{\text{walk}}}{v_{\text{walk}}}. \quad (4.13)$$

A parking lot is considered feasible only if it satisfies the availability, walking-distance, and parking-cost requirements:

$$a_{p_i} > 0, \quad d_{p_i}^{\text{walk}} \leq W_{\max}, \quad C_{p_i} \leq C_{\max}. \quad (4.14)$$

After feasibility filtering, dominance filtering is applied to remove the inferior candidates. A parking lot p_i is dominated if another parking lot p_j is no worse in driving time, walking distance, and parking cost, and is strictly better in at least one of these criteria:

$$t_{p_j}^{\text{drive}} \leq t_{p_i}^{\text{drive}}, \quad d_{p_j}^{\text{walk}} \leq d_{p_i}^{\text{walk}}, \quad C_{p_j} \leq C_{p_i}. \quad (4.15)$$

For each remaining candidate, the total travel time is computed as the sum of the driving time and the walking time:

$$t_{p_i}^{\text{total}} = t_{p_i}^{\text{drive}} + t_{p_i}^{\text{walk}}. \quad (4.16)$$

The final heuristic score is then derived as

$$H_{p_i} = \alpha t_{p_i}^{\text{total}} + (1 - \alpha) C_{p_i}. \quad (4.17)$$

The recommended parking lot is the feasible non-dominated candidate with the minimum heuristic score:

$$p^* = \arg \min_{p_i \in F} H_{p_i}. \quad (4.18)$$

Since the proposed heuristic computes the shortest-path costs once from the current node and then ranks the feasible parking lots directly, it can be recomputed efficiently as the traffic condition changes during route progression.

4.5.2 Meta-Heuristic Baseline Design

To benchmark the proposed heuristic, three meta-heuristic baselines are implemented, namely GA, PSO, and ACO. These methods follow the same feasibility constraints and the same objective structure as the proposed heuristic. Before any meta-heuristic search is applied, the candidate parking lots are filtered using availability, the walking connection, the walking-distance threshold, and the parking-cost threshold. Each feasible solution consists of a selected parking lot and a corresponding driving route from the current node to that lot. The final walking component is then obtained from the parking-to-destination walking connection.

Unlike the proposed heuristic, which first computes Dijkstra-based driving times and then ranks the parking lots directly, the meta-heuristic baselines perform a joint search over parking-lot selection and route construction. Each method generates parking-lot choices together with route-preference information. A guided Depth First Search (DFS) and backtracking decoder then constructs a valid driving route to the selected parking lot by exploring the admissible outgoing branches in the road network. If a branch cannot reach the selected lot, the decoder backtracks and explores the next admissible continuation.

The quality of each decoded route-and-parking solution is evaluated using the common weighted objective:

$$H_{p_i} = \alpha (t_{p_i}^{\text{drive}} + t_{p_i}^{\text{walk}}) + (1 - \alpha) C_{p_i}. \quad (4.19)$$

Here, $t_{p_i}^{\text{drive}}$ is obtained from the decoded driving route, $t_{p_i}^{\text{walk}}$ is derived from the parking-to-destination walking connection, and C_{p_i} is the fixed cost of the selected parking lot.

4.5.3 Genetic Algorithm

GA is implemented as an evolutionary baseline. Each chromosome contains a parking-lot gene and a set of node-priority genes. The parking-lot gene identifies the selected candidate parking lot, while the node-priority genes define the route-preference order used by the guided DFS. During each generation, the chromosomes are decoded into route-and-parking solutions and evaluated using Equation (4.19). Better chromosomes are favored during selection, while crossover and mutation explore new parking-lot and route combinations. Elitism is used to preserve the best solution found so far. The procedure is summarized in Algorithm 2.

Algorithm 2 Genetic Algorithm Baseline

```

1: Data: Road network  $G$ , source or current node, destination  $D$ , feasible parking lots  $P$ , edge travel
   times, parking costs, walking distances,  $W_{\max}$ ,  $C_{\max}$ ,  $\alpha$ , population size  $N_{\text{pop}}$ , generations  $G_{\max}$ 
2: Result: Best parking lot  $p^*$  and corresponding route
3:  $\mathcal{P}^{(0)} \leftarrow \text{InitializePopulation}(N_{\text{pop}}, P)$ 
4:  $(R^*, p^*, f^*) \leftarrow (\emptyset, \emptyset, \infty)$ 
5: for  $g \leftarrow 1$  to  $G_{\max}$  do
6:   for each chromosome  $c \in \mathcal{P}^{(g-1)}$  do
7:      $R_c \leftarrow \text{GuidedDFSBacktracking}(c)$ 
8:      $f_c \leftarrow \text{EvaluateObjective}(R_c, \alpha)$ 
9:     if  $f_c < f^*$  then
10:       $(R^*, p^*, f^*) \leftarrow (R_c, \text{ParkingLot}(c), f_c)$ 
11:    end if
12:  end for
13:   $\mathcal{P}' \leftarrow \text{TournamentSelect}(\mathcal{P}^{(g-1)})$ 
14:   $\mathcal{P}'' \leftarrow \text{Crossover}(\mathcal{P}')$ 
15:   $\mathcal{P}''' \leftarrow \text{Mutate}(\mathcal{P}'')$ 
16:   $\mathcal{P}^{(g)} \leftarrow \text{PreserveElite}(\mathcal{P}''', R^*, p^*, f^*)$ 
17: end for
18:
19: return  $(R^*, p^*)$ 

```

4.5.4 Particle Swarm Optimization

PSO is implemented as a swarm-based baseline. Each particle contains a parking-position variable that maps to a feasible parking lot, together with node-priority values that guide the route construction. At each iteration, every particle is decoded into a route-and-parking solution and evaluated using Equation (4.19). Each particle updates its personal-best solution when an improved solution is found, and the swarm maintains a global-best solution. The particle positions are then updated using the personal-best and global-best information. The procedure is summarized in Algorithm 3.

Algorithm 3 Particle Swarm Optimization Baseline

```
1: Data: Road network  $G$ , source or current node, destination  $D$ , feasible parking lots  $P$ , edge travel
   times, parking costs, walking distances,  $W_{\max}$ ,  $C_{\max}$ ,  $\alpha$ , swarm size  $N_{\text{swarm}}$ , iterations  $I_{\max}$ 
2: Result: Best parking lot  $p^*$  and corresponding route
3:  $\mathcal{S} \leftarrow \text{InitializeSwarm}(N_{\text{swarm}}, P)$ 
4:  $(gbest_R, gbest_p, gbest_f) \leftarrow (\emptyset, \emptyset, \infty)$ 
5: for each particle  $u \in \mathcal{S}$  do
6:    $R_u \leftarrow \text{GuidedDFSBacktracking}(u)$ 
7:    $f_u \leftarrow \text{EvaluateObjective}(R_u, \alpha)$ 
8:    $(pbest_R[u], pbest_p[u], pbest_f[u]) \leftarrow (R_u, \text{ParkingLot}(u), f_u)$ 
9:   if  $f_u < gbest_f$  then
10:     $(gbest_R, gbest_p, gbest_f) \leftarrow (R_u, \text{ParkingLot}(u), f_u)$ 
11:   end if
12: end for
13: for  $iter \leftarrow 1$  to  $I_{\max}$  do
14:   for each particle  $u \in \mathcal{S}$  do
15:      $R_u \leftarrow \text{GuidedDFSBacktracking}(u)$ 
16:      $f_u \leftarrow \text{EvaluateObjective}(R_u, \alpha)$ 
17:     if  $f_u < pbest_f[u]$  then
18:        $(pbest_R[u], pbest_p[u], pbest_f[u]) \leftarrow (R_u, \text{ParkingLot}(u), f_u)$ 
19:     end if
20:     if  $f_u < gbest_f$  then
21:        $(gbest_R, gbest_p, gbest_f) \leftarrow (R_u, \text{ParkingLot}(u), f_u)$ 
22:     end if
23:   end for
24:    $\mathcal{S} \leftarrow \text{UpdateParticles}(\mathcal{S}, pbest, gbest)$ 
25: end for
26:
27: return  $(gbest_R, gbest_p)$ 
```

4.5.5 Ant Colony Optimization

ACO is implemented as a pheromone-based baseline. Each ant selects a feasible parking lot and constructs a driving route using pheromone-guided edge preferences together with the guided DFS and backtracking decoder. After all ants have constructed their route-and-parking solutions, each solution is evaluated using Equation (4.19). The best solution found so far is updated, the pheromone values are evaporated on all driving edges, and the feasible ant solutions deposit pheromone on the edges they use. The procedure is summarized in Algorithm 4.

Algorithm 4 Ant Colony Optimization Baseline

```
1: Data: Road network  $G$ , source or current node, destination  $D$ , feasible parking lots  $P$ , edge travel  
   times, parking costs, walking distances,  $W_{\max}$ ,  $C_{\max}$ ,  $\alpha$ , number of ants  $N_{\text{ants}}$ , iterations  $I_{\max}$   
2: Result: Best parking lot  $p^*$  and corresponding route  
3:  $\tau \leftarrow \text{InitializePheromones}(E)$   
4:  $(R^*, p^*, f^*) \leftarrow (\emptyset, \emptyset, \infty)$   
5: for  $iter \leftarrow 1$  to  $I_{\max}$  do  
6:    $\mathcal{A} \leftarrow \emptyset$   
7:   for  $k \leftarrow 1$  to  $N_{\text{ants}}$  do  
8:      $p_k \leftarrow \text{SelectParkingLot}(P)$   
9:      $R_k \leftarrow \text{GuidedDFSBacktracking}(G, \tau, p_k)$   
10:     $f_k \leftarrow \text{EvaluateObjective}(R_k, \alpha)$   
11:     $\mathcal{A} \leftarrow \mathcal{A} \cup \{(R_k, p_k, f_k)\}$   
12:    if  $f_k < f^*$  then  
13:       $(R^*, p^*, f^*) \leftarrow (R_k, p_k, f_k)$   
14:    end if  
15:  end for  
16:   $\tau \leftarrow \text{EvaporatePheromones}(\tau)$   
17:   $\tau \leftarrow \text{UpdatePheromones}(\tau, \mathcal{A})$   
18: end for  
19:  
20: return  $(R^*, p^*)$ 
```

4.6 Results and Discussion

4.6.1 Simulation Setup, Performance Metrics, and Parameters

This section evaluates the proposed DPLOF framework under synthetic and fixed-network road scenarios. In all experiments, the same two-level recommendation structure is used, in which the driver first travels from the source node to a selected parking lot and then walks from that lot to the destination. This common setting allows all methods to be compared under identical route-and-parking conditions.

The experiments are divided into three groups. First, small-scale instances with 10, 20, 30, 40, and 50 nodes are used to compare the proposed heuristic with the optimization model. Second, medium-scale and large-scale synthetic networks with 500, 1000, 1500, 2000, and 2500 nodes are used to compare the heuristic with ACO, PSO, and GA. Third, a fixed SUMO-based urban road network with 2055 nodes is used to validate the framework on a realistic directed topology.

Unless otherwise stated, the baseline configuration uses a directed road network with a parking supply equal to approximately 5% of the total nodes, a maximum walking distance $W_{\max} = 0.50$ km, an objective weight $\alpha = 0.50$, a parking budget $C_{\max} = 20.5$, and a parking-cost range from 8.0 to 20.5. Under low-traffic conditions, vehicle speeds are sampled from 50 to 70 km/h, while the walking speed is fixed at $v_{\text{walk}} = 5.0$ km/h.

The parameter-sensitivity analysis follows a controlled one-factor-at-a-

time design, which is summarized in Table 4.2. In each experiment, a single parameter is varied while the remaining parameters are kept at their baseline values. The walking-distance analysis uses $W_{\max} = 0.25, 0.50, 0.75$, and 1.00 km. The parking-supply analysis uses approximately 5%, 10%, 15%, and 20% candidate parking lots. The parking-budget analysis uses $C_{\max} = 10.0, 15.0, 20.5$, and 25.0 . The parking-cost analysis uses the ranges $8.0\text{--}20.5$, $10.0\text{--}22.5$, $12.5\text{--}25.0$, and $15.0\text{--}30.0$. The traffic analysis considers very low, low, medium, and high traffic using speed ranges of $60\text{--}80$, $50\text{--}70$, $30\text{--}49$, and $10\text{--}29$ km/h, respectively. For the edge-density analysis, the horizontal axis reports the actual number of directed edges in each scenario.

Table 4.2: Controlled parameter settings used in the parameter-sensitivity analysis.

Analysis	Parameter Setting
Node size	10–50 nodes for small-scale validation and 500–2500 nodes for medium- and large-scale scalability
Walking distance	$W_{\max} = 0.25, 0.50, 0.75, 1.00$ km
Parking supply	Approximately 5%, 10%, 15%, 20% candidate parking lots
Parking budget	$C_{\max} = 10.0, 15.0, 20.5, 25.0$
Parking-cost range	$8.0\text{--}20.5, 10.0\text{--}22.5, 12.5\text{--}25.0, 15.0\text{--}30.0$
Traffic level	Very low $60\text{--}80$, low $50\text{--}70$, medium $30\text{--}49$, high $10\text{--}29$ km/h
Edge density	Increasing road-network connectivity reported using the actual number of directed edges

The evaluation is based on five performance indicators, namely driving time, walking time, parking cost, objective value, and runtime. The driving time measures the travel from the source node to the selected parking lot, while the walking time measures the movement from that lot to the final destination. These two terms together represent the total travel burden. The parking cost captures the monetary cost of the selected parking option. The objective value combines the total travel time and the parking cost using the weight α , as defined in Equation (4.2), and is therefore the single quantity that the methods minimize. The runtime measures the computational efficiency. For every scenario, all methods are tested on the same road network, source-destination pair, parking-lot candidates, walking-distance limit, parking-budget limit, and objective-weight setting. Therefore, the observed differences can be attributed to the solution methods rather than to changes in the underlying test instance.

4.6.2 Small-Scale Evaluation

The small-scale experiments compare the proposed heuristic with the optimization model on networks containing 10 to 50 nodes. These instances are used to examine whether the heuristic can preserve the quality of the optimum while requiring substantially lower runtime. Since the optimization model explores the feasible route-and-parking space exactly, it serves as the reference solution on these smaller instances. In the objective-component comparison, the plotted small-scale node-size analysis is shown for the 10-, 20-, 30-, and 40-node cases,

while the 50-node case is discussed separately in the runtime analysis because the exact optimization run did not complete within practical memory and time limits.

Fig. 4.2 compares the runtime of the optimization model, the proposed heuristic, and the meta-heuristic baselines. As the network size increases, the computational cost of exact optimization grows sharply, while the heuristic remains computationally inexpensive. The 50-node exact case did not finish within practical limits, whereas the heuristic still returned a solution quickly. This result confirms the practical motivation for using the heuristic instead of exact optimization beyond the smallest instances.

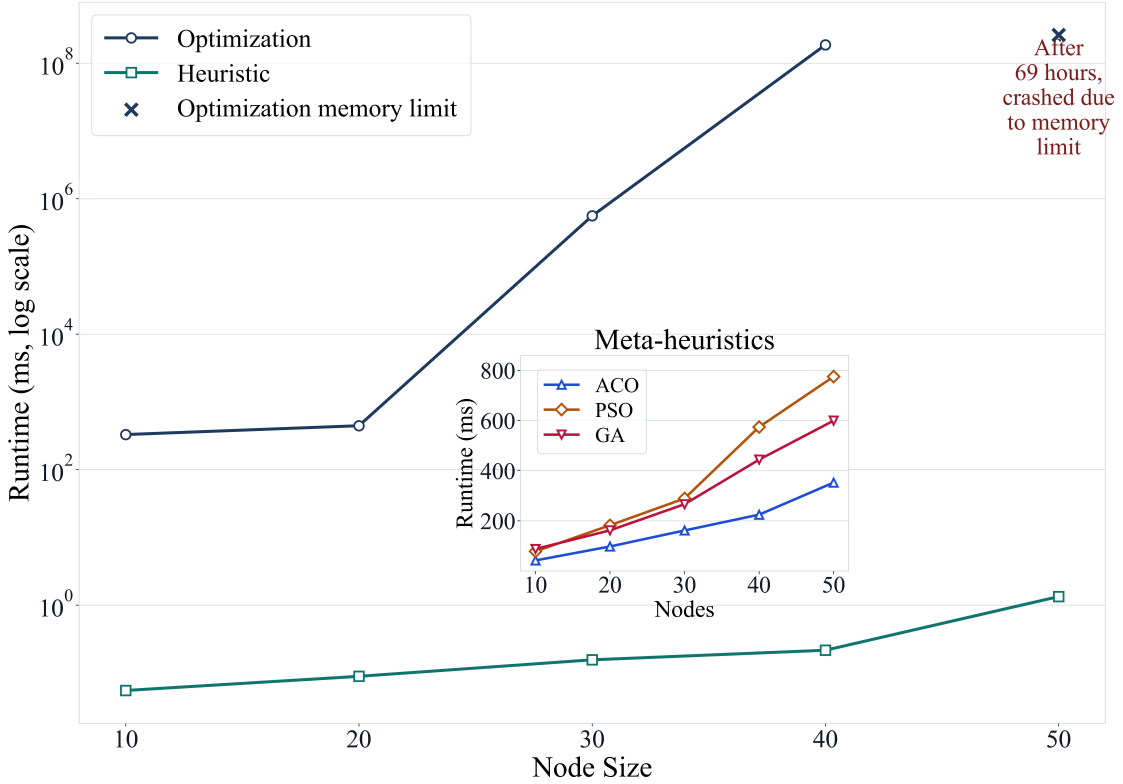


Figure 4.2: Small-scale runtime scalability of the optimization model, the proposed heuristic, and the meta-heuristic methods.

Effect of Network Size

Fig. 4.3 shows the effect of network size on the objective values. The optimization model and the heuristic return the same objective value at 10 and 20 nodes, namely 6.362 and 6.920, respectively. At 30 nodes, the optimization model achieves 6.880 while the heuristic gives 7.149, which is the only visible gap in this comparison. At 40 nodes, both methods again return the same value, namely 7.840. These results show that the heuristic remains very close to the optimum across the plotted small-scale cases. The only mismatch appears at 30

nodes, where the difference comes from the driving-time component, while the walking-time and parking-cost components remain the same.

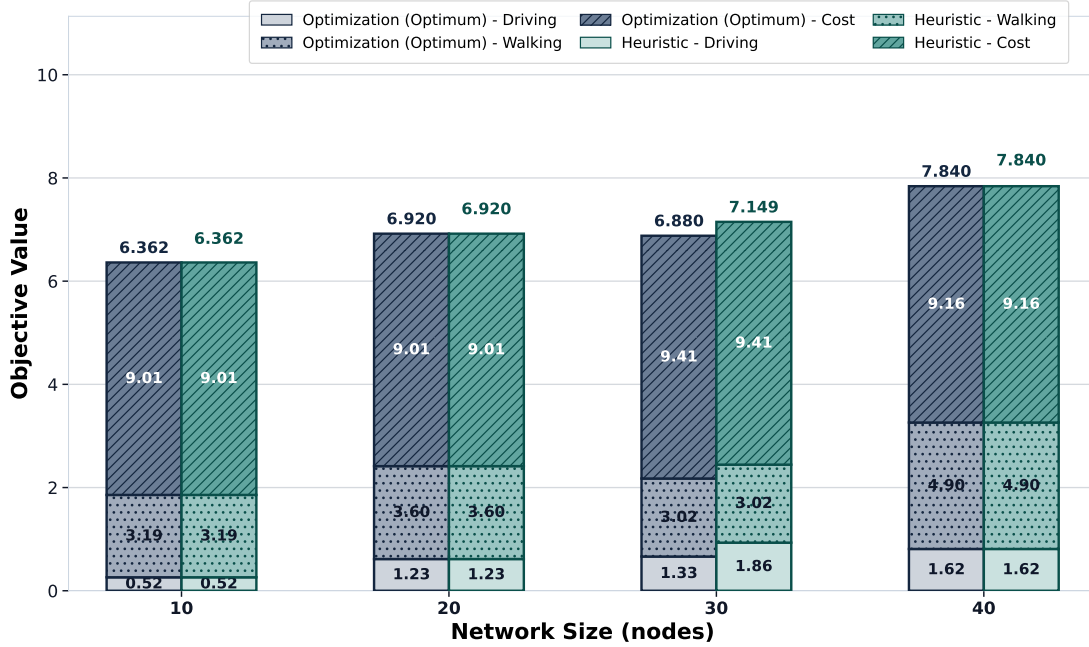


Figure 4.3: Performance comparison between the optimization model and the proposed heuristic as a function of network size.

Effect of Edge Density

Fig. 4.4 shows the effect of edge density. As the connectivity increases from 180 to 192 and 204 directed edges, the objective value decreases from 11.110 to 10.558 and then to 10.399. The optimization model and the heuristic produce the same plotted values throughout this sweep. The improvement is driven mainly by the driving-time component, which falls from 1.62 minutes to 0.52 minutes and then to 0.20 minutes, while the walking-time and parking-cost components remain unchanged. This indicates that denser road connectivity improves route quality without changing the selected parking lot.

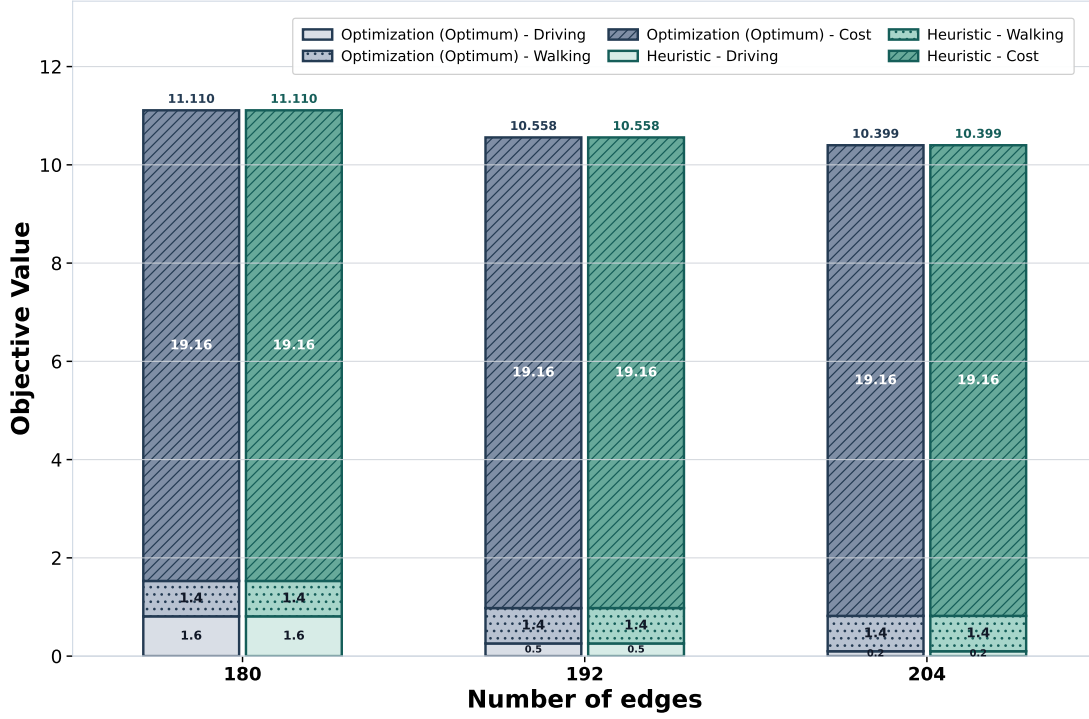


Figure 4.4: Performance comparison between the optimization model and the proposed heuristic as a function of edge density.

Effect of Maximum Walking Distance

Fig. 4.5 shows the results for the maximum walking-distance constraint. The objective value decreases from 11.110 at $W_{\max} = 0.25$ km to 9.989 at 0.50 km and to 9.509 at 0.75 km, before increasing slightly to 9.899 at 1.00 km. This pattern shows that relaxing the walking-distance constraint improves the route-and-parking trade-off up to a point, but the improvement is not indefinite. Across the same sweep, the walking-time component increases from 1.44 minutes to 3.72, 6.72, and 9.84 minutes, while the parking cost decreases from 19.16 to 14.80, 10.90, and 8.80. Therefore, the best trade-off in this plotted small-scale case occurs at 0.75 km, where the reduction in parking cost and driving time still outweighs the additional walking burden.

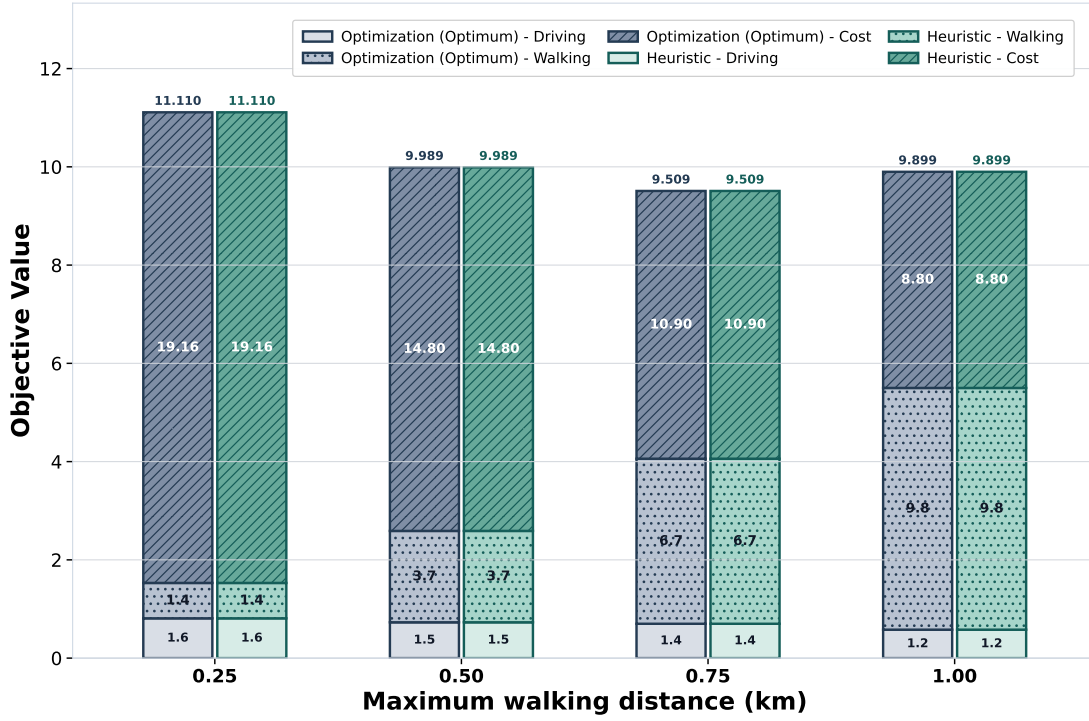


Figure 4.5: Performance comparison between the optimization model and the proposed heuristic as a function of the maximum walking distance.

Effect of Traffic Level

Fig. 4.6 shows the impact of traffic conditions. The objective value increases from 10.870 under very low traffic to 10.879 under low traffic, then to 12.385 under medium traffic, and finally to 20.170 under high traffic. The optimization model and the heuristic again produce the same plotted values. This trend is expected because congestion directly worsens the driving-time component. In particular, the driving-time component rises from 0.60 minutes in the very-low-traffic case to 1.16 minutes under low traffic, 3.63 minutes under medium traffic, and 19.74 minutes under high traffic, while the walking-time and parking-cost components remain nearly unchanged.

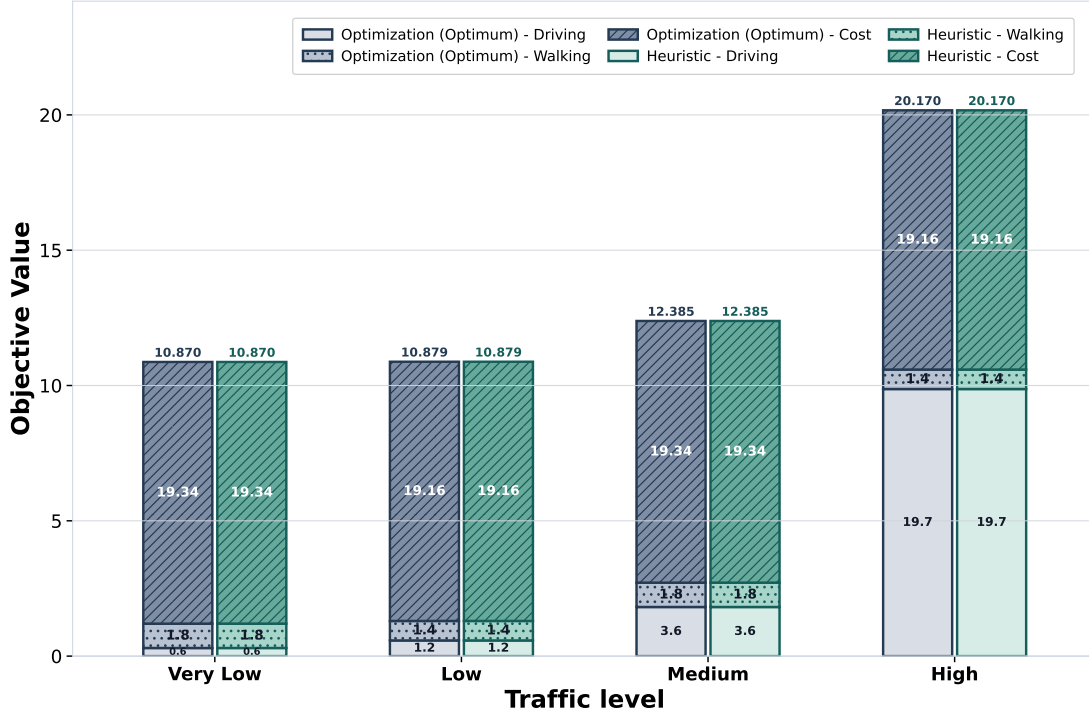


Figure 4.6: Performance comparison between the optimization model and the proposed heuristic as a function of the traffic level.

Effect of Maximum Parking Budget

Fig. 4.7 shows the effect of the parking-budget limit. In this sweep, the objective value changes only slightly, from 6.701 at $C_{\max} = 10.0$ to 6.720, 6.730, and 6.740 as the budget is relaxed to 15.0, 20.5, and 25.0, respectively. At the same time, the driving-time and walking-time components both decrease slightly, while the parking-cost component increases from 10.00 to 10.60. This means that the additional budget allows the framework to accept somewhat more expensive parking in exchange for slightly shorter travel and walking times. However, the overall effect is small, so the parking-budget parameter has only a limited influence at this scale.

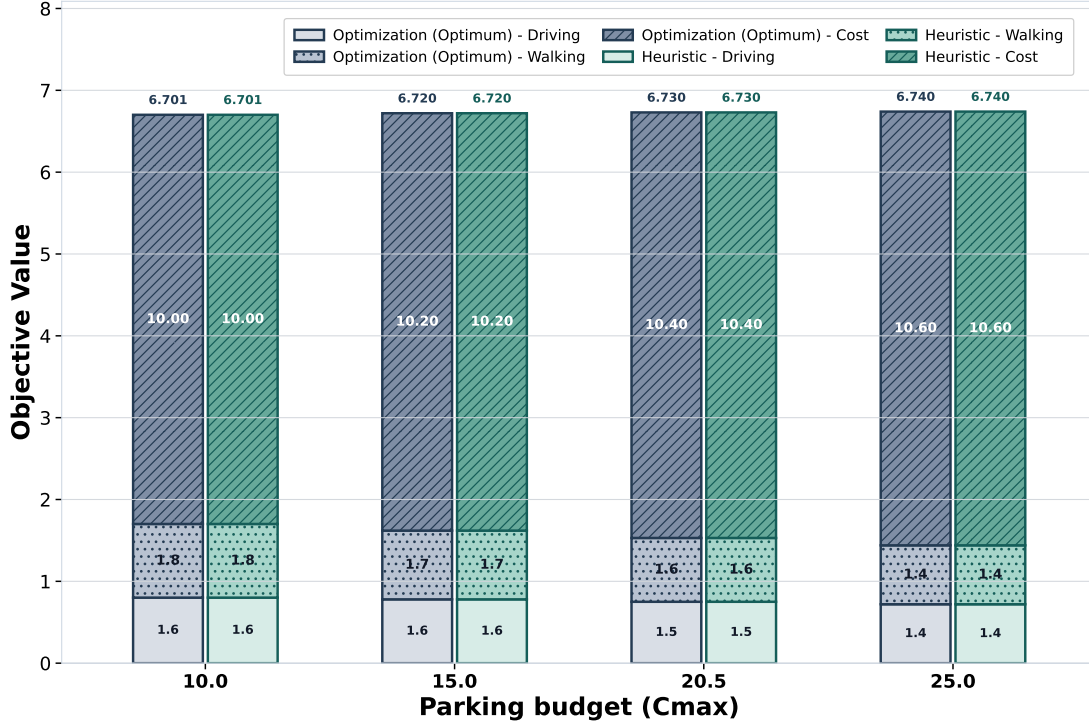


Figure 4.7: Performance comparison between the optimization model and the proposed heuristic as a function of the maximum parking budget.

Effect of Parking Supply

Fig. 4.8 shows the effect of parking supply. As the candidate parking supply increases from 5% to 10%, 15%, and 20%, the objective value decreases from 11.110 to 8.369, 6.819, and 5.730, respectively. This is accompanied by a reduction in walking time from 1.44 minutes to 1.08, 0.84, and 0.60 minutes, and by a reduction in parking cost from 19.16 to 14.20, 11.40, and 9.60. The driving-time component also improves slightly. Therefore, increasing the number of candidate parking lots substantially improves the recommendation quality in the small-scale setting.

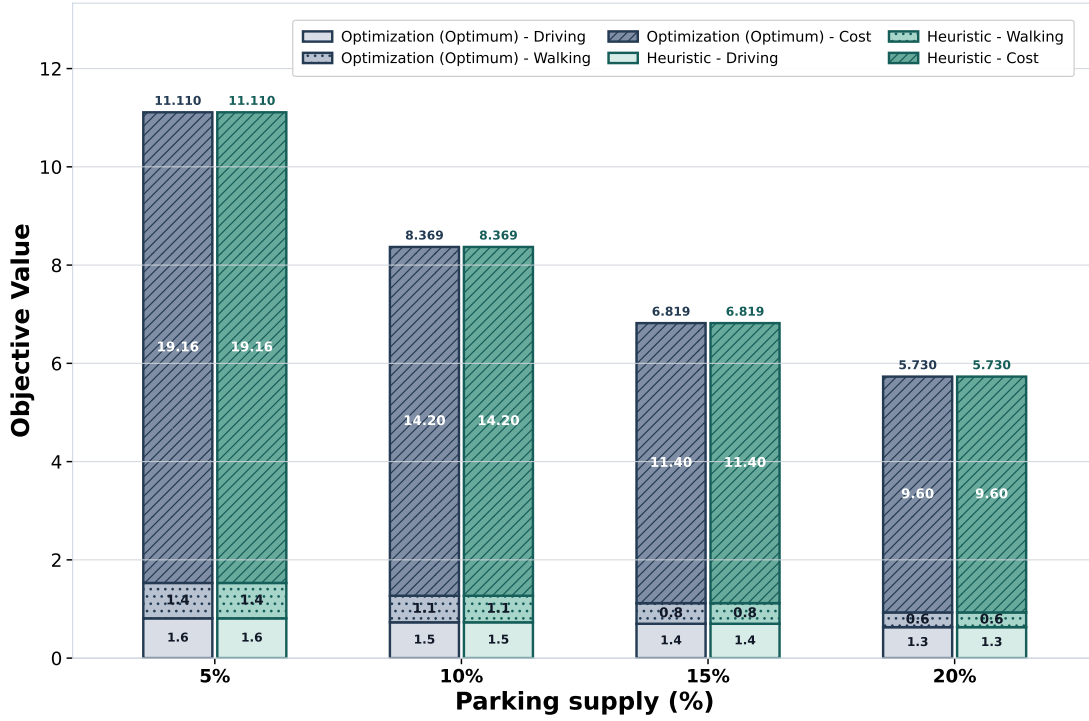


Figure 4.8: Performance comparison between the optimization model and the proposed heuristic as a function of the parking supply.

Effect of Parking-Cost Range

Fig. 4.9 presents the parking-cost analysis. As the assigned parking-cost range shifts upward from 8.0–20.5 to 10.0–22.5, 12.5–25.0, and 15.0–30.0, the objective value rises from 5.980 to 7.301, 8.955, and 10.725, respectively. The main reason is the increase in the parking-cost component from 8.90 to 11.20, 14.25, and 17.55. The driving-time and walking-time components change only modestly across the sweep. This shows that the parking-cost-range experiment primarily affects the monetary part of the objective rather than the route structure.

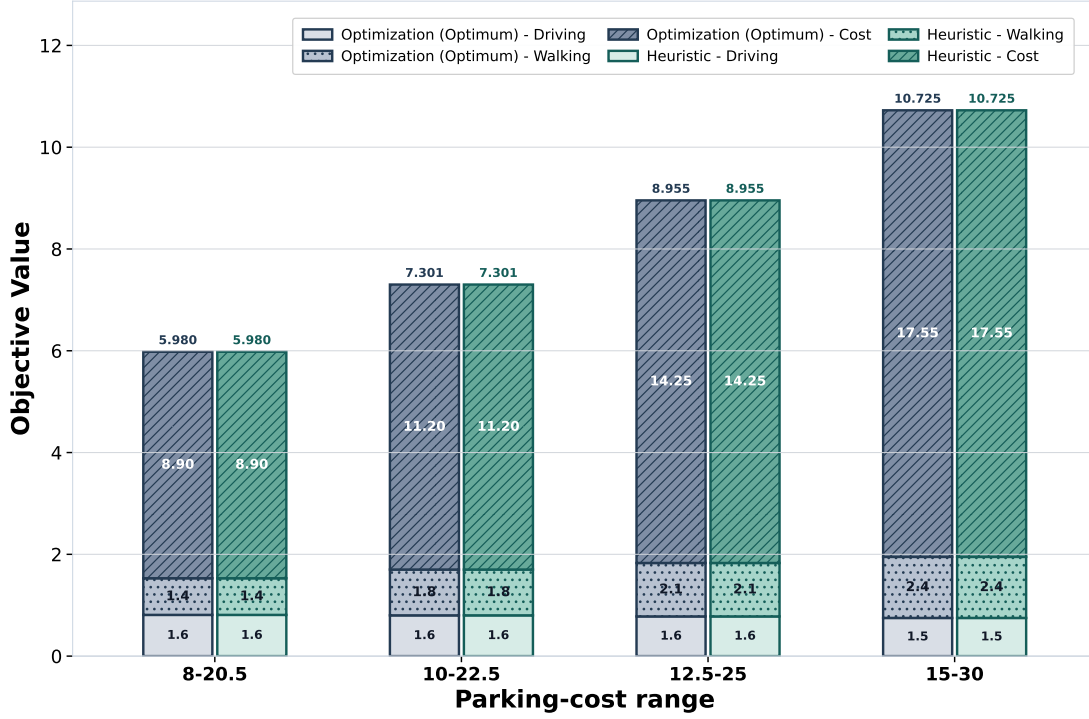


Figure 4.9: Performance comparison between the optimization model and the proposed heuristic as a function of the parking-cost range.

Summary

Overall, the small-scale evaluation shows that the proposed heuristic remains closely aligned with the optimization model across the plotted experiments. In most cases, both methods return exactly the same objective value and the same objective components, which indicates that they select the same parking lot and the same overall route-and-parking structure. The only visible gap appears in the 30-node node-size case, where the heuristic has a slightly worse driving-time component. The observed trends are also consistent across the small-scale analysis: greater edge density improves route quality, heavier traffic worsens the objective, greater parking supply improves the recommendation, a larger walking allowance improves the result up to a point, a higher parking budget has only a small effect, and a higher parking-cost range worsens the cost component of the objective.

4.6.3 Medium- and Large-Scale Networks: Scalability and Parameter Analysis

For medium-scale and large-scale synthetic networks, the evaluation compares the proposed heuristic with ACO, PSO, and GA. Network sizes of 500, 1000, 1500, 2000, and 2500 nodes are used for the scalability analysis, while the remaining parameter-sensitivity experiments are conducted on the fixed 2500-node instance. This separation allows the node-size experiment to isolate scalability and the

remaining experiments to isolate the effect of each individual parameter.

Effect of Network Size

Fig. 4.10 shows the effect of network size on the objective values. The heuristic gives objective values of 9.960, 11.613, 12.686, 12.686, and 11.279 at 500, 1000, 1500, 2000, and 2500 nodes, respectively. ACO follows almost the same pattern, with 9.960, 11.538, 12.686, 12.686, and 11.357. In contrast, PSO and GA are consistently worse, especially at 2000 and 2500 nodes. These results show that the heuristic and ACO remain the strongest methods in the scalability comparison, while PSO and GA become less stable as the search space grows.

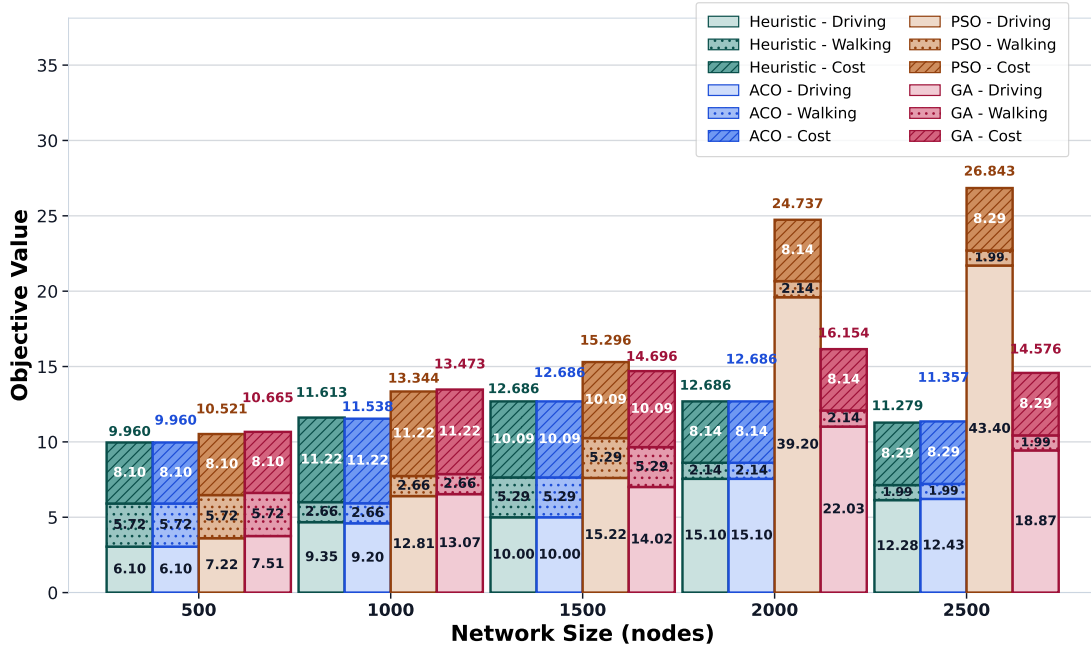


Figure 4.10: Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of network size.

Effect of Edge Density

Fig. 4.11 shows the edge-density results. For the heuristic, the objective value decreases from 11.279 at 7446 directed edges to 9.284 at 8278 edges and to 7.937 at 9110 edges. GA also improves across this sweep, from 14.576 to 10.334 and 9.041. PSO improves sharply from a very poor baseline value of 26.843 to 12.406 and 10.732, while ACO is not monotonic because it rises to 12.497 at 8278 edges before falling to 7.949 at 9110 edges. Thus, the clearest and most stable improvement with edge density is observed for the heuristic, while the other methods remain more sensitive to the route-decoding and search process.

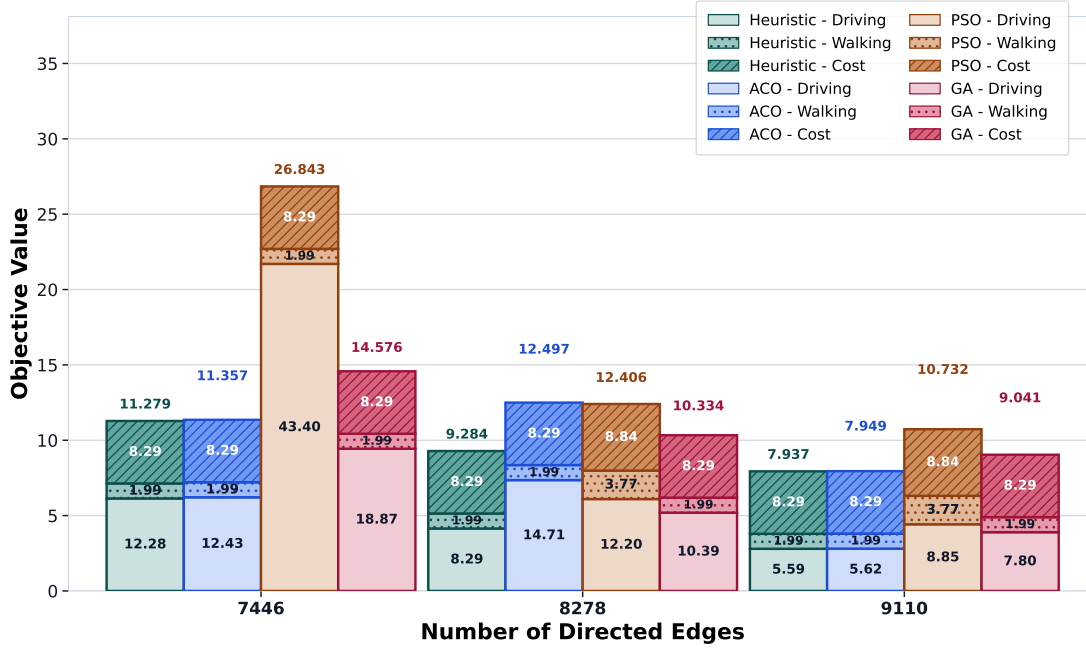


Figure 4.11: Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of edge density.

Effect of Maximum Walking Distance

Fig. 4.12 shows the effect of the maximum walking-distance constraint. For the heuristic, the objective value is 12.360 at 0.25 km, increases slightly to 12.995 at 0.50 km, decreases to 11.934 at 0.75 km, and then rises again to 13.469 at 1.00 km. ACO follows almost the same pattern. Therefore, the plotted medium/large walking-distance sweep does not show a simple monotonic improvement as the walking limit is relaxed. Instead, changing $W_{\max}$ alters the feasible parking set and can lead to different route-and-parking trade-offs, some of which improve the objective and some of which worsen it. In the plotted 2500-node case, the best heuristic result occurs at 0.75 km.

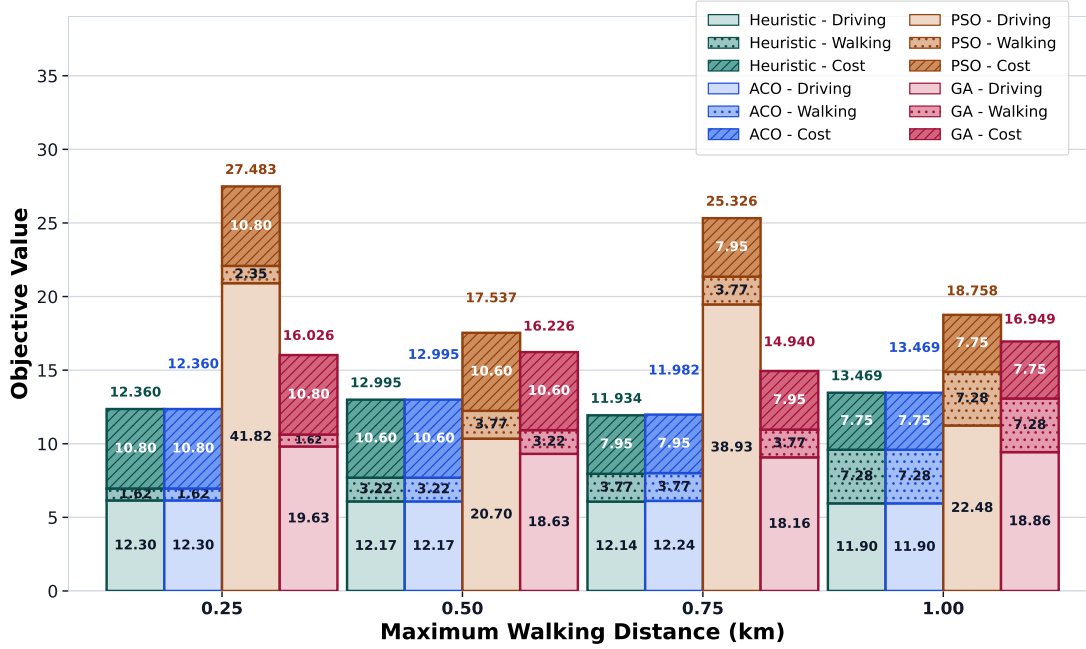


Figure 4.12: Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of the maximum walking distance.

Effect of Traffic Level

Fig. 4.13 reports the effect of the traffic level. For the heuristic, the objective value increases from 9.983 under very low traffic to 10.820 under low traffic, then to 13.838 under medium traffic, and finally to 23.732 under high traffic. ACO follows nearly the same pattern, whereas PSO and GA remain clearly worse over the full sweep. This confirms that heavier traffic mainly worsens the driving-time component while the walking-time and parking-cost components remain comparatively stable.

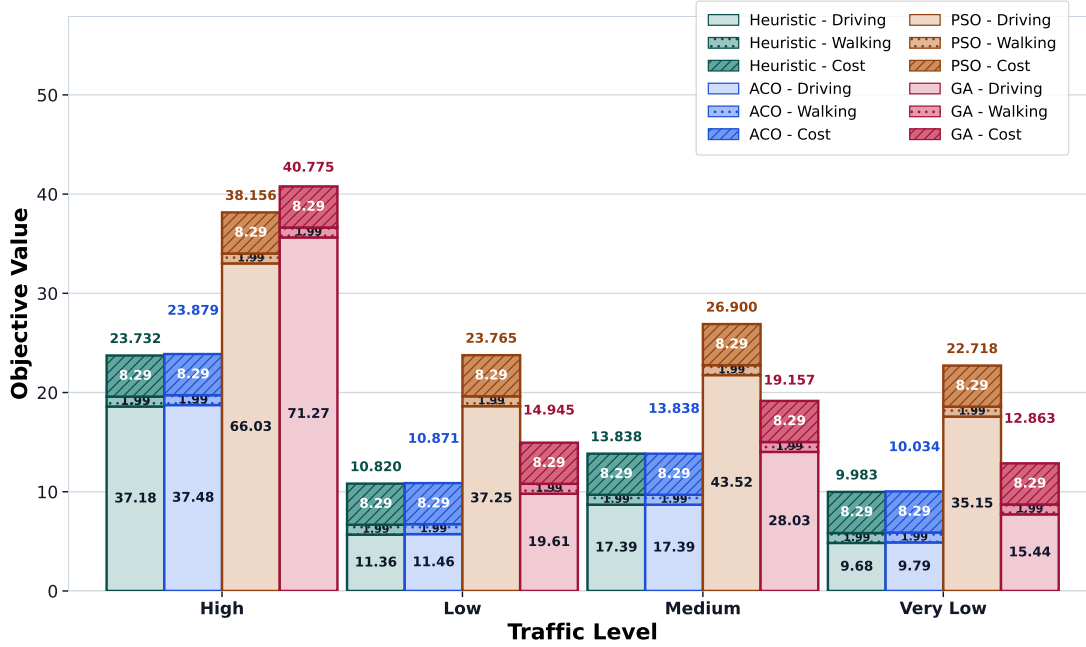


Figure 4.13: Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of the traffic level.

Effect of Maximum Parking Budget

Fig. 4.14 shows the effect of the parking-budget limit. For the heuristic, the objective value drops from 13.143 at $C_{\max} = 10.0$ to 11.279 at 15.0, and then remains unchanged at 20.5 and 25.0. ACO behaves almost identically, while PSO and GA remain substantially worse. This indicates that the main benefit of relaxing the budget occurs when the tightest budget is first loosened. Beyond 15.0, further budget expansion does not materially change the selected solution for the heuristic.

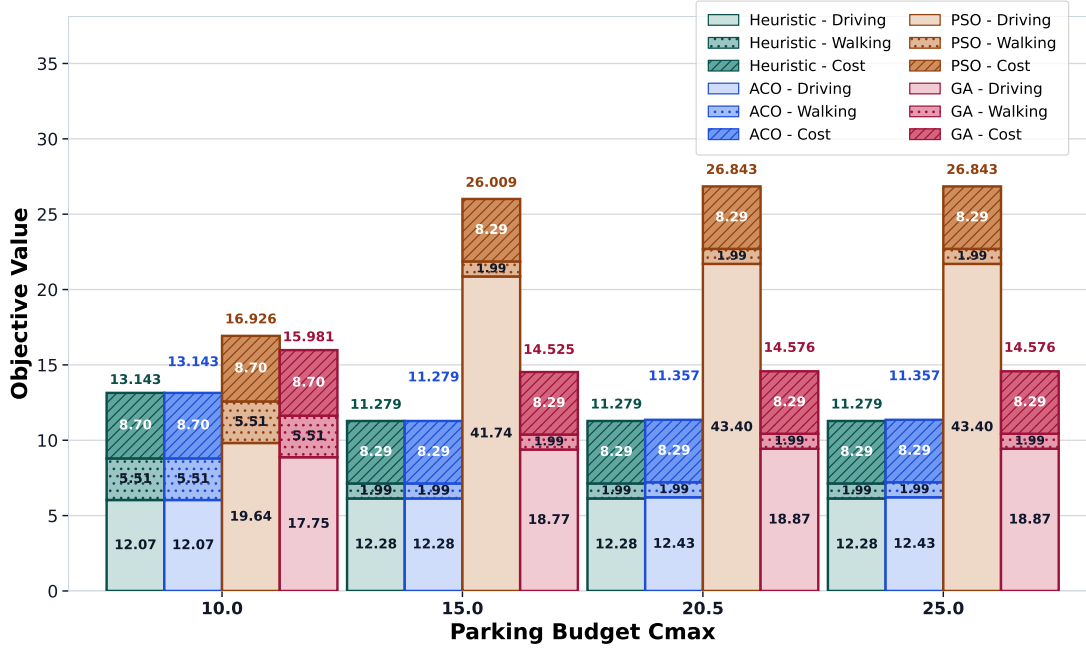


Figure 4.14: Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of the maximum parking budget.

Effect of Parking Supply

Fig. 4.15 presents the parking-supply results. For the heuristic, the objective value decreases from 13.693 at 5% parking supply to 12.709 at 10%, 11.584 at 15%, and 11.334 at 20%. ACO follows the same overall trend, while PSO and GA remain much higher. The improvement comes from a combination of reduced walking time and slightly lower parking cost as the number of candidate parking lots increases. Therefore, increasing parking supply improves the probability of finding a better route-and-parking-lot combination.

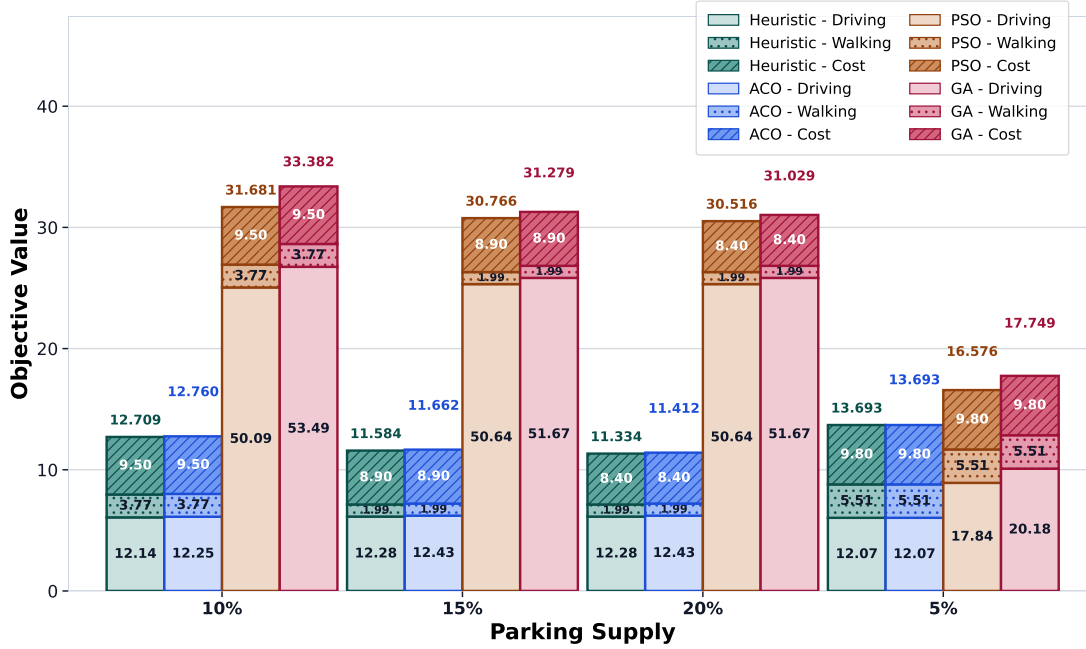


Figure 4.15: Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of the parking supply.

Effect of Parking-Cost Range

Fig. 4.16 shows the effect of the parking-cost range. For the heuristic, the objective value increases from 11.279 in the 8.0–20.5 setting to 12.264, 13.564, and 14.859 as the range shifts upward. ACO follows a nearly identical trend, while PSO and GA remain consistently worse. Across these cases, the driving-time and walking-time components remain almost unchanged, whereas the parking-cost component increases steadily. Hence, this experiment primarily affects the cost component of the objective rather than the road-travel structure.

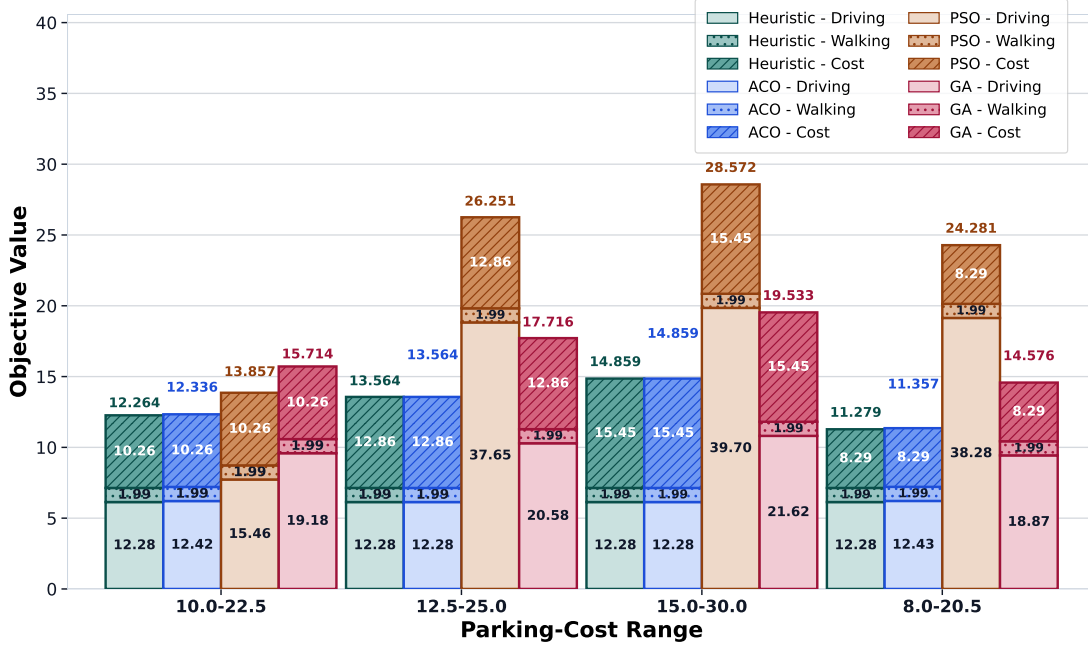


Figure 4.16: Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of the parking-cost range.

Summary

Overall, the medium- and large-scale evaluation shows that the proposed heuristic remains one of the strongest methods across the synthetic larger-scale experiments. In most plots, the heuristic and ACO are the closest competitors, whereas PSO and GA are more variable and often substantially worse. However, the parameter trends are not always monotonic. Edge density and parking supply generally improve the result, traffic congestion worsens it, parking-cost increases worsen it, and the parking budget mainly matters when the constraint is initially tight. By contrast, the walking-distance analysis shows that relaxing $W_{\max}$ does not guarantee a monotonic objective improvement, because the selected parking lot and the resulting driving-walking-cost trade-off may change nonlinearly.

4.6.4 Realistic Urban Network Evaluation Using SUMO

The final evaluation uses a SUMO-based road network extracted from the Waterloo–Kitchener urban area using OpenStreetMap data. In contrast to the synthetic experiments, this stage does not vary the network size. Instead, all experiments are conducted on a fixed 2055-node directed urban road network, and the analysis focuses on controlled variations in edge density, maximum walking distance, traffic level, parking budget, parking supply, and parking-cost range.

This fixed-network evaluation is important because strong performance on synthetic graphs does not necessarily reflect the performance on a realistic urban topology with irregular connectivity, one-way roads, and a non-uniform

road structure. The SUMO experiments therefore serve as a realistic validation of the framework rather than as an additional scalability benchmark.

Effect of Fixed-Network Baseline Performance

Fig. 4.17 provides the baseline comparison among the heuristic, ACO, PSO, and GA on the fixed real-road network. In the baseline scenario, the heuristic achieves an objective value of 11.847, compared with 14.979 for ACO, 17.556 for PSO, and 13.818 for GA. Thus, the heuristic gives the best baseline result among the scalable methods on the fixed SUMO road network.

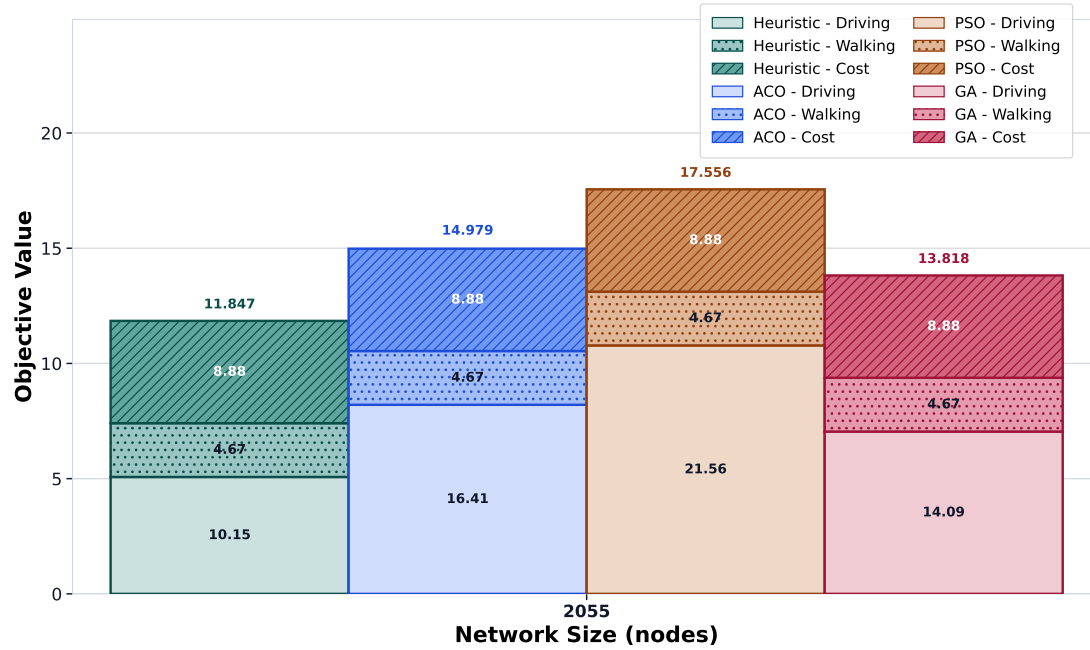


Figure 4.17: Baseline performance comparison between the proposed heuristic and the meta-heuristic methods on the fixed 2055-node Waterloo–Kitchener road network.

Effect of Edge Density

Fig. 4.18 shows the effect of edge density on the fixed urban network. For the heuristic, the objective value decreases from 11.847 at 5093 directed edges to 7.223 at 5777 edges and to 5.165 at 6461 edges. ACO matches the heuristic at the two denser settings, whereas PSO and GA remain slightly worse. These results show that increased connectivity improves route quality substantially on the realistic network, mainly through the driving-time component.

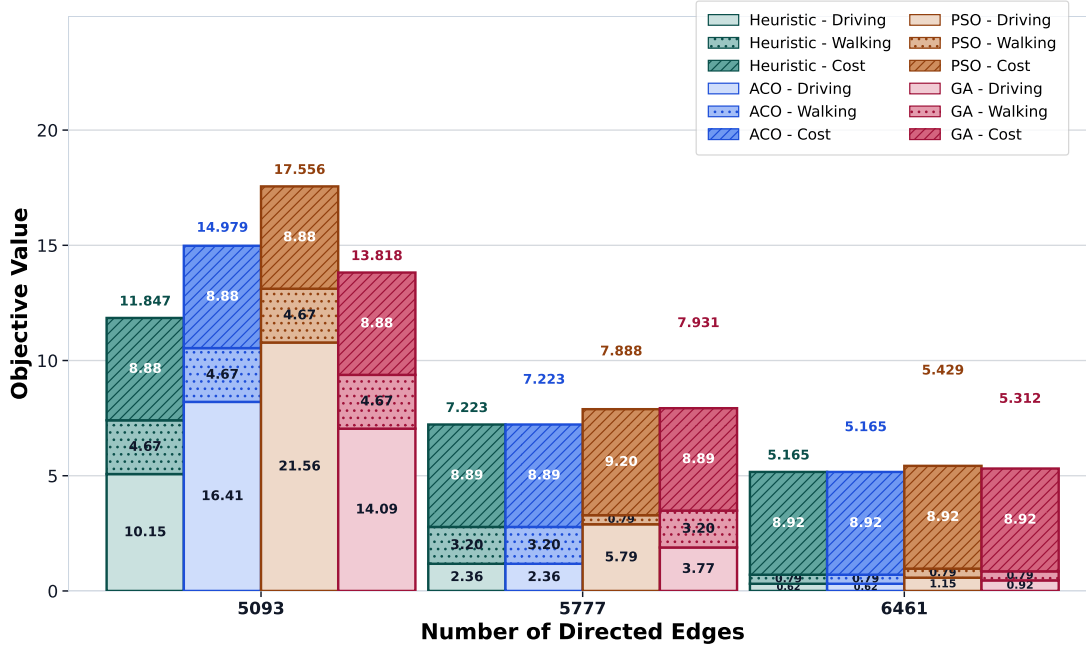


Figure 4.18: Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of edge density on the fixed real-road network.

Effect of Maximum Walking Distance

Fig. 4.19 shows the effect of the walking-distance limit on the realistic network. For the heuristic, the objective value is 11.635 at 0.25 km, then increases to 12.132 at 0.50 km, 13.607 at 0.75 km, and 13.232 at 1.00 km. Therefore, unlike the small-scale synthetic case, the plotted SUMO walking-distance sweep does not improve when the walking limit is relaxed. Instead, the additional walking burden is not fully compensated by the changes in route and parking cost. In this fixed real-road-network case, the best heuristic result occurs at the strictest walking-distance setting, namely 0.25 km.

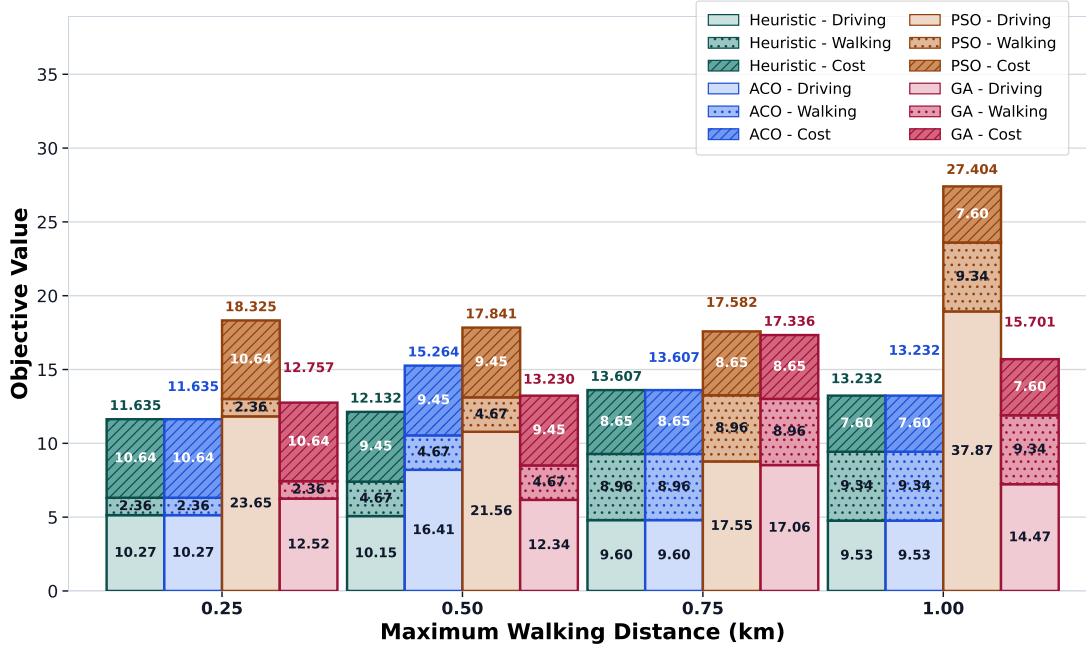


Figure 4.19: Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of the maximum walking distance on the fixed real-road network.

Effect of Traffic Level

Fig. 4.20 shows the traffic-level analysis on the fixed urban network. For the heuristic, the objective value increases sharply from 6.352 under very low traffic to 10.041 under low traffic, 17.393 under medium traffic, and 87.987 under high traffic. ACO produces the same plotted values at very low, medium, and high traffic, and a nearly identical value under low traffic. This confirms that heavy congestion strongly worsens the route quality in the realistic urban network, with the driving-time component becoming dominant under high traffic.

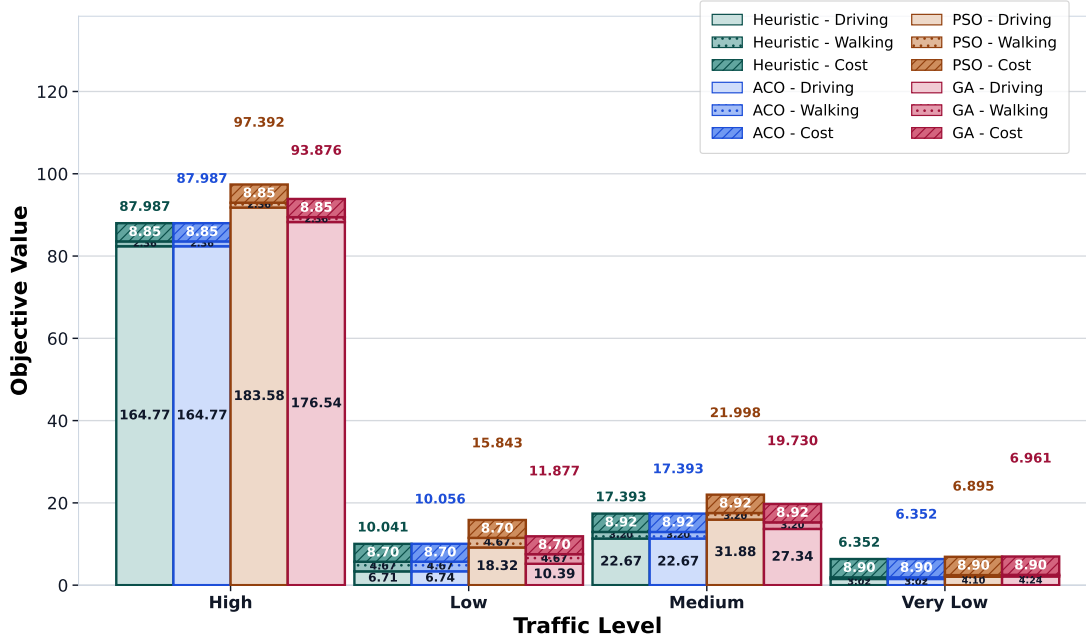


Figure 4.20: Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of the traffic level on the fixed real-road network.

Effect of Maximum Parking Budget

Fig. 4.21 shows the effect of the parking-budget limit on the realistic network. For the heuristic, the objective value decreases from 12.588 at $C_{\max} = 10.0$ to 11.387 at 15.0, and then to 9.998 at 20.5, before rising again to 10.765 at 25.0. Thus, the best plotted heuristic result occurs at $C_{\max} = 20.5$, not at the largest budget. This indicates that relaxing the budget initially helps by making better parking options feasible, but a further budget increase can shift the selected trade-off in a way that does not improve the total objective.

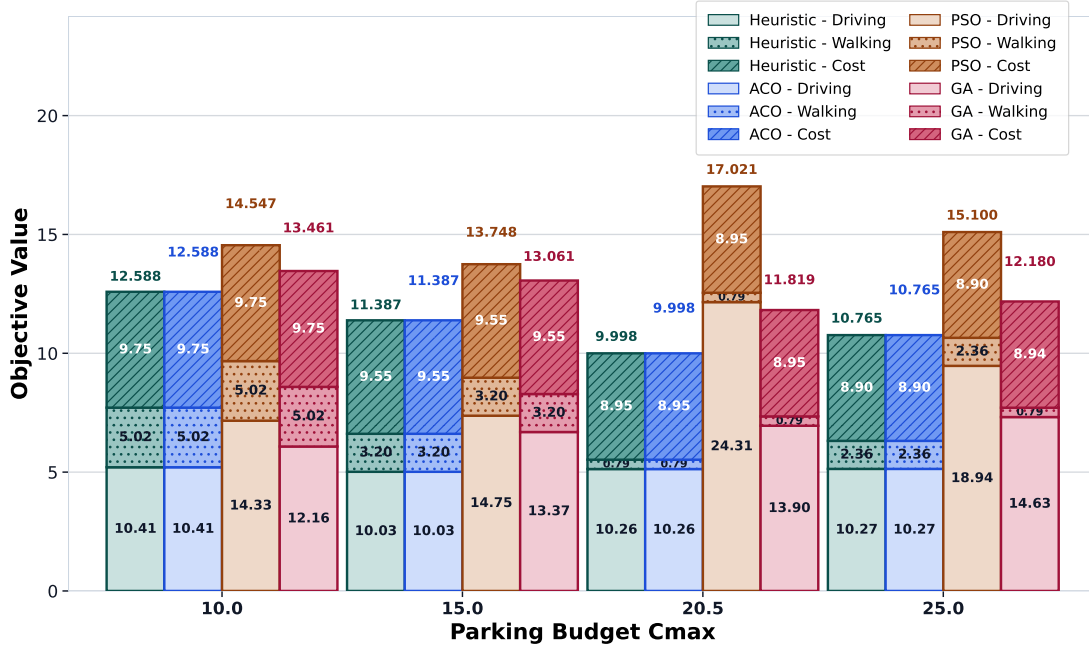


Figure 4.21: Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of the maximum parking budget on the fixed real-road network.

Effect of Parking Supply

Fig. 4.22 shows the effect of parking supply. For the heuristic, the objective value decreases from 13.434 at 5% parking supply to 12.964 at 10%, 11.387 at 15%, and 9.898 at 20%. This is accompanied by lower walking time and lower parking cost. Therefore, increasing the number of candidate parking lots improves the likelihood of finding a better parking-lot recommendation on the realistic network.

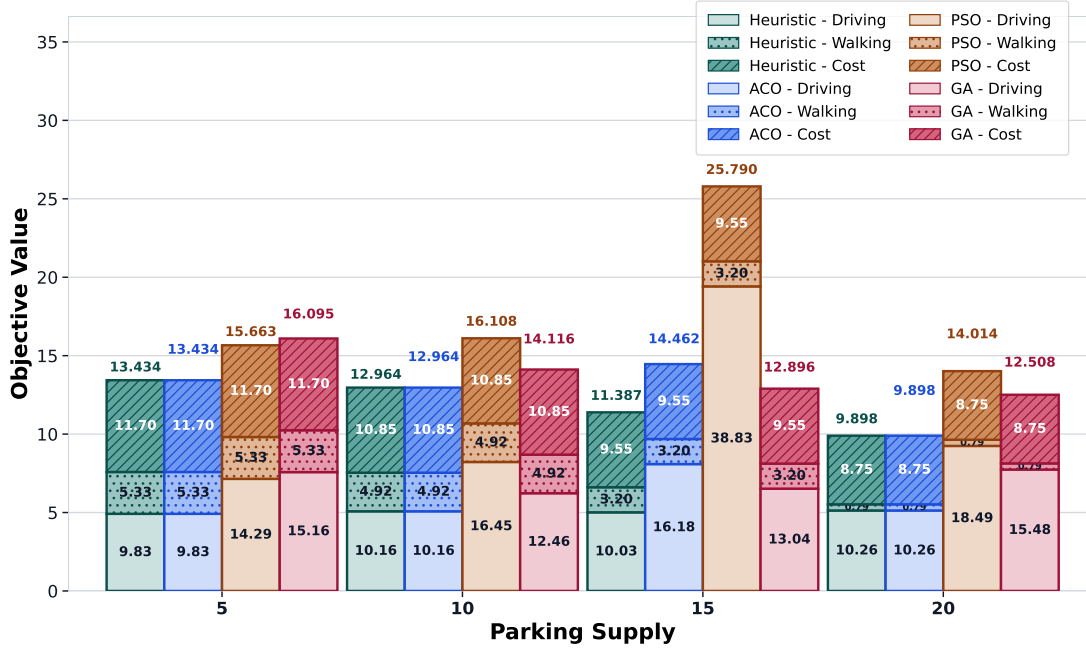


Figure 4.22: Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of the parking supply on the fixed real-road network.

Effect of Parking-Cost Range

Fig. 4.23 shows the parking-cost analysis for the fixed real-road network. For the heuristic, the objective value rises from 9.898 in the 8.0–20.5 case to 11.537, 12.615, and 15.107 as the parking-cost range shifts upward. The same overall trend is observed for ACO, while PSO and GA remain worse. The increase is driven mainly by the parking-cost component, whereas the driving-time and walking-time components change less dramatically across the sweep.

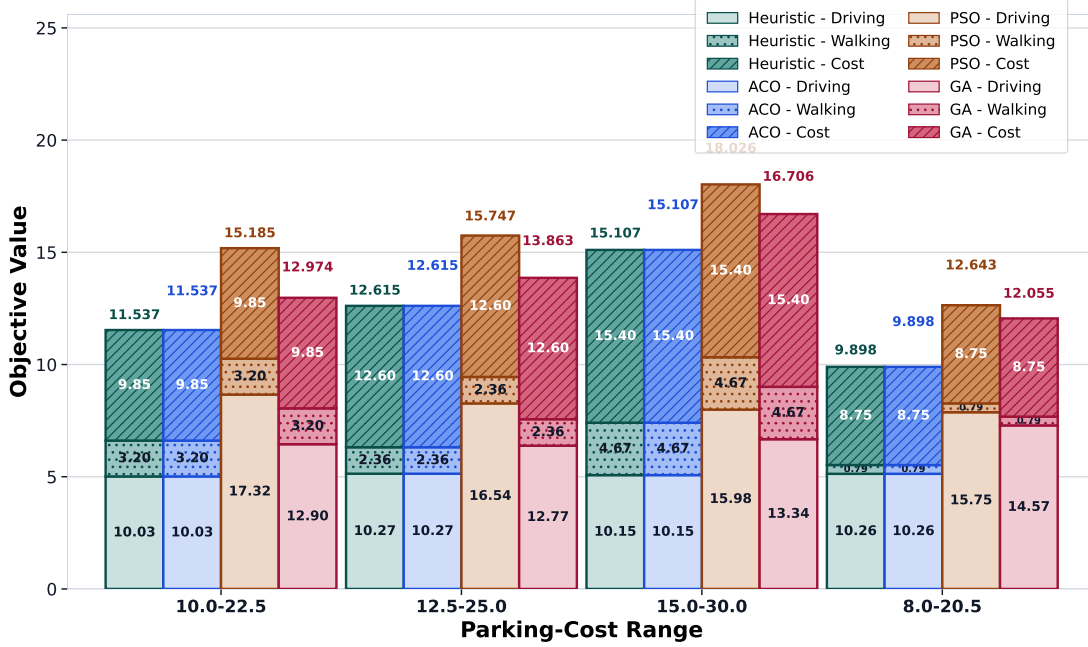


Figure 4.23: Performance comparison between the proposed heuristic and the meta-heuristic methods as a function of the parking-cost range on the fixed real-road network.

Summary

Overall, the SUMO-based evaluation confirms that the framework remains effective on a realistic directed urban road network. In the baseline fixed-network case, the heuristic produces the best plotted result among the scalable methods. Across most SUMO parameter sweeps, the heuristic and ACO remain the strongest methods, while PSO and GA are generally worse. The most stable trends are that greater edge density improves the result, heavier traffic worsens it sharply, larger parking supply improves the result, and higher parking-cost ranges worsen it. However, the walking-distance and parking-budget sweeps are not monotonic in the SUMO setting, which shows that realistic route-and-parking trade-offs can behave differently from the synthetic cases.

4.7 Conclusion

This chapter presented a Dynamic Parking Lot Optimization Framework (DPLOF) for real-time parking lot-level recommendation in urban road networks. The framework formulates parking recommendation as a two-level route-and-parking optimization problem, in which the driver first travels to a selected parking lot and then walks to the destination. The model jointly considers total travel time, walking distance, parking cost, parking availability, and the user-defined constraints on walking distance and parking cost. The optimization model was used for small-scale validation, which showed that exact solving is useful as a benchmark

but becomes computationally expensive as the network size increases. Therefore, the proposed heuristic was developed for larger instances and compared with GA, PSO, and ACO. The results show that the proposed heuristic provides stable and computationally efficient recommendations while remaining competitive with ACO, whereas GA and PSO are more sensitive to route-decoding complexity and parameter variation. The SUMO-based fixed-road-network instance further shows that DPLOF can be applied to a realistic urban road topology with directed links. Overall, the findings indicate that DPLOF is suitable for scalable real-time parking lot-level recommendation while balancing travel efficiency, walking burden, and parking cost. Chapter 5 extends this work by moving the decision from the parking lot to the individual parking spot inside the selected lot.

Chapter 5

Parking Spot-level recommendation Framework

Urban parking systems require real-time decision-making that reduces parking spot search time, improves space utilization, and provides reliable guidance under changing traffic and occupancy conditions. Existing parking recommendation systems commonly select a parking lot based on travel time, walking distance, cost, and availability. However, these approaches often ignore the internal structure of the parking lot, where individual spots may differ in availability, cost, distance from the entrance, and walking time to the pedestrian exit. To address this limitation, this chapter proposes a Dynamic Parking Spot Optimization Framework (DPSOF) for smart urban parking environments. DPSOF extends parking recommendation from parking lot-level selection to parking spot-level recommendation by representing each parking spot as a decision option with its own cost, availability, and route characteristics. The framework solves the routing and spot-selection problem dynamically by updating the driver's current node and recomputing the best feasible parking spot as the vehicle progresses node by node through the network. The total cost of selecting a spot considers external driving time, internal parking-lot travel time, walking time to the destination, and spot-specific cost, while unavailable spots are excluded before optimization. The problem minimizes a weighted combination of total travel time and parking cost under user-defined walking-distance and budget constraints. Meta-heuristic methods, namely Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Ant Colony Optimization (ACO), are considered for scalable comparison under larger urban scenarios. By selecting a specific parking spot before arrival, the proposed framework reduces post-arrival uncertainty and improves parking-space utilization in comparison with conventional lot-level recommendation approaches.

5.1 Introduction

The rapid growth of population and vehicle ownership in dense urban areas has made parking spot search a major challenge in modern smart transportation systems. Drivers often spend a considerable amount of time searching for an available parking space near their destination, which increases congestion, travel delay, fuel consumption, emissions, and user frustration [21]. Therefore, efficient parking management is required for improving road-network performance and for

reducing unnecessary vehicle movement in urban areas. Smart Parking Systems (SPS) have been developed to improve parking monitoring, availability detection, reservation, pricing, and guidance by using real-time sensing, communication, and intelligent decision-making methods [22], [23].

Existing parking allocation and recommendation systems commonly consider factors such as travel time, walking distance, parking cost, and parking availability. Multi-objective parking allocation methods have been used to balance different parking-related objectives during the recommendation [24], and genetic-algorithm-based approaches have been applied to parking-lot selection and allocation performance analysis [25]. In addition, data-driven and experience-based optimization methods have been proposed to improve adaptive decision-making in smart parking environments [26]. Other studies have considered dynamic parking guidance using ant colony optimization [27], memetic optimization for street parking allocation [28], shared parking reservation and allocation [29], and parking allocation under peak demand conditions [30]. These studies show that intelligent optimization can improve parking recommendation when traffic conditions, parking demand, and parking availability vary over time.

Recent SPS research has further improved parking management through sensing, automation, reservation, dynamic pricing, and user-centered allocation. Reservation-based valet parking studies show that reservation can improve reliability, but internal movement inside the parking facility is still not fully considered [31]. Existing surveys on smart-city parking solutions show that most systems focus on availability detection, pricing, reservation, and facility-level recommendation rather than on a detailed parking-spot-level path [32]. Similarly, IoT-based parking surveys focus on real-time sensing and parking management, but internal movement inside a parking lot is usually not treated as a core optimization component [33]. Other studies have highlighted the importance of efficient space utilization [34], dynamic fee design with rematching strategies [36], dynamic pricing and allocation under changing demand [37], and adaptive pricing with virtual voting for parking-slot allocation [38].

Although these studies improve parking recommendation and allocation, many existing frameworks still operate mainly at the parking-lot level. In these approaches, a parking lot is usually treated as a single candidate location, and the individual parking spots inside the lot are not represented explicitly. As a result, after entering a recommended facility, the driver may still need to search for a suitable available parking space. Competition-based parking reservation systems show that reservation services can reduce uncertainty, but spot-level recommendation still requires detailed modeling of the individual parking spaces [39]. IoT and machine-learning-based parking systems improve detection and prediction, but they often focus on whether a space is available rather than on how the internal driving and walking inside the facility affect the quality of the recommendation [40]. Optimization models that consider fuel, time, and cost further show the importance of evaluating several parking factors together [41], while real-time space detection and path-finding systems motivate the need to connect parking availability with route planning [42].

Another important limitation is that many parking recommendation systems solve the route-and-parking decision only once, usually from the driver’s initial location. However, in a real urban road network, the traffic condition and the parking availability may change while the driver is traveling. A parking spot that is suitable at the beginning of the route may become less suitable later because of congestion, occupancy changes, or route updates. Therefore, an effective parking recommendation framework should update the driver’s current node, recompute the route, and re-evaluate the feasible parking spots as the vehicle progresses through the network. This node-by-node recomputation allows the parking decision to adapt to the current traffic and parking state instead of relying only on the initial recommendation.

To address this gap, this chapter proposes a Dynamic Parking Spot Optimization Framework (DPSOF) for smart urban parking environments. Unlike existing parking lot-level recommendation methods, DPSOF extends the parking decision from parking lot-level selection to individual parking spot-level recommendation. Each parking spot is modeled as a separate graph node with its own cost, availability, and route characteristics. Artificial-intelligence-based parking guidance studies show the importance of intelligent parking decision support [43], and parking management systems that consider occupancy prediction and customer preferences further support the need for more detailed user-aware models [44]. Automated real-time parking management systems also show the importance of timely parking-space information for practical SPS deployment [45].

In the proposed framework, the complete path from the driver’s current location to the final destination is represented using both road-network links and parking-facility links. For each candidate parking spot, the total cost of selecting that spot includes the driving time to the parking-lot entrance, the internal driving time to the spot, the walking time from the spot to the pedestrian exit, the walking time from the exit to the destination, and the spot-specific parking cost. Online parking allocation, dynamic parking modeling, and parking-space-view-based reservation studies support the importance of real-time decision-making, optimization, and individual parking-space consideration in changing parking environments [46], [47], [48]. In the proposed model, the input graph is extended with parking-spot nodes and internal route links without changing the underlying optimization structure. Unavailable spots are removed during input generation, so only the currently available spots are evaluated as the traffic and occupancy conditions change. This framework supports downtown parking applications by guiding the driver to both a suitable parking lot and a specific available spot inside it. To evaluate scalability, the optimization model and the meta-heuristic approaches are considered under varying traffic, cost, and occupancy conditions, supported by comparative studies of GA and PSO in intelligent parking systems [49]. The proposed framework is designed for real-world parking applications, where the driver requires guidance not only to a parking lot but also to the most convenient available parking spot inside that lot.

The main contributions of this chapter are summarized as follows:

- A Dynamic Parking Spot Optimization Framework (DPSOF) is proposed by extending parking recommendation from parking lot-level selection to individual parking spot-level recommendation.
- Each parking spot is modeled as a separate graph node with its own parking identifier, spot-specific cost, availability status, and traversal characteristics.
- Internal parking-facility traversal is incorporated by adding weighted links from parking-lot entrances to parking spots and from parking spots to pedestrian exits.
- The total cost of selecting a parking spot considers external driving time, internal driving time, internal walking time, external walking time, and spot-specific parking cost.
- Dynamic parking availability is handled at the input-generation stage by considering only the currently available parking spots at each optimization step.
- The optimization model and the meta-heuristic approaches are considered for scalable comparison under varying traffic, cost, and parking occupancy conditions.
- The performance evaluation is conducted using both synthetic urban-road-network environments and a SUMO-based real-road-network case study, in order to assess the proposed framework under controlled and realistic traffic conditions.

The remainder of this chapter is organized as follows. Section 5.2 reviews related work on smart parking systems, parking allocation, pricing, reservation, and intelligent optimization methods. Section 5.3 presents the system model and the graph representation of the proposed framework. Section 5.4 formulates the dynamic parking spot recommendation problem. Section 5.5 describes the meta-heuristic solution approaches. Section 5.6 presents the simulation setup and the performance evaluation. Finally, Section 6 concludes the chapter and outlines future research directions.

5.2 Related Work

SPS have emerged as an effective solution for reducing the time drivers spend searching for parking in urban areas. Traditional parking systems provide limited parking information and often require the driver to search manually for an available parking space, which increases congestion, fuel consumption, emissions, travel delay, and inefficient use of limited parking facilities. Aljohani et al. [21] reviewed parking solutions for smart cities and discussed sensing, parking guidance, reservation, pricing, and parking management methods. Alam et al. [22] surveyed IoT-driven smart parking management systems and highlighted sensing technologies, system limitations, and future research directions. Agrawal et al. [23] reviewed smart parking techniques and innovations for maximizing urban

space, while Almalki and Ragab [24] discussed smart parking challenges and future perspectives in sustainable environments. These survey studies show that SPS have progressed through sensing, communication, automation, and intelligent parking-management technologies. However, many systems still focus on facility-level recommendation, availability detection, or general parking management rather than on detailed parking spot-level recommendation.

Several studies have focused on real-time parking allocation and optimization. Abdelmagid et al. [25] proposed a dynamic modeling framework for real-time parking allocation. Kulkarni et al. [26] developed a smart automated and real-time parking management system. Amidu et al. [27] presented a primal-dual algorithm for online parking allocation problems. Zheng et al. [28] studied parking-space allocation under peak demand in intelligent parking lots. Lian [29] proposed a shared parking-spot reservation and allocation model that prioritizes commuting demand. Hu et al. [30] developed a multi-objective intelligent optimization algorithm for parking-space allocation. These works show that allocation, reservation, and real-time decision-making are important for improving parking-space utilization. However, they generally do not represent the internal route from the parking-facility entrance to the selected spot and from that spot to a pedestrian exit.

Pricing, reservation, and demand-responsive parking strategies have also been widely studied. Nakazato et al. [31] investigated parking-lot allocation using rematching and dynamic parking-fee design. Wang et al. [32] proposed PRPSV to improve parking efficiency and reservation service based on a parking-space view. Saharan et al. [33] proposed DyPARK, a dynamic pricing and allocation scheme for smart on-street parking systems. These studies show that reservation, rematching, and pricing can improve parking utilization and reduce uncertainty. However, most of these approaches still focus on facility-level or space-view-based management rather than on complete spot-level routing with internal vehicle movement and walking components.

Meta-heuristic and search-based methods have also been applied to smart parking problems. Cheng and Nian [34] studied parking-lot optimization using a genetic-algorithm-based approach. Cheng and Kang [35] applied a genetic algorithm to path planning for automatic navigation to parking spaces inside parking lots. Abo-Zahhad and Abo-Zahhad [36] proposed an IoT and machine-learning-based smart parking system. Dimlo et al. [37] introduced a GA-RBF-based parking management model to optimize fuel, time, and cost. Dong et al. [38] developed a chaotic particle swarm optimization method for parking-space guidance. Wang et al. [39] applied parallel PSO for parking guidance and showed that swarm-based search can support route selection in complex parking environments. Adeyiga et al. [40] compared genetic algorithm and particle swarm optimization techniques in intelligent parking systems. These studies support the use of GA, PSO, and learning-based methods for parking optimization. However, their main focus is on parking guidance, path planning, fuel-time-cost optimization, or general parking-space recommendation, rather than on treating each currently available spot as a graph node with its own internal travel and walking characteristics.

Other adaptive and intelligent parking methods further show the importance of real-time and scalable decision-making. Xie et al. [41] proposed a self-optimization algorithm for multi-style smart parking based on experience, knowledge, and data. Hassoune et al. [42] developed a dynamic parking guidance architecture using ant colony optimization and multi-agent systems. Bhimavarapu et al. [43] presented an end-to-end system for real-time space detection and path finding. Akandwanaho and Govender [44] applied a memetic algorithm for dynamic street parking-space optimization. Baikani et al. [45] proposed NeuroPark Guide as an artificial-intelligence-based parking guidance solution. He et al. [46] developed a smart parking management system that considers occupancy-rate prediction and customer preferences. These studies show the value of adaptive optimization, ACO, real-time detection, path finding, intelligent guidance, and user-aware parking management. However, they mostly focus on detection, prediction, route guidance, or general allocation rather than on detailed spot-level recommendation using internal parking-facility traversal links.

Recent studies have also examined adaptive pricing, autonomous valet parking, and reservation-based services. Hassija et al. [47] proposed a parking-slot allocation framework using virtual voting and adaptive pricing. Zhang et al. [48] studied reservation-enhanced autonomous valet parking under practical constraints. Hu et al. [49] proposed CRPark, a smart parking system with competition and reservation services. These studies show that adaptive pricing, reservation, competition, and valet-parking services can reduce uncertainty and improve parking-space allocation. However, spot-level recommendation still requires explicit modeling of each available parking spot, its cost, its internal vehicle travel time, its internal walking time, and its connection to the final destination.

Overall, existing research has improved SPS through sensing, prediction, automation, allocation, pricing, reservation, path planning, and meta-heuristic optimization. However, most studies still focus on parking-lot recommendation, parking-space detection, pricing, reservation, or general allocation. Limited attention has been given to frameworks that represent the currently available parking spots as part of the routing network, with separate spot nodes, internal vehicle links, pedestrian-exit links, and spot-specific costs. Motivated by this gap, DP-SOF extends parking recommendation from parking lot-level selection to parking spot-level recommendation. The proposed framework preserves the core route-and-parking optimization structure while extending the environment graph with available parking spots and internal parking-facility traversal links. This allows the optimization model to evaluate the full cost of selecting a parking spot, including external driving time, internal vehicle travel time, internal walking time, external walking time, and the spot-specific parking cost.

5.3 System Model and Problem Description

This section describes the system model of the proposed DPSOF. The framework consists of a driver, a centralized recommendation server, real-time traffic infor-

mation, parking-spot information, parking-cost information, and an urban road network with parking-facility traversal links. These components work together to select a suitable parking spot under changing traffic, parking-availability, and parking-cost conditions. In SPS, parking prices may change with demand, so the cost of a parking lot and the cost of an individual parking spot can vary over time. Unlike conventional parking recommendation systems that select only a parking lot, DPSOF recommends a specific parking spot inside a parking facility by considering both external road-network travel and internal parking-facility movement.

The urban environment is modeled as a directed graph $G = (N, E)$, where N denotes the set of graph nodes and E denotes the set of directed edges. The nodes represent road intersections, the driver's current node A , the destination node D , the parking-facility entrance nodes, the vehicle exit nodes, the pedestrian exit nodes, and the individual parking-spot nodes. Each directed edge $(i, j) \in E$ is associated with a traversal weight that represents either vehicle travel time or pedestrian walking time. Therefore, the graph includes both external road-network links and internal parking-facility links. The road-network structure and the parking-facility layout remain fixed during the recommendation interval, while the traffic-dependent travel times and the parking-spot availability may change over time.

Let $P = \{p_1, p_2, \dots, p_m\}$ denote the set of m parking facilities, where $p_i \in P$ represents parking facility i . The driver's current vehicle location is denoted by A . At the initial decision step, A corresponds to the driver's source node. As the vehicle moves through the road network, A is updated to the driver's current node, and the routing and parking-spot decision is recomputed from that node.

Each parking facility p_i contains a set of currently available parking spots,

$$S_i = \{s_{i1}, s_{i2}, \dots, s_{ik_i}\},$$

where k_i denotes the number of currently available candidate spots in facility p_i at decision time t . Since parking availability changes over time, unavailable or occupied spots are not included in the candidate input set. Instead, the centralized recommendation server retrieves the current parking status and provides only the currently available parking spots to the optimization model.

DPSOF follows a spot-level route-and-parking decision process. For each available parking spot s_{ij} , the system evaluates a complete path from the driver's current location to the final destination. At the first level, the driver travels from the current vehicle location A to the entrance of parking facility p_i . At the second level, the vehicle moves inside the facility from the entrance to the selected parking spot s_{ij} . At the third level, after parking, the driver walks from the selected spot to the pedestrian exit of the facility. At the final level, the driver walks from the pedestrian exit to the destination D .

Let T_i^{road} denote the estimated driving time from the current vehicle location A to the entrance of parking facility p_i under the current traffic condition.

The internal vehicle travel time from the facility entrance to parking spot s_{ij} is denoted by T_{ij}^{veh} . The internal walking time from parking spot s_{ij} to the pedestrian exit is denoted by $T_{ij}^{walk,in}$, and the walking time from the pedestrian exit of facility p_i to the destination D is denoted by $T_i^{walk,out}$. Each parking spot also has a spot-specific parking cost C_{ij}^s , which may vary across the spots within the same facility.

The total travel-time component associated with selecting parking spot s_{ij} is therefore expressed as

$$T_{ij}(t) = T_i^{road} + T_{ij}^{veh} + T_{ij}^{walk,in} + T_i^{walk,out}. \quad (5.1)$$

This formulation allows the system to distinguish between different parking spots inside the same parking facility. For example, two spots in the same lot may have the same external driving time from the current vehicle location to the facility entrance, but they may have different internal vehicle travel times, internal walking times, and parking costs, depending on their location inside the facility.

As the vehicle moves through the road network, the centralized recommendation server updates the traffic state and reconstructs the set of available parking spots. When the vehicle reaches a new decision point, the system retrieves the updated road and parking information, evaluates the candidate parking spots using the objective function, and updates the recommended route and the recommended spot. This repeated recomputation continues until the vehicle reaches the selected parking spot. The model uses current traffic and parking-availability information only and does not forecast future traffic conditions or future parking availability.

Fig. 5.1 illustrates the proposed system model. Panel (A) shows the external road network, where the driver travels from the source toward the candidate parking facilities $P1$ and $P2$. The centralized server receives the updated road-network and parking information and evaluates the feasible parking alternatives. Each parking facility has a vehicle entrance, a vehicle exit, parking-cost information, and a pedestrian connection from the facility exit to the destination. Panel (B) shows the internal structure of parking facility $P1$, where the selected spot is represented inside the lot. The internal vehicle path from the entrance to the selected spot, the internal pedestrian path from the selected spot to the pedestrian exit, and the external walking path from the pedestrian exit to the destination are all represented explicitly.

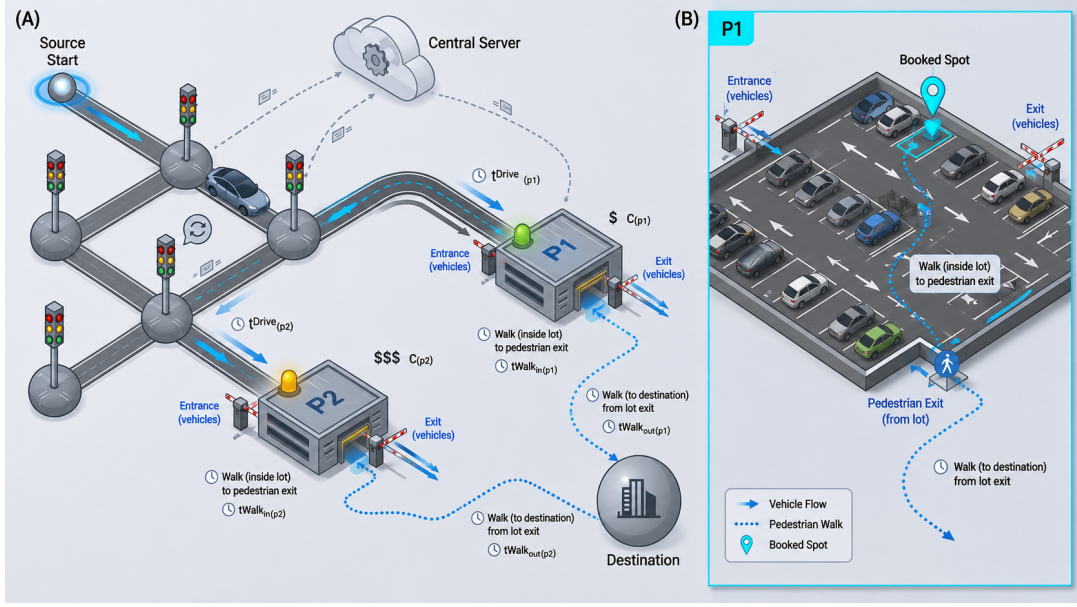


Figure 5.1: System model of the Dynamic Parking Spot Optimization Framework with (A) candidate parking facilities and (B) internal structure of $P1$.

Each available parking spot s_{ij} is characterized by its internal vehicle travel time T_{ij}^{veh} , its internal walking time $T_{ij}^{walk,in}$, its external walking time $T_i^{walk,out}$, and its spot-specific parking cost C_{ij}^s . A parking spot is excluded from consideration if it is unavailable or occupied at decision time t . In addition, a candidate spot is considered feasible only if it satisfies the user-defined walking and parking-cost constraints. The walking constraint is applied to the external walking segment from the pedestrian exit of the facility to the destination, while the internal walking time from the selected spot to the pedestrian exit is included in the objective calculation. This distinction allows the proposed model to capture the walking movement inside the parking facility, which is usually ignored in parking lot-level recommendation models.

At each decision step, the system evaluates the tuple

$$\left(T_i^{road}, T_{ij}^{veh}, T_{ij}^{walk,in}, T_i^{walk,out}, C_{ij}^s \right)$$

for every available parking spot $s_{ij} \in S_i$ and every parking facility $p_i \in P$. The parking spots that satisfy the walking and parking-cost constraints are retained as feasible options. These feasible spots are then evaluated by the optimization model and the meta-heuristic baselines under the same system model. Once a spot is selected and confirmed, the centralized server reserves that spot for a predefined reservation threshold τ . During this interval, the selected spot is blocked from being recommended to any other vehicle. If the driver arrives within τ , the spot status changes to occupied. Otherwise, the reservation expires and the spot is returned to the available candidate set.

Several assumptions are made in the proposed system model. The urban road-network structure, the parking-facility layout, and the internal traversal links remain fixed during the recommendation interval, while the road-edge travel times may change according to the current traffic condition. The parking-spot availability is updated at each decision step, and only the currently available spots are included in the optimization input. The model assumes that the centralized server can retrieve current traffic, parking-cost, and parking-availability information. The framework uses current system information only, and future traffic conditions and future parking availability are not forecasted. In addition, the proposed framework extends the parking lot-level recommendation model by adding parking-spot nodes and internal weighted links while preserving the overall route-and-parking optimization structure. The notations used throughout the system model and the problem formulation are summarized in Table 5.1.

Table 5.1: Notation used in the DPSOF system model and problem formulation

Parameters	
$G = (N, E)$	Directed graph representing the road network and parking-facility traversal links
N	Set of graph nodes
E	Set of directed edges
A	Driver's current vehicle location, initialized to the source node
D	Driver's destination node
P	Set of parking facilities
p_i	Parking facility i , where $p_i \in P$
m	Number of parking facilities
S_i	Set of currently available parking spots in facility p_i at decision time t
s_{ij}	Parking spot j in facility p_i
k_i	Number of currently available candidate spots in facility p_i
$(i, j) \in E$	Directed edge from node i to node j
t	Decision time
T_i^{road}	Driving time from the current vehicle location A to the entrance of facility p_i
T_{ij}^{veh}	Internal vehicle travel time from the facility entrance to spot s_{ij}
$T_{ij}^{walk,in}$	Internal walking time from spot s_{ij} to the pedestrian exit
$T_i^{walk,out}$	Walking time from the pedestrian exit of facility p_i to destination D
$T_{ij}(t)$	Total travel-time component for selecting spot s_{ij}
C_{ij}^s	Spot-specific parking cost of spot s_{ij} at decision time t
$W_{\max}$	Maximum acceptable walking threshold
$C_{\max}$	Maximum allowable parking-cost threshold
α	Objective-weight parameter balancing travel time and parking cost
τ	Reservation threshold duration
$q_{ij}(t)$	State of parking spot s_{ij} at time t
Z	Objective function value
Decision Variable	
z_{ij}	Binary variable equal to 1 if spot s_{ij} is selected, and 0 otherwise

5.4 Problem Formulation

This section formulates the dynamic parking spot recommendation problem based on the system model presented in the previous section. The objective is to select one feasible parking spot for the driver by jointly considering the external road-network travel time, the internal vehicle traversal time inside the parking facility, the internal walking time from the selected spot to the pedestrian exit, the external walking time from the facility exit to the destination, the spot-specific parking cost, and the user-defined constraints. At the initial decision step, the current vehicle location A corresponds to the driver's source node. During route progression, the same formulation is re-evaluated by updating A to the driver's

current node whenever the vehicle reaches a new decision point. In this structure, the destination is reached after the driver first travels to a selected parking facility, parks at the selected spot, and then walks from that spot to the destination through the pedestrian exit.

5.4.1 Network Representation and Decision Variables

Based on the network representation described in the system model, the parking environment is extended from parking lot-level candidates to parking spot-level candidates. The same route-and-parking structure used in the parking lot-level recommendation model is preserved, but the final parking decision is made at the individual parking-spot level. Therefore, the graph $G = (N, E)$ contains both external road-network edges and internal parking-facility traversal edges, including the links from parking-facility entrances to available parking spots and the links from parking spots to pedestrian exits.

The selected parking spot is identified using the binary decision variable z_{ij} . For each currently available parking spot s_{ij} , the binary decision variable in Equation (5.2) is defined as follows:

$$z_{ij} = \begin{cases} 1, & \text{if parking spot } s_{ij} \text{ is selected,} \\ 0, & \text{otherwise.} \end{cases} \quad (5.2)$$

Here, $z_{ij} = 1$ means that parking spot s_{ij} in parking facility p_i is recommended to the driver. Since unavailable or occupied spots are removed during the input-generation stage, the optimization model evaluates only the currently available parking-spot nodes in the updated environment graph.

5.4.2 Objective Function

The objective function in Equation (5.3) minimizes a weighted combination of the total travel time and the spot-specific parking cost:

$$\min Z = \sum_{i=1}^m \sum_{j=1}^{k_i} z_{ij} [\alpha T_{ij}(t) + (1 - \alpha) C_{ij}^s], \quad (5.3)$$

where $T_{ij}(t)$ denotes the total travel time associated with selecting parking spot s_{ij} , C_{ij}^s is the spot-specific parking cost at decision time t , and $\alpha \in [0, 1]$ is the objective-weight parameter that controls the trade-off between travel time and parking cost. Equation (5.3) therefore balances travel efficiency and parking cost while reflecting the user preferences. A larger value of α gives more importance to travel-time minimization, while a smaller value gives more importance to parking-cost reduction.

5.4.3 Parking-Spot Selection Constraint

The model has to select exactly one currently available parking spot for the driver. This requirement is enforced using Equation (5.4):

$$\sum_{i=1}^m \sum_{j=1}^{k_i} z_{ij} = 1. \quad (5.4)$$

Equation (5.4) ensures that the solution selects one parking spot from the candidate set generated at decision time t . Since unavailable and occupied spots are removed before optimization, the selected spot necessarily belongs to the currently available set.

5.4.4 Walking and Parking-Cost Constraints

To satisfy the user's walking preference, the selected parking spot has to satisfy the maximum acceptable walking threshold. Since this constraint is applied to the external walking segment from the parking-facility exit to the destination, it is written as

$$z_{ij} T_i^{walk,out} \leq W_{\max}, \quad \forall p_i \in P, \forall s_{ij} \in S_i. \quad (5.5)$$

Here, $T_i^{walk,out}$ denotes the walking time from the pedestrian exit of facility p_i to the destination D , and $W_{\max}$ denotes the maximum acceptable walking threshold. Equation (5.5) allows a parking spot to be selected only when the external walking segment satisfies the user's walking constraint. The internal walking time from the selected spot to the pedestrian exit is not used as a feasibility threshold in this constraint, but it is included in the total travel-time calculation in Equation (5.1).

The user's maximum parking-cost preference is enforced using Equation (5.6):

$$z_{ij} C_{ij}^s \leq C_{\max}, \quad \forall p_i \in P, \forall s_{ij} \in S_i. \quad (5.6)$$

Equation (5.6) ensures that the selected parking spot satisfies the maximum allowable parking-cost threshold. Since the cost is defined at the spot level, different spots inside the same parking facility may have different parking costs.

Finally, the decision variable is restricted to be binary:

$$z_{ij} \in \{0, 1\}, \quad \forall p_i \in P, \forall s_{ij} \in S_i. \quad (5.7)$$

Together, Equations (5.4) to (5.7) ensure that exactly one currently available parking spot is selected and that it satisfies the user-defined walking and parking-cost constraints.

5.4.5 Dynamic Availability and Reservation Update

Parking availability is not modeled as a separate decision variable in the optimization formulation. Instead, the centralized server updates the parking occupancy information at each decision step and reconstructs the candidate set by excluding the unavailable or occupied spots. Therefore, the optimization model remains deterministic for the current system state while still supporting changing parking conditions.

Once a parking spot is selected, the reservation mechanism blocks that spot for a predefined reservation threshold τ . If $z_{ij} = 1$, then the state of the selected spot is updated as

$$q_{ij}(t) = \text{reserved}. \quad (5.8)$$

At the end of the reservation window, the state of the selected spot is updated using Equation (5.9):

$$q_{ij}(t + \tau) = \begin{cases} \text{occupied}, & \text{if the driver arrives within } \tau, \\ \text{available}, & \text{otherwise.} \end{cases} \quad (5.9)$$

This reservation update prevents the same parking spot from being recommended to another driver during the reservation interval. If the driver does not arrive within the threshold duration, the spot is returned to the available candidate set.

At each decision step, the centralized server updates the road travel times, the parking costs, and the parking-spot states. It then constructs the available parking-spot candidate set, computes the complete travel-time and cost components for each candidate spot, removes the infeasible spots, solves the optimization model, and reserves the selected parking spot.

5.4.6 Interpretation of the Selected Solution

Under the proposed formulation, a solution consists of a selected parking facility, a selected parking spot inside that facility, internal vehicle movement from the facility entrance to that spot, and walking movement from the spot to the destination through the pedestrian exit. This structure ensures that the destination is reached after the driver parks at a specific available spot rather than after selecting only a general parking facility.

The formulation therefore combines travel-time evaluation, spot-level parking-cost control, walking feasibility, dynamic availability filtering, and reservation update within a single optimization model. Since the recommendation can be recomputed repeatedly as the vehicle progresses through the road network, the computational complexity may increase as the number of parking facilities, available parking spots, and road-network links increases. For this reason, the optimization model is mainly suitable for small-scale validation, while scalable meta-heuristic approaches are used for medium-scale and large-scale scenarios.

5.5 Meta-Heuristic Baselines

This section presents the meta-heuristic baselines used to evaluate the proposed DPSOF. Since the optimization formulation has already been defined in the previous section, the purpose of this section is to describe how GA, PSO, and ACO search for a feasible parking spot using the same objective structure and the same constraints. The spot-level recommendation problem becomes computationally challenging as the number of parking facilities, available parking spots, road-network links, and internal parking-facility traversal links increases. Therefore, GA, PSO, and ACO are used as comparative baselines for evaluating scalable spot-level recommendation.

Unlike the parking lot-level framework in Chapter ??, no problem-specific heuristic is developed for spot-level recommendation. This is because the spot-level decision is made after the external route to the facility entrance has already been evaluated, so the candidate set at each decision step is much smaller than the full route-and-parking search space of the lot-level problem. The comparison at this level is therefore made between the optimization model and the three meta-heuristic baselines.

All baseline methods follow the same route-and-parking structure. At each decision step, the centralized server updates the road travel times, the parking costs, and the parking-spot states. Only the currently available parking spots are included in the candidate set. A candidate solution represents the selection of one available parking spot $s_{ij} \in S_i$, together with its complete travel-time and cost components. Candidate spots that violate the walking threshold $W_{\max}$ or the parking-cost threshold $C_{\max}$ are excluded before evaluation. The remaining feasible candidates are evaluated using the same weighted objective in Equation (5.3) and the same travel-time definition in Equation (5.1). Therefore, GA, PSO, and ACO are compared under the same cost structure, feasibility constraints, and dynamic candidate set as the optimization model.

5.5.1 Genetic Algorithm

GA is implemented as an evolutionary baseline. Each chromosome represents one candidate solution and encodes the selected parking facility together with the selected available parking spot inside that facility. During each generation, the chromosomes are evaluated using Equation (5.3). Chromosomes with lower objective values are preferred, because they represent parking options with a lower combined travel time and parking cost.

The initial population is generated from the available parking-spot candidate set. Selection is applied to favor the better candidates, while crossover and mutation are used to explore alternative parking facilities and parking spots. Elitism is used to preserve the best solution found so far. The procedure is summarized in Algorithm 5.

Algorithm 5 Genetic Algorithm Baseline

```
1: Data: Parking facilities  $P$ , available parking spots  $S_i$ , travel-time components, spot-specific parking costs,  $W_{\max}$ ,  $C_{\max}$ ,  $\alpha$ , population size  $N_{\text{pop}}$ , generations  $G_{\max}$ 
2: Result: Selected parking spot  $s^*$ 
3: Generate an initial population from the currently available parking spots
4:  $s^* \leftarrow \emptyset$ ,  $Z^* \leftarrow \infty$ 
5: for  $g \leftarrow 1$  to  $G_{\max}$  do
6:   for each chromosome representing a candidate spot  $s_{ij}$  do
7:     Evaluate the total travel time using Equation (5.1)
8:     Evaluate the objective value using Equation (5.3)
9:     if  $s_{ij}$  satisfies  $W_{\max}$  and  $C_{\max}$ , and the objective value is less than  $Z^*$  then
10:       $s^* \leftarrow s_{ij}$ 
11:      Update  $Z^*$  with the current objective value
12:     end if
13:   end for
14:   Apply selection, crossover, and mutation to generate the next population
15:   Preserve the best solution using elitism
16: end for
17:
18: return  $s^*$ 
```

5.5.2 Particle Swarm Optimization

PSO is implemented as a swarm-based baseline. Each particle represents one candidate solution. The particle position maps to an available parking spot, while the particle velocity guides the movement of the search across the candidate parking-spot space. At each iteration, every particle is evaluated using Equation (5.3).

Each particle maintains its personal-best solution, while the swarm maintains a global-best solution. After evaluation, the particle positions and velocities are updated using the personal-best and global-best information. The final global-best particle is returned as the PSO-based parking spot. The procedure is summarized in Algorithm 6.

Algorithm 6 Particle Swarm Optimization Baseline

```
1: Data: Parking facilities  $P$ , available parking spots  $S_i$ , travel-time components, spot-specific parking costs,  $W_{\max}$ ,  $C_{\max}$ ,  $\alpha$ , swarm size  $N_{\text{swarm}}$ , iterations  $I_{\max}$ 
2: Result: Selected parking spot  $s^*$ 
3: Initialize particle positions using the currently available parking spots
4: Initialize particle velocities
5:  $s^* \leftarrow \emptyset$ ,  $Z^* \leftarrow \infty$ 
6: for  $iter \leftarrow 1$  to  $I_{\max}$  do
7:   for each particle do
8:     Map the particle position to a candidate parking spot  $s_{ij}$ 
9:     Evaluate the total travel time using Equation (5.1)
10:    Evaluate the objective value using Equation (5.3)
11:    Update the particle's personal best if the current solution improves it
12:    if  $s_{ij}$  satisfies  $W_{\max}$  and  $C_{\max}$ , and the objective value is less than  $Z^*$  then
13:       $s^* \leftarrow s_{ij}$ 
14:      Update  $Z^*$  with the current objective value
15:    end if
16:  end for
17:  Update particle velocities and positions using the personal-best and global-best solutions
18: end for
19:
20: return  $s^*$ 
```

5.5.3 Ant Colony Optimization

ACO is implemented as a pheromone-based baseline. Each ant constructs a candidate solution by selecting one available parking spot from the feasible candidate set. The probability of selecting a candidate spot depends on the pheromone intensity and the heuristic desirability. In this work, the heuristic desirability is based on the inverse of the objective value, so the spots with a lower combined travel time and parking cost are more likely to be selected.

After all ants have constructed their solutions, each selected spot is evaluated using Equation (5.3). Pheromone evaporation is applied to reduce the effect of weaker previous choices, and pheromone is deposited on the higher-quality parking spots. The best parking spot found across all iterations is returned as the ACO-based solution. The procedure is summarized in Algorithm 7.

Algorithm 7 Ant Colony Optimization Baseline

```
1: Data: Parking facilities  $P$ , available parking spots  $S_i$ , travel-time components, spot-specific parking costs,  $W_{\max}$ ,  $C_{\max}$ ,  $\alpha$ , number of ants  $N_{\text{ants}}$ , iterations  $I_{\max}$ 
2: Result: Selected parking spot  $s^*$ 
3: Initialize pheromone values for the currently available parking spots
4:  $s^* \leftarrow \emptyset$ ,  $Z^* \leftarrow \infty$ 
5: for  $iter \leftarrow 1$  to  $I_{\max}$  do
6:   for each ant do
7:     Select a candidate parking spot  $s_{ij}$  using the pheromone intensity and the heuristic desirability
8:     Evaluate the total travel time using Equation (5.1)
9:     Evaluate the objective value using Equation (5.3)
10:    if  $s_{ij}$  satisfies  $W_{\max}$  and  $C_{\max}$ , and the objective value is less than  $Z^*$  then
11:       $s^* \leftarrow s_{ij}$ 
12:      Update  $Z^*$  with the current objective value
13:    end if
14:  end for
15:  Apply pheromone evaporation
16:  Deposit pheromone on the high-quality parking spots
17: end for
18:
19: return  $s^*$ 
```

5.6 Performance Evaluation

5.6.1 Simulation Setup, Performance Metrics, and Parameters

The experimental evaluation is conducted on controlled synthetic urban-like road networks generated at multiple scales, together with a SUMO-based real-road network for fixed-network validation. In this framework, the parking recommendation problem is modeled at the parking-spot level rather than only at the parking-lot level. Each parking lot is expanded into multiple individual parking spots, and the recommendation process accounts for spot-specific availability, internal access time, internal walking time, and parking cost. This formulation is intended to reflect realistic parking behavior more closely, since the driver is not only choosing a parking facility but is effectively looking for a feasible and convenient parking spot inside that facility.

The evaluation follows the multi-level route-and-parking structure described in the system model. The driver first travels from the source node to the entrance of a selected parking lot, then moves internally to the selected spot, and finally walks from that spot to the destination through the pedestrian exit. In contrast to the lot-level formulation, this framework models the differences among spots inside the same lot explicitly. Therefore, even when two candidate spots belong to the same parking lot, they may differ in availability, internal driving effort, internal walking effort, and final parking charge.

The datasets are divided into small-scale, medium-scale, and large-scale

instances. The small-scale instances contain 10, 20, 30, 40, and 50 nodes and are used to compare the optimization model with the meta-heuristic baselines. The medium-scale and large-scale scalability analyses use 500, 1000, 1500, 2000, and 2500 nodes. The remaining parameter-sensitivity experiments are conducted on the fixed 2500-node instance. The SUMO-based evaluation uses a fixed 2055-node real-road network for realistic fixed-network validation.

The baseline road-network setting uses a directed urban-road graph, a parking-lot supply of approximately 5% of the nodes, a maximum walking distance $W_{\max} = 0.50$ km, an objective weight $\alpha = 0.50$, a parking-cost cap $C_{\max} = 20.5$, and a parking-cost range from 8.0 to 20.5. Low-traffic driving speeds are sampled from 50 to 70 km/h, and the walking speed is fixed at $v_{\text{walk}} = 5.0$ km/h. On top of this baseline, a spot-level parking layer is generated for every parking lot.

In the default setting, each parking lot is assigned a variable number of parking spots. Rather than using a fixed count for all lots, the number of spots per lot is generated within the range of 20 to 40 spots. Spot availability is modeled explicitly, and approximately 10% of the generated spots are marked as available in the default setting, while each parking lot is guaranteed to contain at least one available spot. This avoids degenerate cases in which a parking lot is included in the candidate set but contains no feasible spot.

To represent realistic internal parking behavior, the generated spots are divided into three classes, namely near, mid, and far. Near spots have a relatively low internal driving time and a low internal walking time, mid spots have moderate internal access characteristics, and far spots have higher internal access times. The spot costs are derived from the parent parking-lot cost, so that near spots are generally more expensive, mid spots remain close to the parent lot cost, and far spots are generally cheaper. In addition, the spot generation process enforces uniqueness in the attribute combinations inside each parking lot, so that no two spots are identical across all of the key generated values.

The evaluation follows a controlled parameter-variation setting, which is summarized in Table 5.2. For each parameter-sensitivity experiment, one parameter is varied while the remaining settings are kept at their baseline values. The walking-distance analysis uses $W_{\max} = 0.25, 0.50, 0.75$, and 1.00 km. The parking-spot-availability analysis uses approximately 10%, 20%, 30%, and 40% available parking spots. The spots-per-lot analysis uses variable spot-count ranges of 10–20, 20–40, 40–60, and 60–80 spots per parking lot. The parking-budget analysis uses $C_{\max} = 10.0, 15.0, 20.5$, and 25.0. The parking-cost variation tests the ranges 8.0–20.5, 10.0–22.5, 12.5–25.0, and 15.0–30.0. The traffic-level analysis considers very low, low, medium, and high traffic using speed ranges of 60–80, 50–70, 30–49, and 10–29 km/h, respectively. For the edge-density variation, the x-axis reports the actual number of directed edges in each scenario.

Table 5.2: Controlled parameter settings used in the spot-level parameter-sensitivity analysis.

Analysis	Parameter Setting
Node size	10–50 nodes for small-scale validation and 500–2500 nodes for medium- and large-scale scalability
Walking distance	$W_{\max} = 0.25, 0.50, 0.75, 1.00$ km
Parking-spot availability	Approximately 10%, 20%, 30%, 40% available parking spots
Spots per lot	Variable, typically 10–20, 20–40, 40–60, 60–80 spots per parking lot
Parking budget	$C_{\max} = 10.0, 15.0, 20.5, 25.0$
Parking-cost range	8.0–20.5, 10.0–22.5, 12.5–25.0, 15.0–30.0
Traffic level	Very low 60–80, low 50–70, medium 30–49, high 10–29 km/h
Edge density	Increasing road-network connectivity reported using the actual number of directed edges

The main performance metrics are driving time, walking time, parking cost, objective value, and runtime. Driving time represents the travel time from the source node to the selected parking lot, while walking time represents the time required to walk from the selected lot to the destination. Together, these components form the total travel time. Parking cost represents the monetary cost associated with the selected parking option, and runtime measures the computational efficiency of each method. For every scenario, all methods are evaluated under the same experimental conditions, including the same road network, source and destination nodes, parking candidates, maximum walking-distance constraint, parking-cost limit, and objective-weight parameter. This ensures that the comparison reflects differences between the methods rather than differences in the underlying test instance.

5.6.2 Small-Scale Evaluation

The small-scale experiments compare the optimization model with the ACO, PSO, and GA baselines on synthetic spot-level instances. These experiments are used to assess whether the meta-heuristic methods can approach the optimization benchmark while avoiding the rapid growth in computational cost that appears when exact spot-level enumeration is applied to larger instances. In the plotted objective-component comparison, the node-size analysis is reported for the 10-, 20-, 30-, and 40-node cases.

At the spot level, the exact optimization model becomes expensive more quickly than in the lot-level framework because the decision space expands from parking-lot selection to individual parking-spot selection. The runtime plot therefore plays an important role in the small-scale analysis. The optimization runtime is 0.56 s at 10 nodes and 0.30 s at 20 nodes, but it rises sharply to 1382.14 s at 30 nodes and 189890.48 s at 40 nodes. By contrast, ACO remains below 0.10 s over the plotted range, PSO remains below 0.25 s, and GA remains below 0.18 s. This result confirms that exact spot-level optimization is useful as a benchmark for small instances, but the meta-heuristic methods are much more practical once the number of feasible parking spots grows.

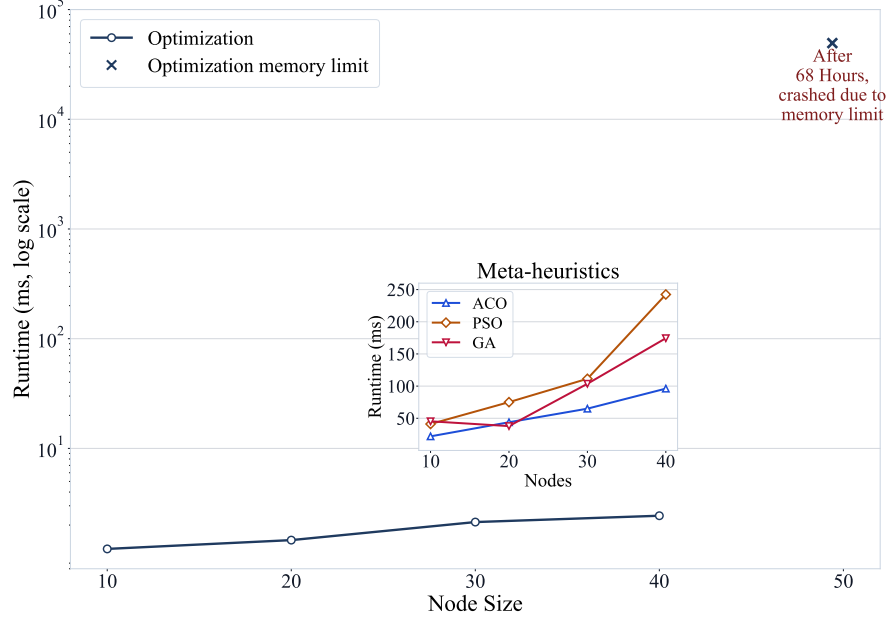


Figure 5.2: Small-scale runtime scalability of the optimization model and the meta-heuristic methods.

Effect of Network Size

Fig. 5.3 shows the effect of network size on the spot-level objective values. The optimization model gives objective values of 3.6789, 3.6533, 3.8840, and 3.7079 at 10, 20, 30, and 40 nodes, respectively. The three meta-heuristic methods remain close to one another, with ACO and GA matching exactly in the plotted values and PSO differing only slightly in the 30- and 40-node cases. At 10 nodes, all three meta-heuristics return 5.4340, and at 20 nodes they all return 4.6450. At 30 nodes, ACO and GA both return 4.7560, while PSO is effectively the same at 4.7560. At 40 nodes, all three remain close at about 4.6320. Thus, the optimization benchmark remains clearly better in the plotted node-size comparison, while the meta-heuristic methods remain tightly clustered with each other.

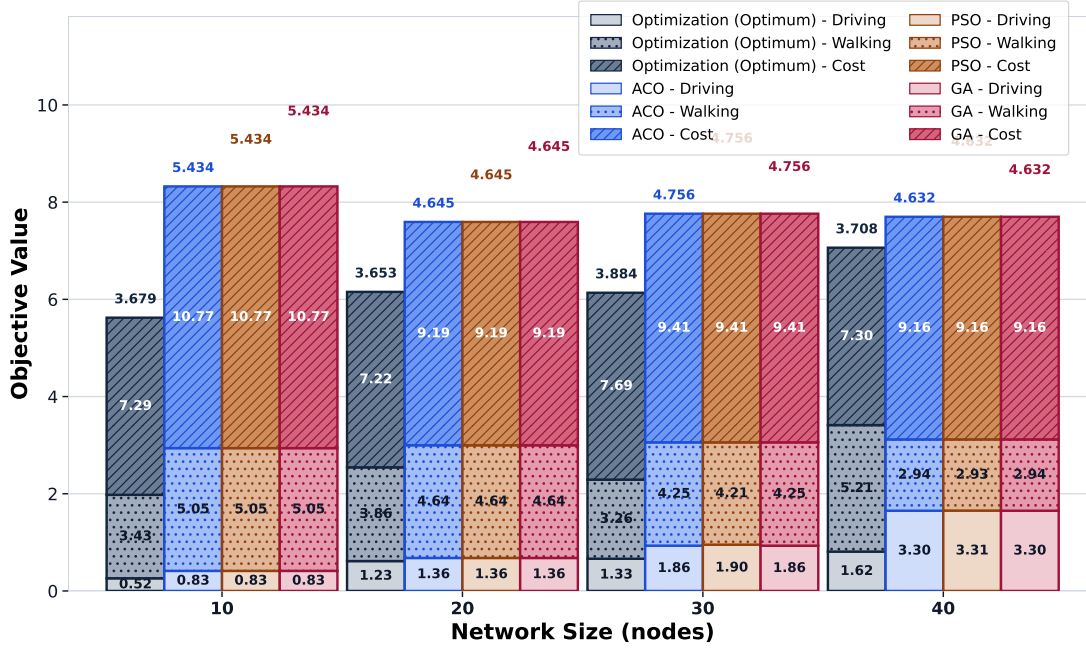


Figure 5.3: Performance comparison between the optimization model and the meta-heuristic methods as a function of network size.

Effect of Edge Density

Fig. 5.4 shows the effect of edge density. The plotted objective values remain very close across all methods, but there is a slight improvement as the number of directed edges increases. For the optimization model, the objective value changes from 9.8626 at 180 edges to 9.8608 at 192 edges and 9.8607 at 204 edges. ACO improves from 9.8626 to 9.8534 and then to 9.8508, while PSO and GA follow nearly the same pattern. The walking-time and parking-cost components remain unchanged across the sweep, so the small improvement comes mainly from better route access to the selected parking spot rather than from a major change in the chosen spot itself.

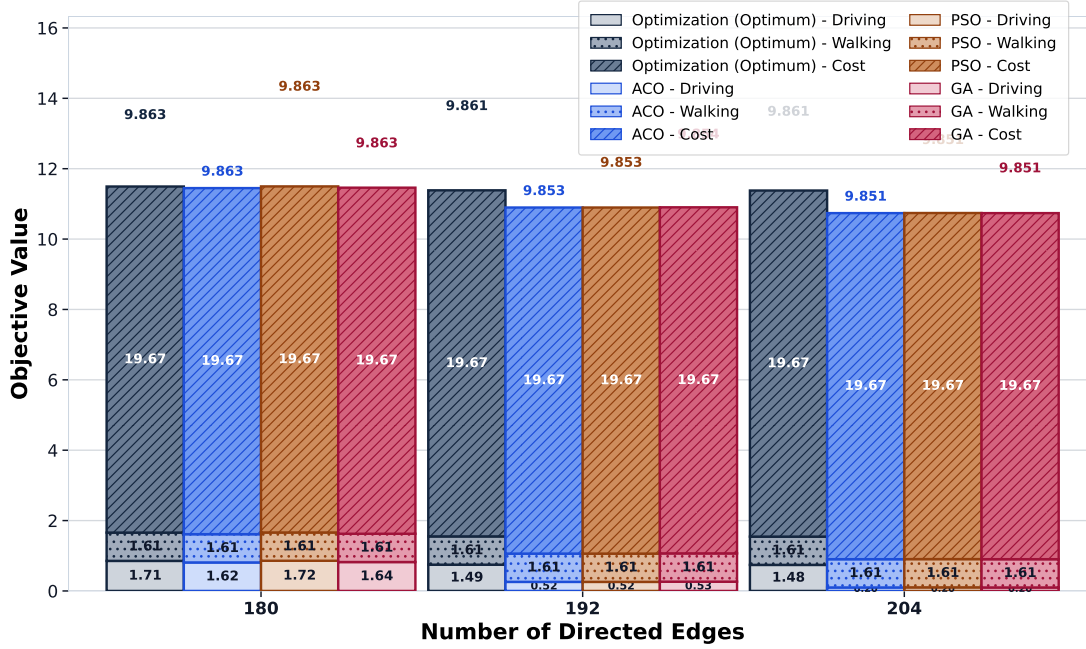


Figure 5.4: Performance comparison between the optimization model and the meta-heuristic methods as a function of edge density.

Effect of Maximum Walking Distance

Fig. 5.5 shows the effect of the maximum walking-distance constraint. All methods improve as $W_{\max}$ is relaxed. For the optimization model, the objective value decreases from 8.8957 at 0.25 km to 7.4501 at 0.50 km, 6.1506 at 0.75 km, and 4.3683 at 1.00 km. The meta-heuristic methods improve even more strongly in the plotted figures, with ACO decreasing from 8.8957 to 5.8594, 3.8279, and 2.5025. PSO and GA follow nearly the same pattern. Across the sweep, the walking-time component increases while the parking-cost component decreases substantially, which indicates that the relaxed walking limit allows the framework to select spots that are cheaper or easier to reach by vehicle even if they require a longer final walk.

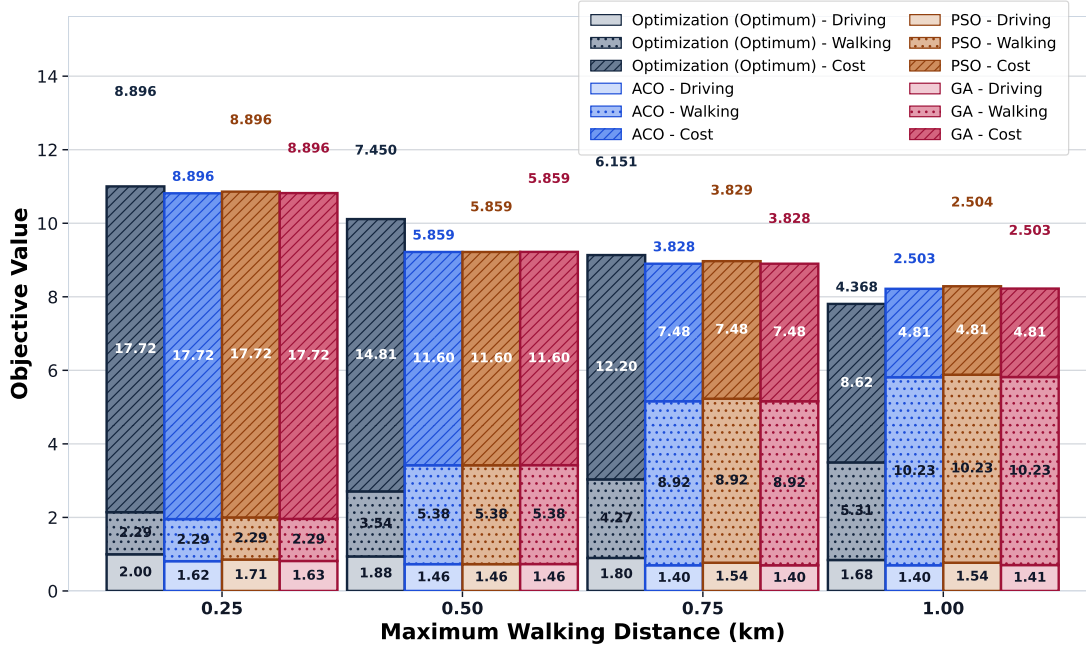


Figure 5.5: Performance comparison between the optimization model and the meta-heuristic methods as a function of the maximum walking distance.

Effect of Traffic Level

Fig. 5.6 shows the impact of traffic conditions. The optimization-model objective values remain within a relatively narrow band, from 9.8630 under very low traffic to 9.9885 under high traffic. ACO, PSO, and GA show the same overall behavior, with the best values under very low and low traffic and the worst values under high traffic. In particular, ACO increases from 9.8542 under very low traffic to 10.0136 under high traffic, while PSO rises from 9.8545 to 10.0380, and GA from 9.8542 to 10.0157. The main change comes from the driving-time component, whereas the walking-time and parking-cost terms remain almost unchanged. Therefore, the traffic-level experiment confirms that spot-level recommendation is still strongly affected by road congestion, even when the chosen spot remains similar.

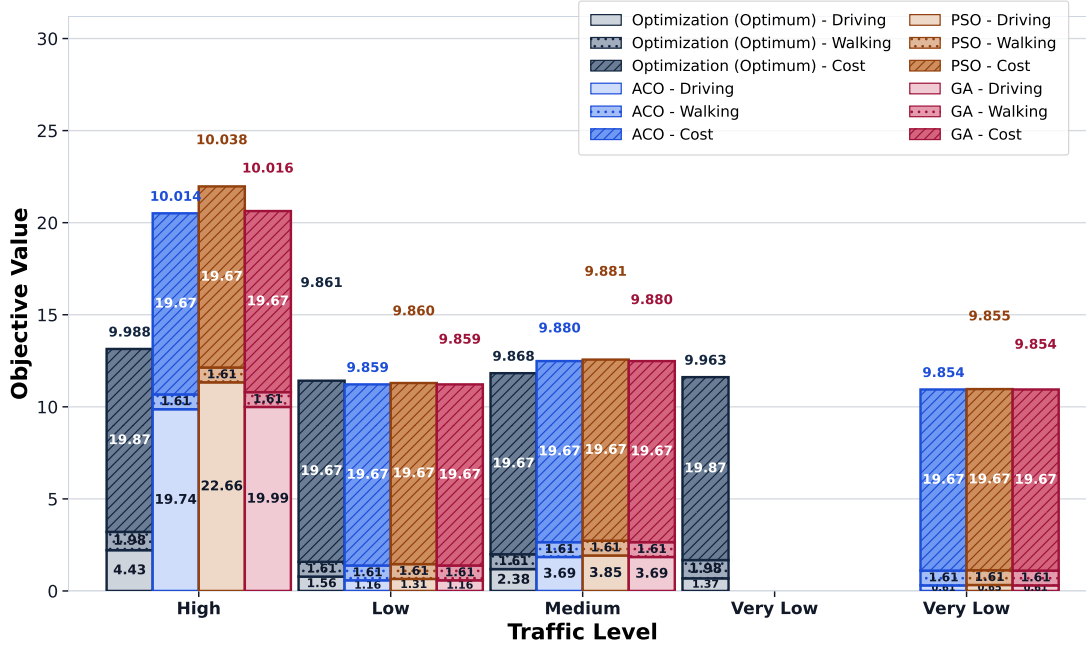


Figure 5.6: Performance comparison between the optimization model and the meta-heuristic methods as a function of the traffic level.

Effect of Maximum Parking Budget

Fig. 5.7 shows the effect of the parking-budget limit. In the plotted spot-level results, the objective value increases as the budget is relaxed. For the optimization model, the value rises from 2.3978 at $C_{\max} = 10.0$ to 2.5670, 2.8238, and 3.0422 at 15.0, 20.5, and 25.0, respectively. ACO, PSO, and GA show the same upward pattern. At the same time, the parking-cost component increases steadily, while the walking-time component decreases. This means that a larger budget allows the selection of more expensive spots that reduce some of the travel burden, but in the plotted cases the increase in cost outweighs the gain in travel-related terms. Consequently, the total objective becomes worse as the budget increases.

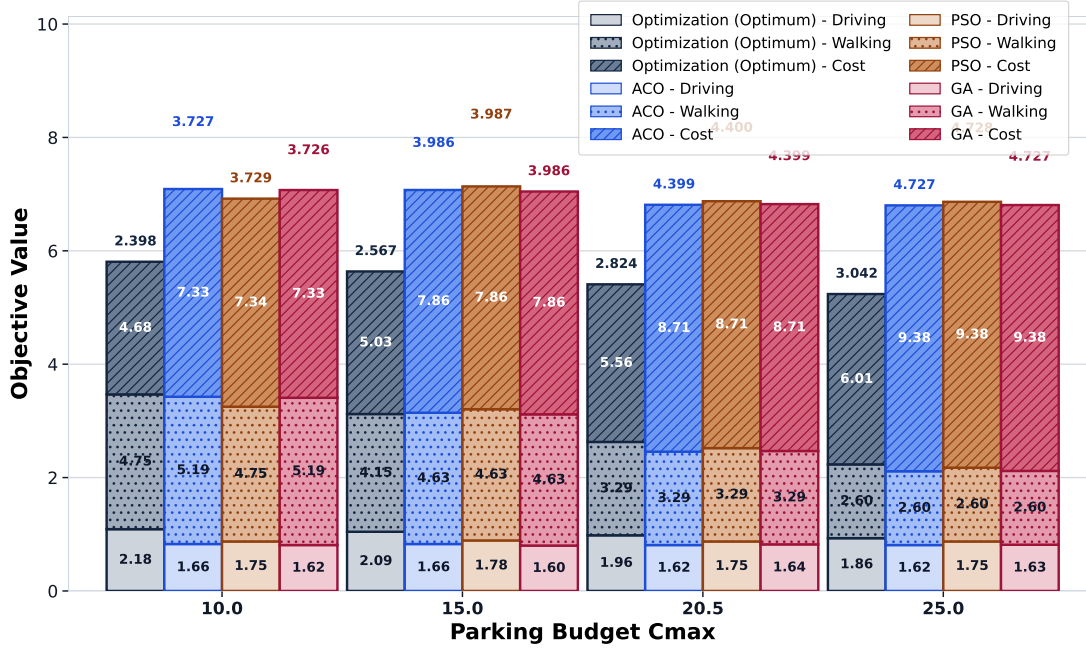


Figure 5.7: Performance comparison between the optimization model and the meta-heuristic methods as a function of the maximum parking budget.

Effect of Parking-Spot Availability

Fig. 5.8 shows the effect of parking-spot availability. The objective value improves as the proportion of available spots increases from 10% to 40%. For the optimization model, the objective decreases from 3.9962 to 3.8266, 3.7186, and 3.7079. ACO exactly matches the optimization values in the plotted figure, while GA and PSO remain extremely close. The improvement comes mainly from reductions in walking time and parking cost, while the driving-time component stays nearly unchanged. Thus, as more spots are available, the model can choose more favorable spot-level assignments within the candidate parking lots.

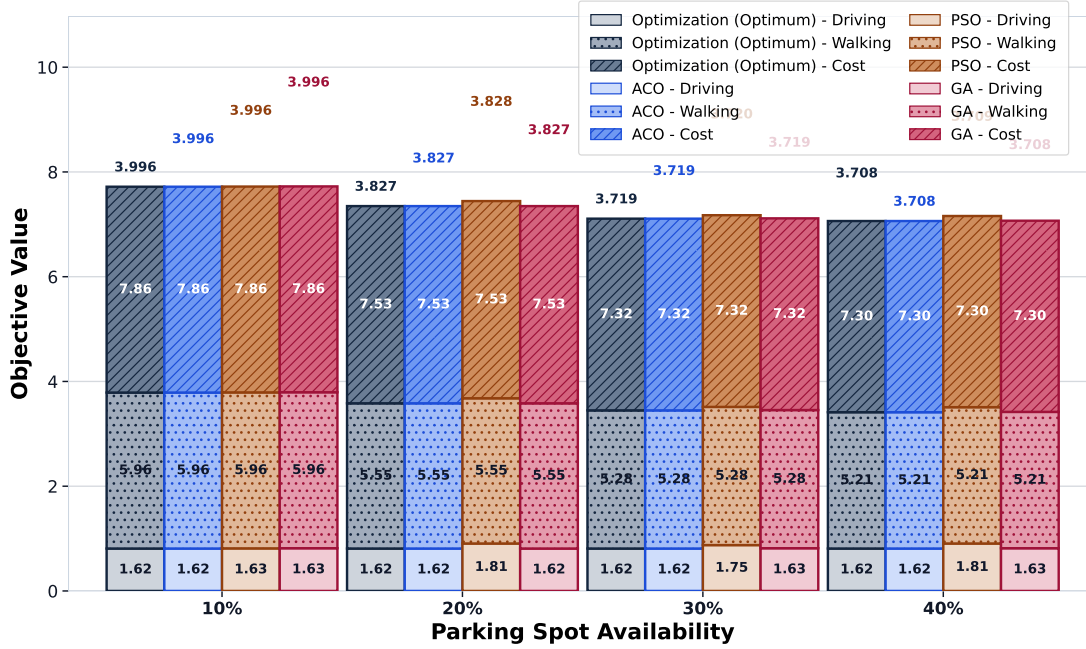


Figure 5.8: Performance comparison between the optimization model and the meta-heuristic methods as a function of the parking-spot availability.

Effect of Number of Spots per Lot

Fig. 5.9 shows the effect of the number of spots per lot. As the spot-count range increases from 10–20 to 60–80 spots per lot, the objective value decreases steadily for all methods. For the optimization model, the objective improves from 3.9484 to 3.7179, 3.4624, and 3.2170. ACO again matches the optimization values in the plotted graph, while PSO and GA remain only slightly worse. The main effect is a gradual reduction in parking cost, with only small changes in driving time and walking time. Therefore, increasing the number of available spot choices inside each lot improves the probability of selecting a better internal spot assignment.

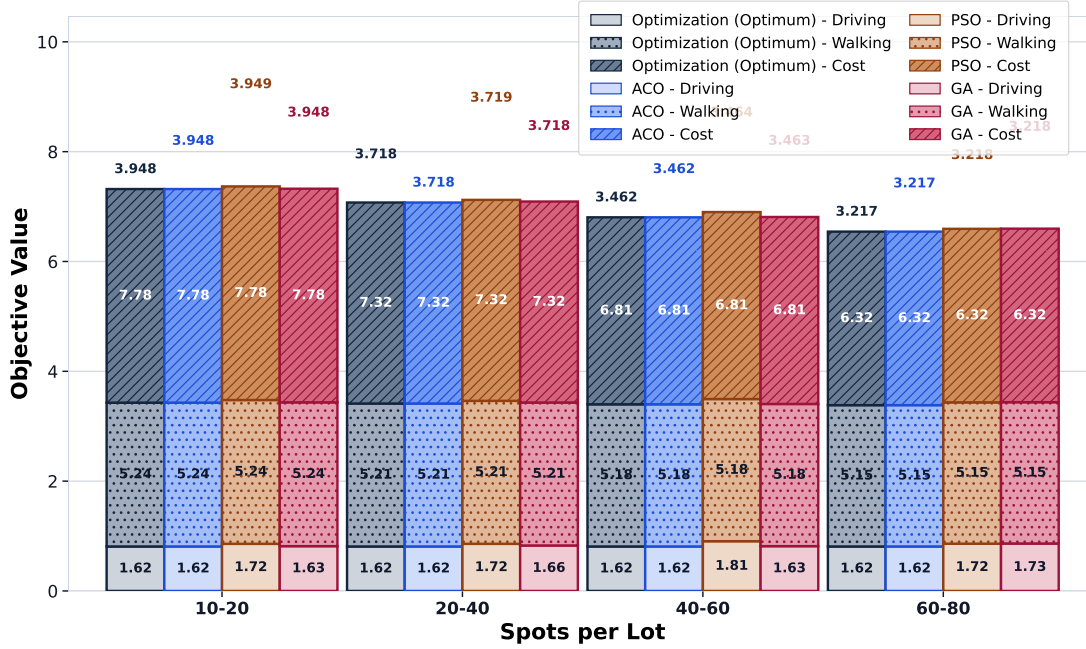


Figure 5.9: Performance comparison between the optimization model and the meta-heuristic methods as a function of the number of parking spots per lot.

Effect of Parking-Cost Range

Fig. 5.10 shows the effect of the parking-cost range. As the spot-level cost range shifts upward, the objective value increases for every method. For the optimization model, the value increases from 3.6976 in the 8.0–20.5 setting to 4.9326, 6.7026, and 9.2307. ACO, PSO, and GA show the same overall pattern, with only very small differences between them. Across the sweep, the parking-cost component increases substantially, while the driving-time and walking-time components change only slightly. Hence, this experiment is dominated by the monetary part of the spot-level objective.

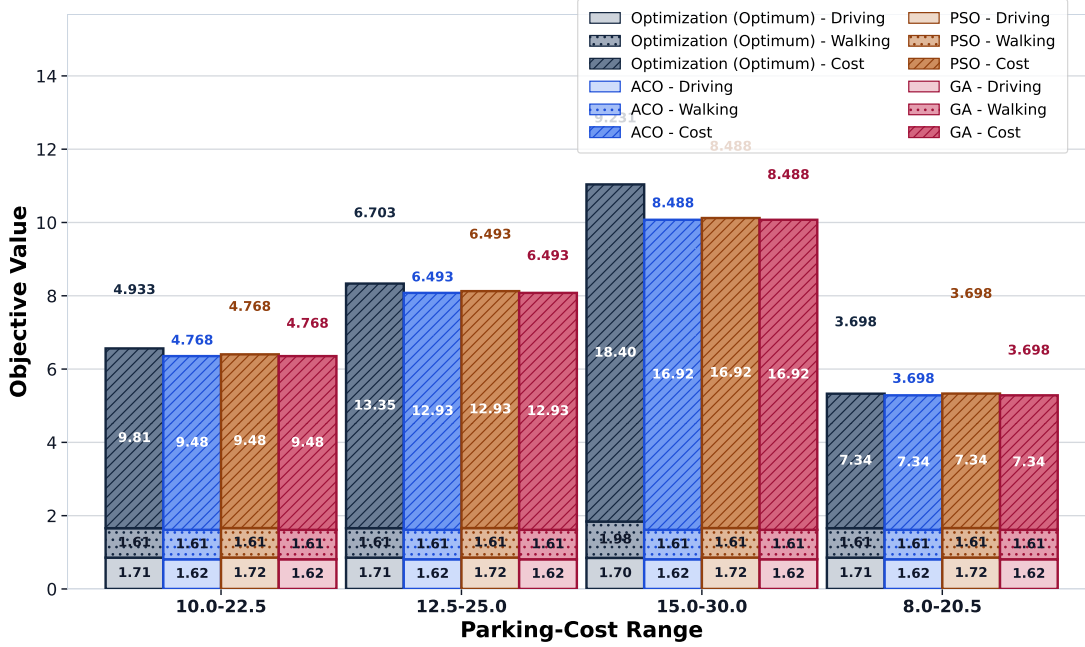


Figure 5.10: Performance comparison between the optimization model and the meta-heuristic methods as a function of the parking-cost range.

Summary

Overall, the small-scale spot-level results show that the optimization model provides the strongest benchmark in the node-size comparison, while the meta-heuristic methods frequently remain very close to one another and often nearly match the optimization trace in the parameter-sensitivity experiments. ACO and GA are especially close across most of the plotted small-scale cases, with PSO also remaining competitive. The main trends are clear: greater walking flexibility, greater spot availability, and more spots per lot improve the objective value, whereas higher parking-cost ranges and larger parking budgets worsen it in the plotted cases. Traffic congestion also worsens performance through the driving-time component, while edge density provides a small but consistent benefit.

5.6.3 Medium- and Large-Scale Networks: Scalability and Parameter Analysis

For medium-scale and large-scale networks, the evaluation compares ACO, PSO, and GA under controlled parameter-sensitivity settings. Network sizes of 500, 1000, 1500, 2000, and 2500 nodes are used for the scalability analysis. The remaining parameter-sensitivity experiments are conducted on the fixed 2500-node instance in order to isolate the effects of edge density, traffic level, walking-distance limit, parking budget, parking-spot availability, number of spots per lot, and parking-cost range. These experiments are required because a method that

performs well on small instances may not remain practical when the road network and the spot-level candidate set expand.

Effect of Network Size

Fig. 5.11 shows the effect of network size. ACO gives objective values of 3.7059, 4.9573, 4.2411, 3.3438, and 3.4462 at 500, 1000, 1500, 2000, and 2500 nodes, respectively. PSO and GA follow a similar non-monotonic pattern, but both remain worse than ACO at every plotted node size. PSO is the weakest overall, while GA is closer to ACO at the smaller sizes and diverges more clearly at 2500 nodes. Thus, ACO is the most stable method in the medium/large node-size comparison.

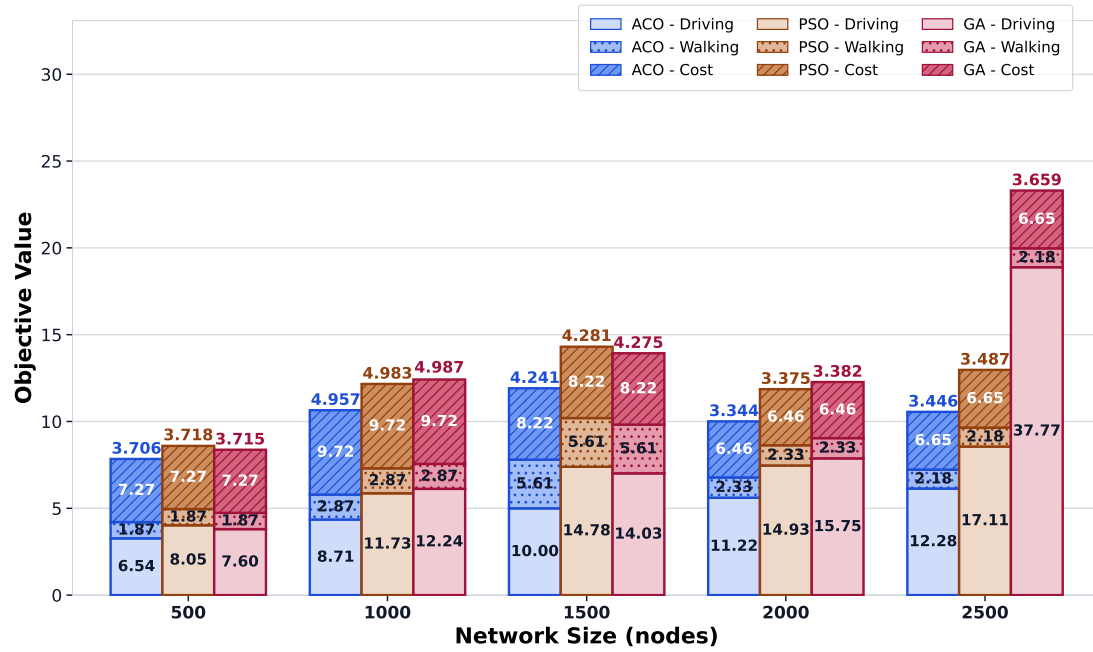


Figure 5.11: Performance comparison between the meta-heuristic methods as a function of network size.

Effect of Edge Density

Fig. 5.12 shows the edge-density results. ACO improves slightly from 3.4462 at 7446 directed edges to 3.4268 at 8278 edges and 3.4252 at 9110 edges. GA also improves overall, from 3.6587 to 3.4495 and 3.4362. PSO improves sharply from 3.4865 to 3.4360 at 8278 edges, but then worsens slightly to 3.5160 at 9110 edges. Therefore, the main improvement with edge density is again seen most clearly in ACO, while PSO remains more variable.

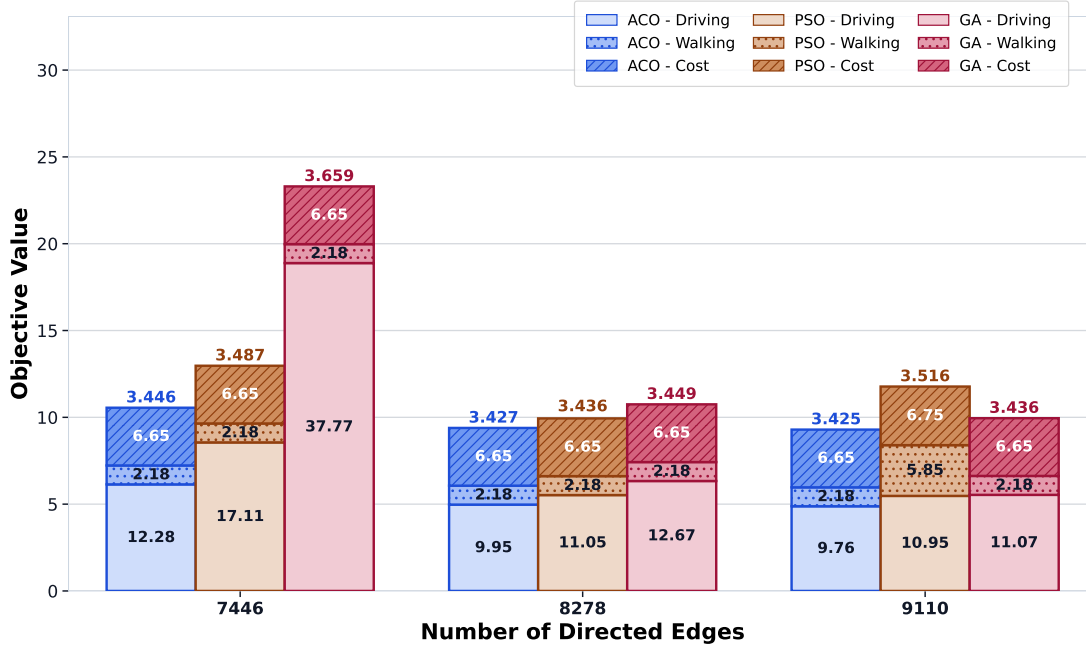


Figure 5.12: Performance comparison between the meta-heuristic methods as a function of edge density.

Effect of Maximum Walking Distance

Fig. 5.13 shows the effect of the maximum walking-distance constraint. All three methods improve as $W_{\max}$ is relaxed. ACO decreases from 5.2858 at 0.25 km to 4.3824, 3.4576, and 2.4549 at 0.50, 0.75, and 1.00 km, respectively. PSO and GA follow the same overall trend, although they remain worse than ACO throughout. The plotted figures show that the relaxed walking constraint allows the selection of cheaper or otherwise more favorable spots, even though the walking-time component itself increases.

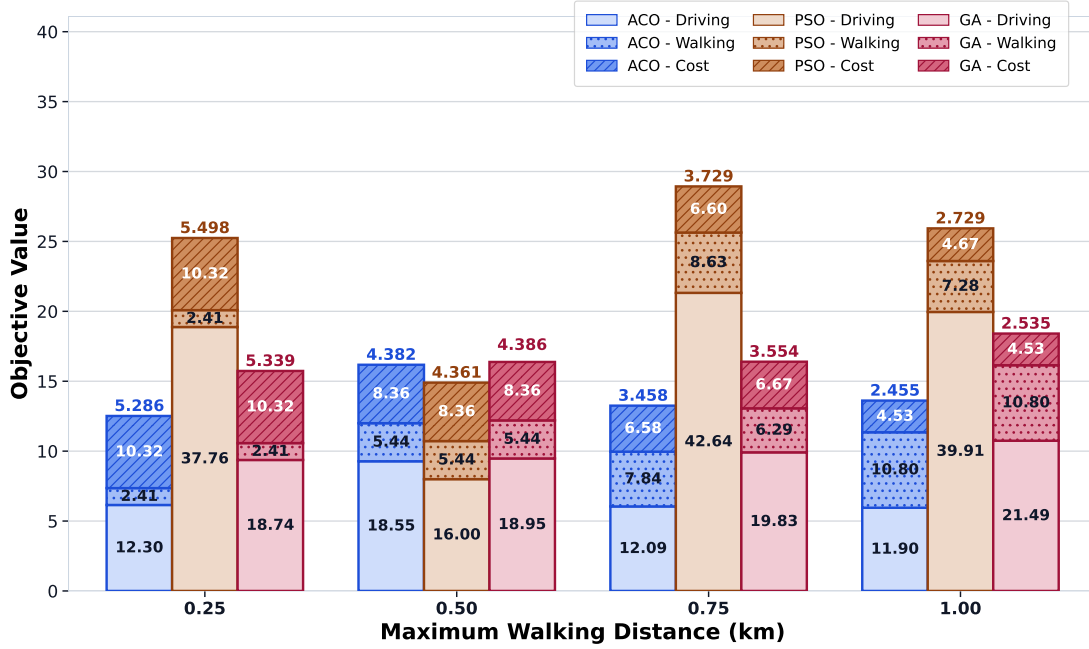


Figure 5.13: Performance comparison between the meta-heuristic methods as a function of the maximum walking distance.

Effect of Traffic Level

Fig. 5.14 shows the traffic-level results. ACO increases from 3.4246 under very low traffic to 3.5173 under low traffic, 3.5667 under medium traffic, and 3.7298 under high traffic. PSO and GA show the same overall increase, with both methods performing worse than ACO at every traffic level. This indicates that at the medium/large spot level, congestion still primarily affects the driving-time component, while the internal walking and parking-cost terms change less dramatically.

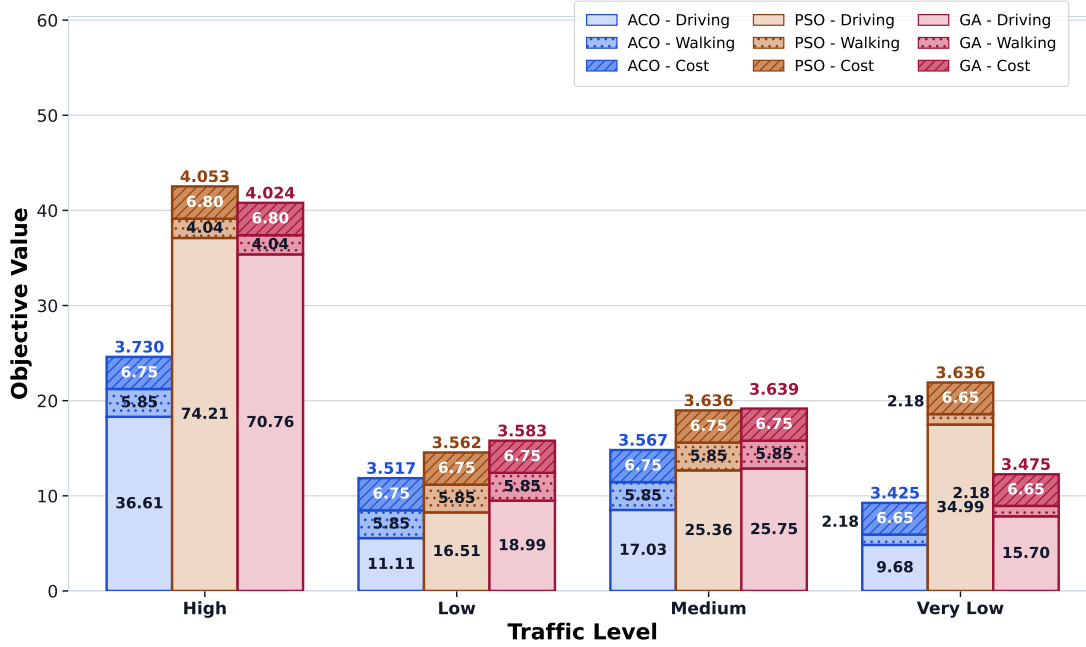


Figure 5.14: Performance comparison between the meta-heuristic methods as a function of the traffic level.

Effect of Maximum Parking Budget

Fig. 5.15 shows the effect of the parking-budget limit. In the plotted spot-level results, the objective value increases steadily as the budget is relaxed. For ACO, the value rises from 2.4554 at $C_{\max} = 10.0$ to 2.9402, 3.8648, and 4.6511. PSO and GA again show the same qualitative pattern but remain worse overall. Thus, under this plotted budget sweep, a looser budget encourages the selection of more expensive spots whose travel-time savings do not compensate for the increased parking cost.

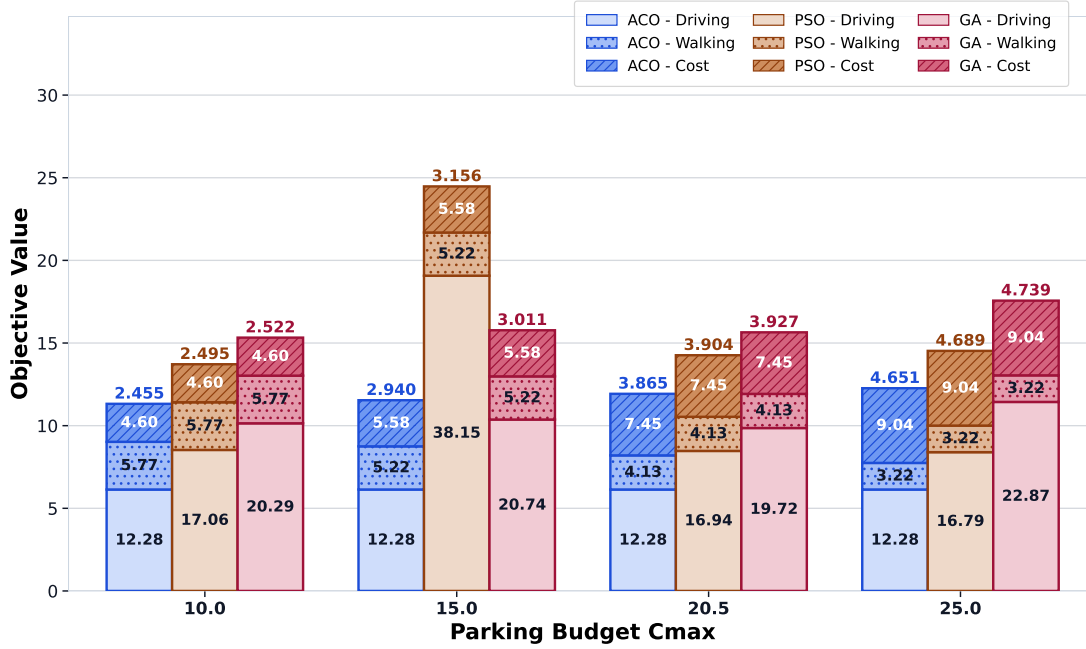


Figure 5.15: Performance comparison between the meta-heuristic methods as a function of the maximum parking budget.

Effect of Parking-Spot Availability

Fig. 5.16 shows the effect of parking-spot availability. ACO improves from 4.3426 at 10% availability to 4.2354, 4.1481, and 4.0559 as availability increases to 20%, 30%, and 40%. PSO and GA also improve overall, although both remain worse than ACO. The reduction in objective value is accompanied by lower walking-time and parking-cost components, which indicates that increased availability allows the recommendation to choose more attractive spots inside the candidate parking lots.

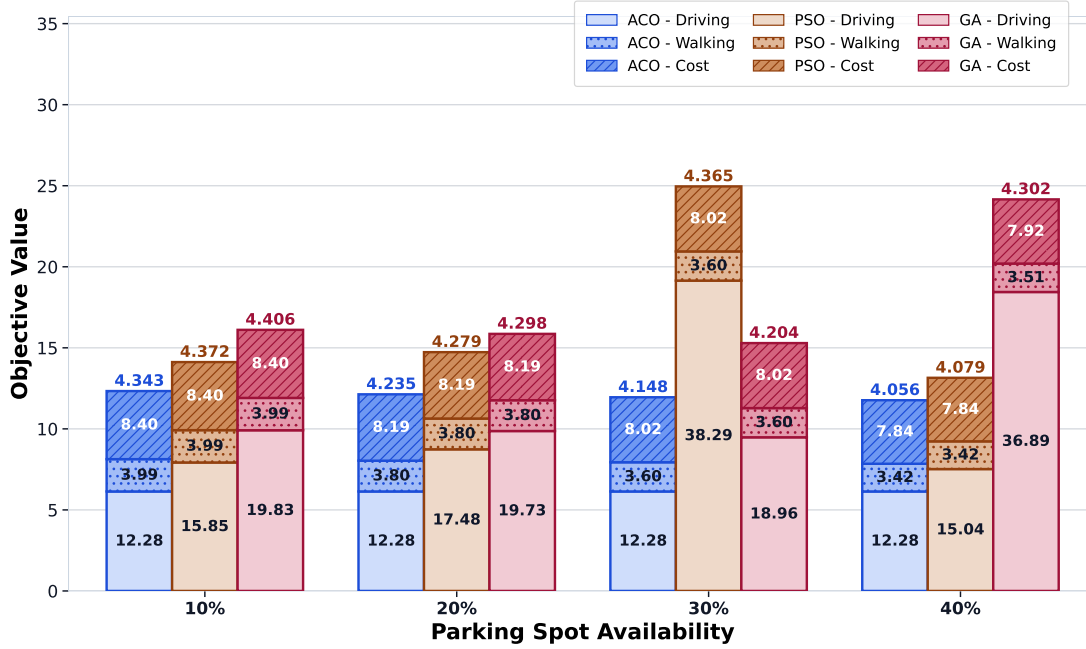


Figure 5.16: Performance comparison between the meta-heuristic methods as a function of the parking-spot availability.

Effect of Number of Spots per Lot

Fig. 5.17 shows the spots-per-lot analysis. ACO improves steadily from 3.7018 for the 10–20 setting to 3.4563, 3.2209, and 2.9705 as the number of spots per lot increases. PSO and GA follow the same downward trend but remain consistently worse. This shows that a richer internal spot set improves the probability of finding a more favorable parking-spot assignment, mainly through a reduction in the parking-cost component.

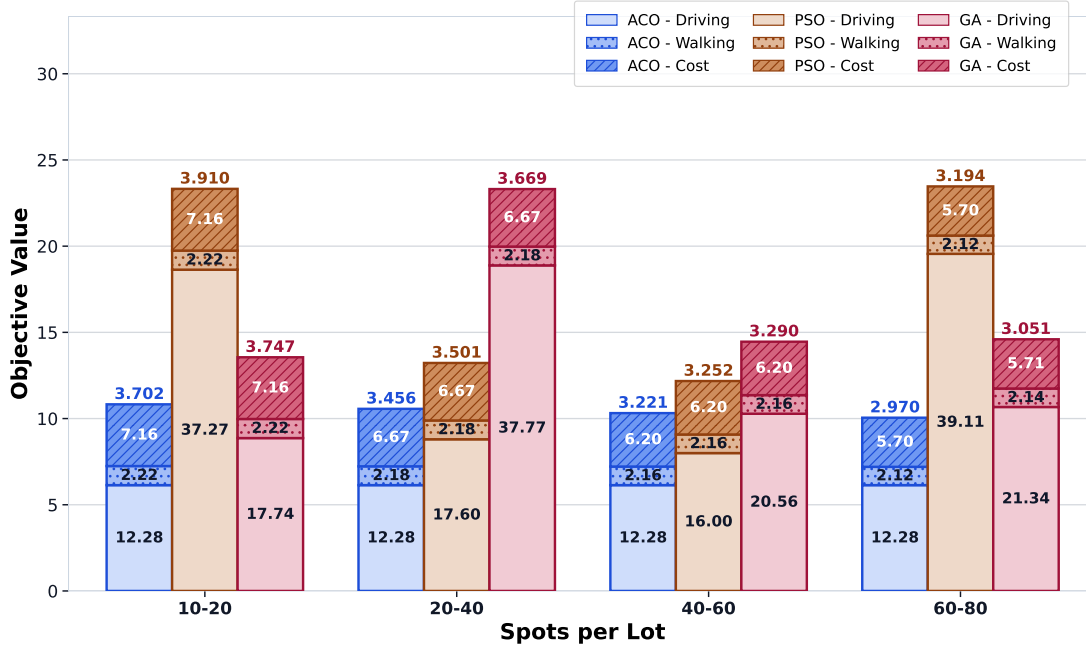


Figure 5.17: Performance comparison between the meta-heuristic methods as a function of the number of parking spots per lot.

Effect of Parking-Cost Range

Fig. 5.18 shows the parking-cost-range analysis. For ACO, the objective increases from 3.4462 in the 8.0–20.5 setting to 4.4190, 5.3361, and 7.2232 as the spot-cost range shifts upward. PSO and GA follow the same trend and remain worse than ACO. The main driver of this increase is the parking-cost component, while the driving-time and walking-time components change more modestly.

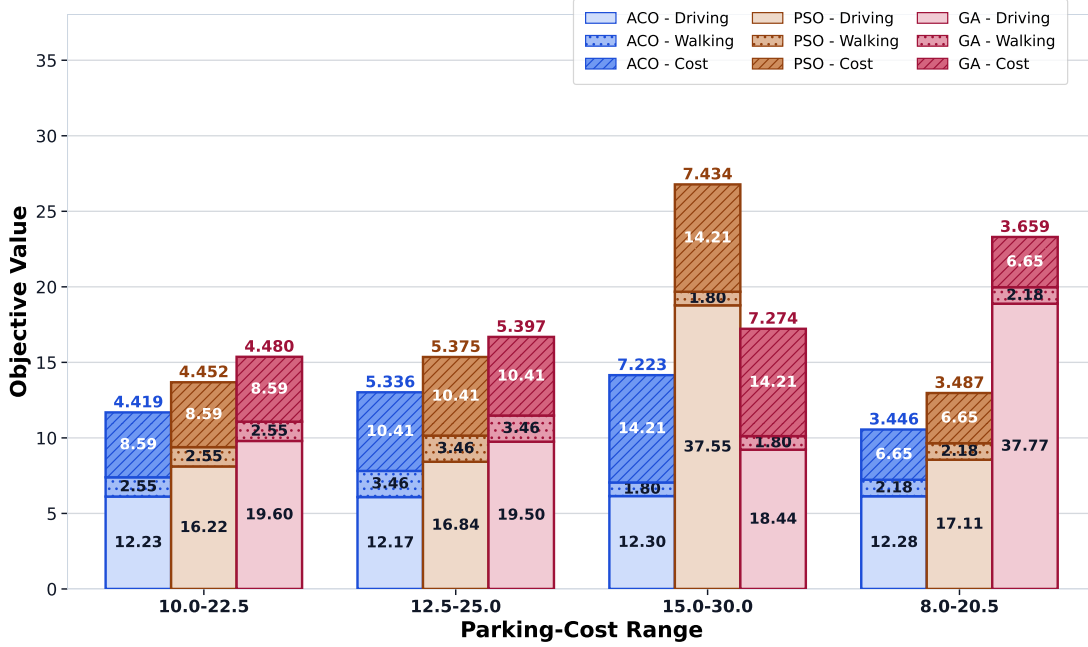


Figure 5.18: Performance comparison between the meta-heuristic methods as a function of the parking-cost range.

Summary

Overall, the medium- and large-scale spot-level results show that ACO is the strongest and most stable method across the plotted synthetic experiments. PSO and GA remain competitive in some cases, but they are generally worse and more variable. The main trends are consistent across the medium/large analysis: greater walking flexibility, greater spot availability, and more spots per lot improve the objective, while heavier traffic, higher parking-cost ranges, and larger parking budgets worsen it in the plotted cases. Edge density provides a small improvement overall, although the gain is modest compared with the effects of the other parameters.

5.6.4 Realistic Urban Network Evaluation Using SUMO

For the realistic urban-road evaluation, the framework is tested on a SUMO-based road network extracted from the Waterloo–Kitchener urban area using OpenStreetMap data. Unlike the medium-scale and large-scale synthetic experiments, this evaluation does not study scalability across multiple network sizes. Instead, all experiments are conducted on a fixed 2055-node directed road network, and the analysis examines how ACO, PSO, and GA behave under controlled variations in traffic level, walking-distance limit, parking-cost range, parking-spot availability, number of spots per lot, parking-budget limit, and edge density.

The SUMO-based evaluation is required because strong performance on synthetic graphs does not guarantee stable behavior on a realistic urban topology

with one-way roads, irregular connectivity, and a real-road travel structure. The fixed-network analysis is therefore used as a realistic validation rather than as a scale-based benchmark.

Effect of Network Size

Fig. 5.19 gives the baseline comparison on the fixed 2055-node SUMO road network. ACO achieves the best plotted objective value, namely 3.6319, while PSO and GA return 3.6781 and 3.6837, respectively. The three methods are therefore very close in the baseline SUMO setting, but ACO remains the best among them.

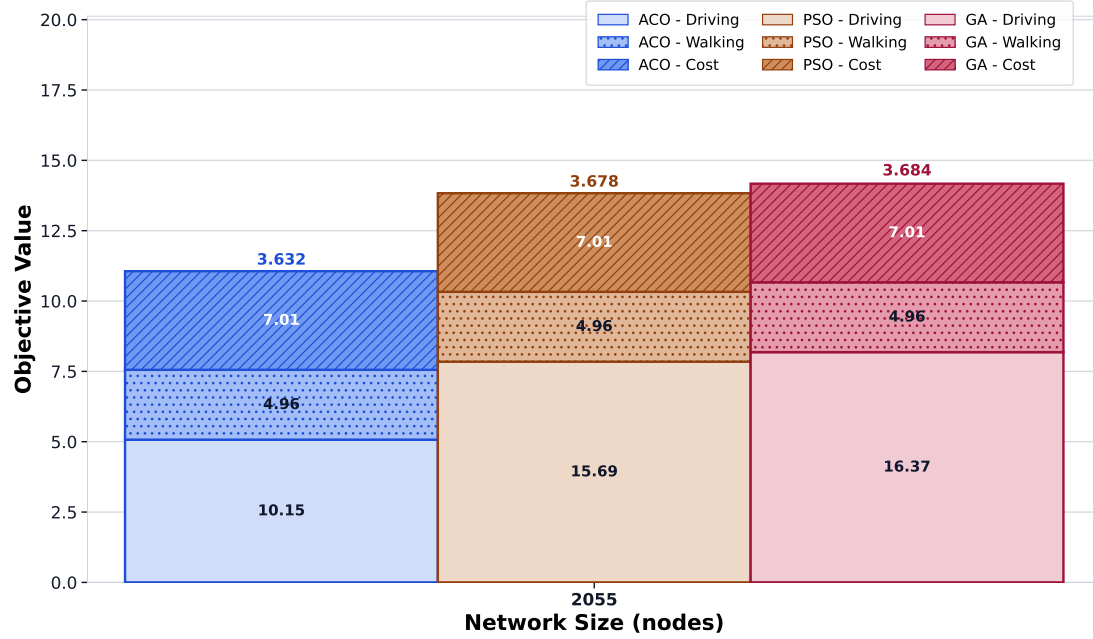


Figure 5.19: Performance comparison between the meta-heuristic methods on the fixed 2055-node SUMO road network.

Effect of Edge Density

Fig. 5.20 shows the edge-density results on the fixed real-road network. ACO changes from 3.6319 at 5093 edges to 3.6341 at 5777 edges and 3.7136 at 6461 edges. PSO and GA follow almost the same pattern. Unlike the synthetic medium/large case, this SUMO sweep does not show a monotonic improvement with denser connectivity. Instead, the best plotted result occurs at the baseline edge-density setting, and the denser settings are slightly worse.

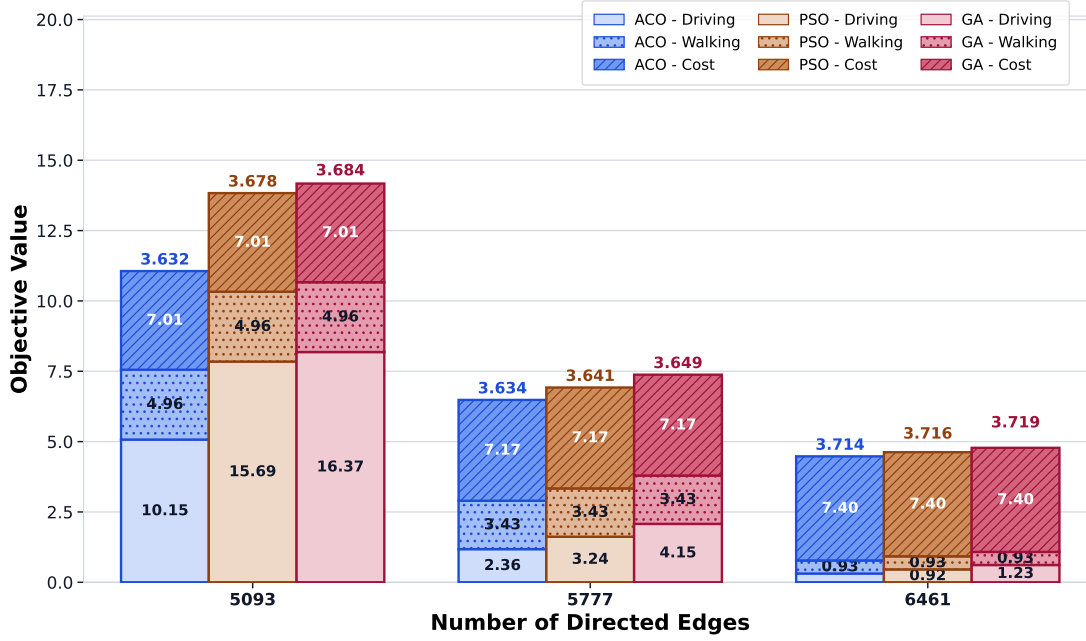


Figure 5.20: Performance comparison between the meta-heuristic methods as a function of edge density.

Effect of Maximum Walking Distance

Fig. 5.21 shows the effect of the walking-distance limit. All three methods improve steadily as $W_{\max}$ increases. For ACO, the objective value decreases from 5.3687 at 0.25 km to 4.2669, 3.2163, and 2.3506 at 0.50, 0.75, and 1.00 km, respectively. PSO and GA follow the same trend, with GA slightly outperforming ACO at 1.00 km by a very small margin. This indicates that on the realistic road network, allowing a longer final walk can substantially improve the overall spot-level recommendation.

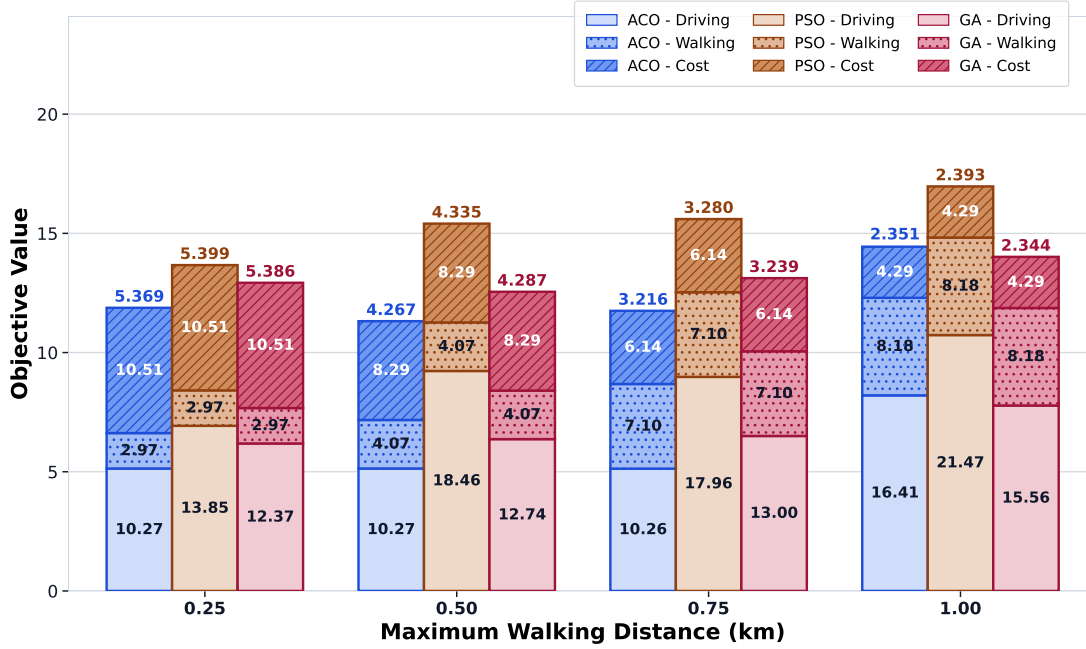


Figure 5.21: Performance comparison between the meta-heuristic methods as a function of the maximum walking distance.

Effect of Traffic Level

Fig. 5.22 shows the traffic-level analysis. ACO increases from 3.7121 under very low traffic to 3.5083 under low traffic, 3.8234 under medium traffic, and 4.9902 under high traffic. Although the low-traffic case is slightly better than the very-low-traffic case, the dominant trend is that heavier congestion worsens the objective, especially in the jump from medium to high traffic. PSO and GA follow the same pattern and remain slightly worse than ACO across the sweep.

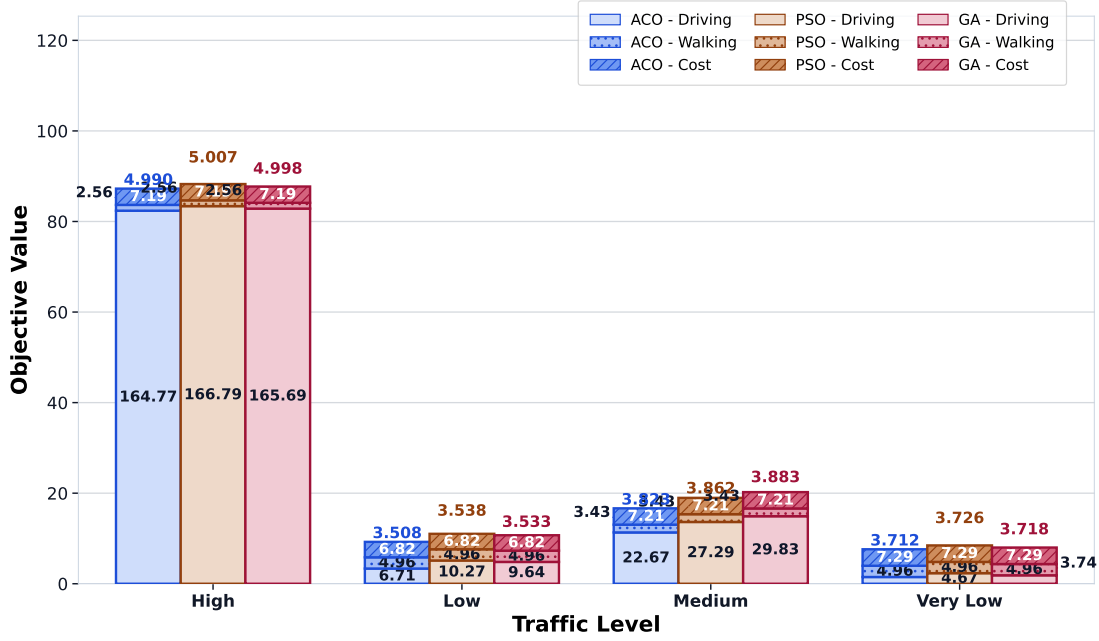


Figure 5.22: Performance comparison between the meta-heuristic methods as a function of the traffic level.

Effect of Maximum Parking Budget

Fig. 5.23 shows the effect of the parking-budget limit. ACO increases from 3.0360 at $C_{\max} = 10.0$ to 3.6296, 4.1669, and 4.8986 as the budget is relaxed. PSO and GA show the same upward pattern. Thus, for the plotted SUMO spot-level cases, a larger parking budget does not improve the objective; instead, it permits more expensive spot selections whose travel-time benefits are not large enough to offset the higher cost.

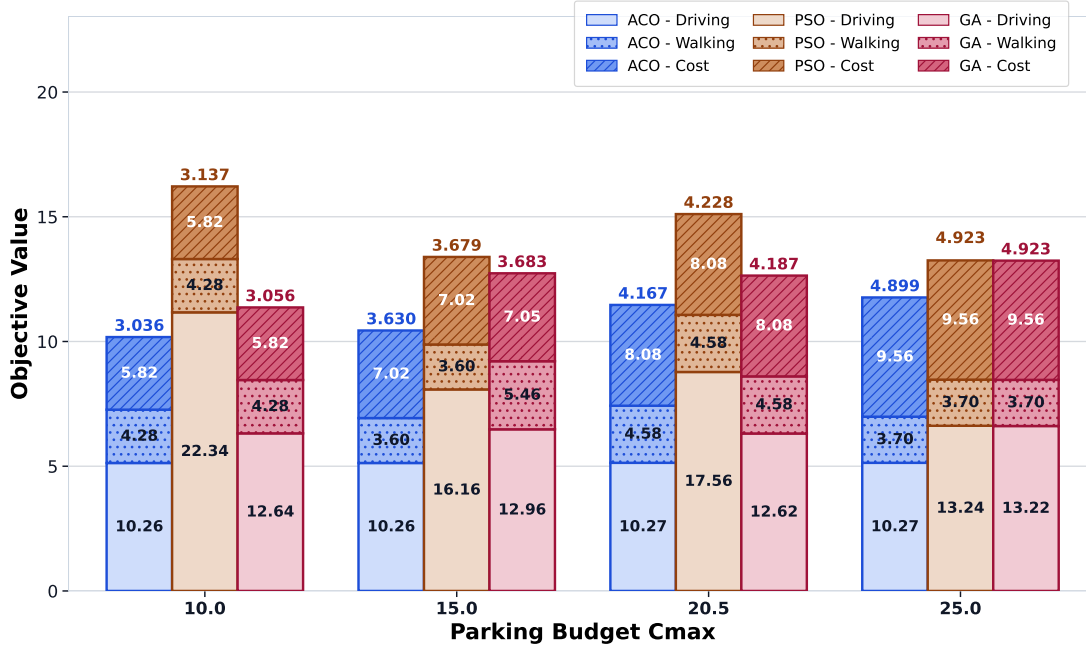


Figure 5.23: Performance comparison between the meta-heuristic methods as a function of the maximum parking budget.

Effect of Parking-Spot Availability

Fig. 5.24 shows the effect of parking-spot availability. ACO improves steadily from 3.9505 at 10% availability to 3.7963, 3.6833, and 3.6328 as the percentage of available spots increases. PSO and GA also improve overall, although GA shows a flat segment between 20% and 30%. The general pattern is that more available spots improve the recommendation by reducing both the walking burden and the effective parking-cost penalty.

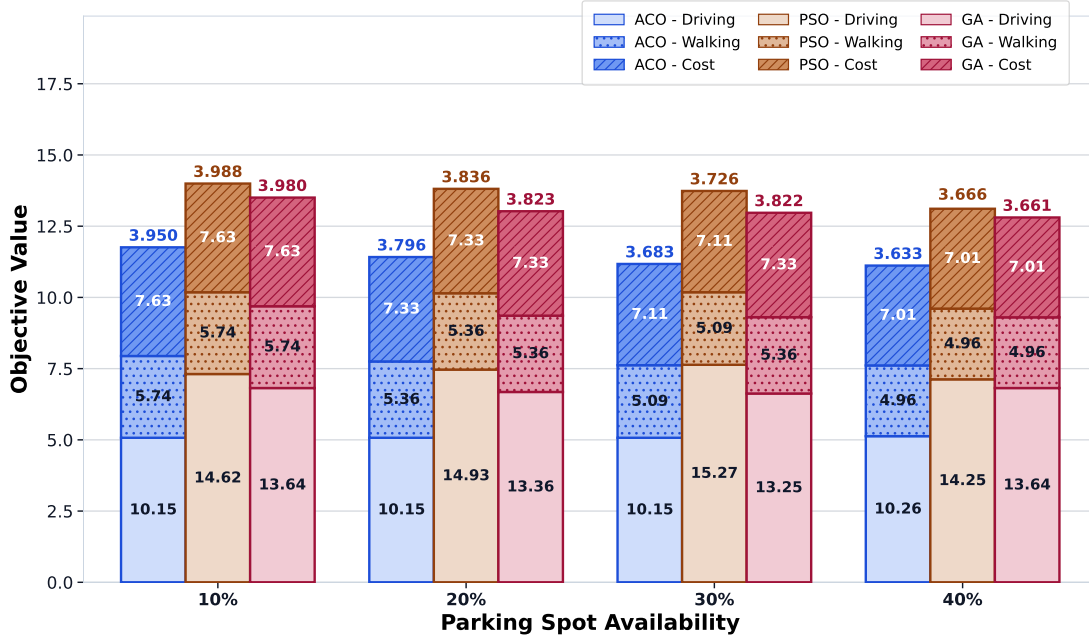


Figure 5.24: Performance comparison between the meta-heuristic methods as a function of the parking-spot availability.

Effect of Number of Spots per Lot

Fig. 5.25 shows the spots-per-lot analysis. ACO improves from 3.8924 in the 10–20 setting to 3.6419, 3.4015, and 3.1713 as the number of spots per lot increases. PSO and GA follow the same downward pattern and remain only slightly worse than ACO. Thus, increasing the number of available internal parking choices improves the spot-level objective on the realistic network as well.

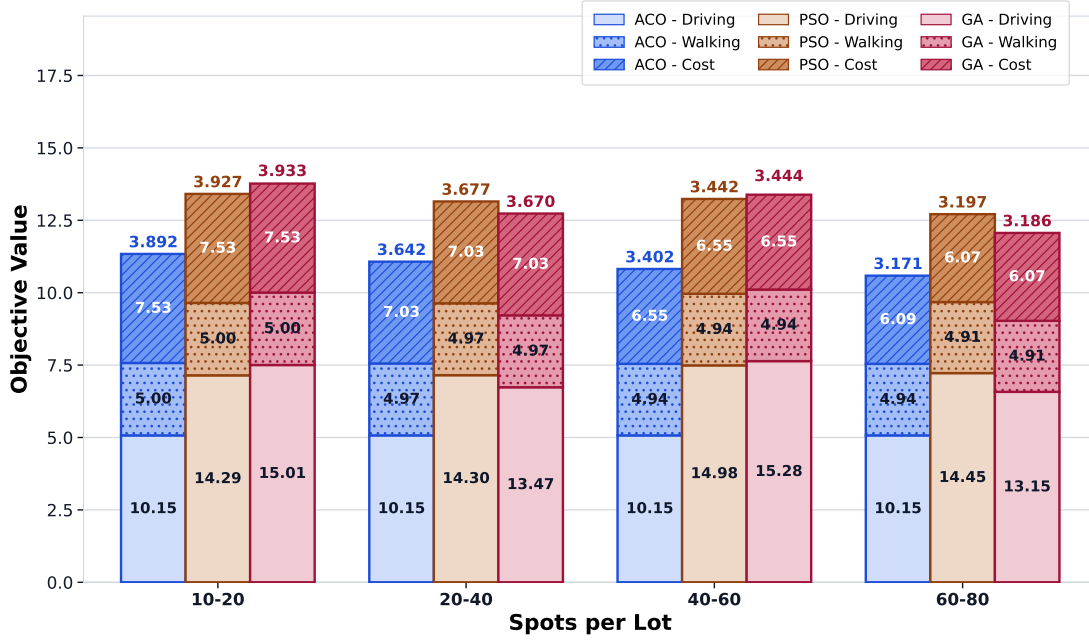


Figure 5.25: Performance comparison between the meta-heuristic methods as a function of the number of parking spots per lot.

Effect of Parking-Cost Range

Fig. 5.26 shows the parking-cost-range analysis. ACO increases from 3.7039 in the 8.0–20.5 setting to 4.2030, 5.7528, and 7.3519 as the cost range shifts upward. PSO and GA behave similarly, with only small differences between the methods. Therefore, the SUMO spot-level cost-range experiment is again dominated by the rise in the parking-cost component.

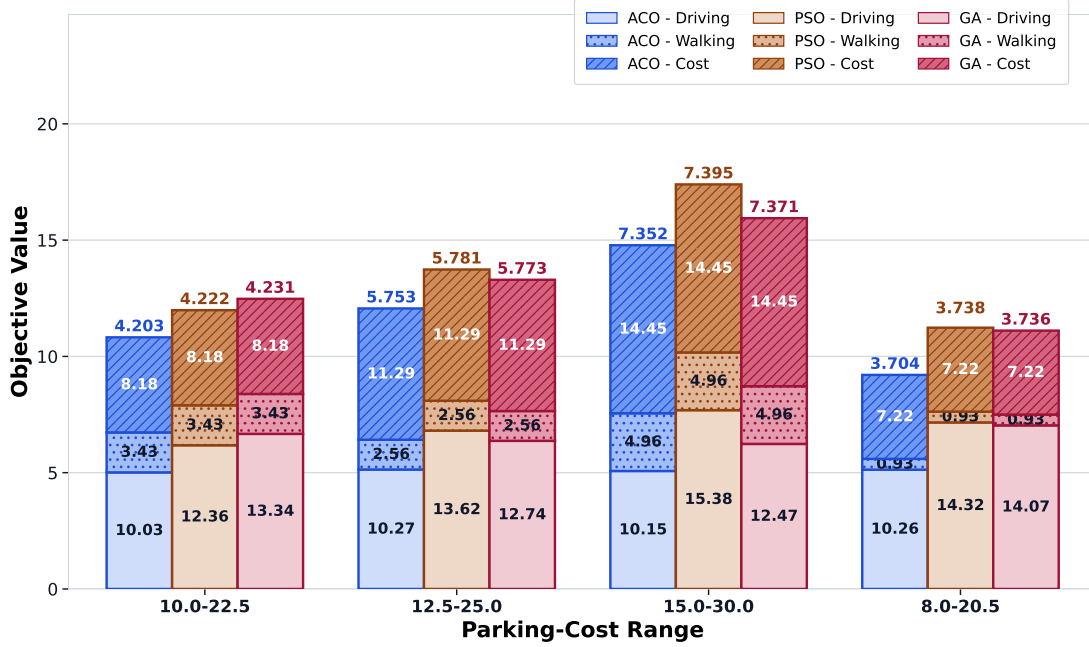


Figure 5.26: Performance comparison between the meta-heuristic methods as a function of the parking-cost range.

Summary

Overall, the SUMO-based spot-level evaluation confirms that the three meta-heuristic methods remain close on the realistic urban road network, with ACO providing the best or near-best result in most of the plotted cases. The most consistent trends are that greater walking flexibility, greater spot availability, and more spots per lot improve the objective value, while larger parking budgets and higher parking-cost ranges worsen it. Traffic congestion also degrades the recommendation quality, especially under high traffic. In contrast to the synthetic medium/large edge-density sweep, the SUMO edge-density experiment does not show a monotonic improvement, which highlights that realistic road topology can interact with spot-level recommendation in more complex ways.

5.7 Conclusion and Future Work

This chapter presented a Dynamic Parking Spot Optimization Framework (DP-SOF) for real-time parking recommendation at the parking-spot level. In contrast to the parking lot-level framework of Chapter 4, the proposed model extends the decision process from selecting a parking lot to selecting an individual feasible parking spot inside that lot. The framework jointly considers external driving time, internal driving time, internal walking time, external walking time, parking cost, and the user-defined constraints on walking distance and parking budget. In this way, the recommendation reflects a more detailed parking decision process

in which the selected spot, and not only the parking facility, affects the overall quality of the solution.

The experimental results show that the spot-level formulation is able to capture meaningful trade-offs across synthetic urban-like road networks and the SUMO-based real-road network. In the small-scale analysis, the optimization model provides a useful benchmark for validating the formulation, while the meta-heuristic methods produce competitive solutions at a much lower computational cost. As the problem size increases, the exact optimization approach becomes impractical because the decision space expands significantly with the number of parking spots, whereas ACO, PSO, and GA remain computationally feasible. Among the compared meta-heuristic methods, ACO is generally the most stable and frequently gives the best or near-best objective values across the medium-scale, large-scale, and SUMO-based experiments. PSO and GA also remain competitive, although they are generally more variable across parameter settings.

The parameter-sensitivity analysis further shows that the spot-level recommendation responds in a consistent way to changes in the experimental settings. Greater parking-spot availability and a larger number of spots per lot generally improve the objective value by increasing the chance of selecting a better feasible spot. A larger walking-distance allowance also tends to improve the objective value by expanding the feasible spot set, although the resulting trade-off depends on the balance between additional walking effort and lower parking cost. Higher parking-cost ranges worsen the objective mainly through the monetary component, while heavier traffic worsens the driving-time component. The parking-budget analysis shows that increasing the budget does not necessarily improve the total objective, because the additional feasible spots may be more expensive without producing enough travel-time savings to offset their cost.

The SUMO-based fixed-road-network evaluation confirms that the spot-level framework remains effective on a realistic directed urban topology. The results show that the three meta-heuristic methods remain close to one another in many SUMO scenarios, with ACO most often giving the strongest overall performance. This realistic-network evaluation is important because it shows that the framework is not limited to synthetic graph settings and can still produce meaningful parking-spot recommendations when the road structure is irregular, directed, and more complex.

Overall, the findings of this chapter indicate that DPSOF is a practical extension of the parking lot-level recommendation framework. By incorporating internal parking-lot movements, spot-specific availability, and spot-level cost variation, the framework provides a more detailed and realistic parking recommendation model while remaining computationally manageable through scalable meta-heuristic solution approaches.

Several directions remain open for future work. First, the framework can be extended to include time-varying parking-spot occupancy and reservation updates, so that a spot may become unavailable while the driver is still traveling toward it. Second, the current framework assumes a single-user recommendation

setting; future work could consider multi-vehicle competition for parking spots and shared-capacity effects across multiple drivers. Third, the internal parking-lot representation could be further refined using more realistic lot geometries, entrance-exit rules, and pedestrian routing structures. Fourth, the framework could be extended to incorporate electric-vehicle-specific considerations such as charging availability, charging duration, and charger-type preference. Finally, future work may investigate adaptive or learning-based search strategies that combine the strengths of optimization and meta-heuristic methods in order to improve both runtime and solution quality on large-scale dynamic urban networks.

Chapter 6

Conclusion

Dynamic parking recommendation has become an increasingly important component of intelligent transportation systems as rapid urbanization and increasing vehicle ownership continue to intensify parking demand in metropolitan areas. Traditional parking guidance systems typically select a parking facility once before the journey begins and assume that traffic conditions, parking availability, and travel costs remain unchanged during the trip. However, these assumptions are rarely valid in real urban environments where congestion, parking occupancy, and travel conditions continuously evolve. Consequently, parking decisions based on static information often lead to unnecessary driving, increased walking distances, higher parking costs, and additional traffic congestion. Motivated by these challenges, this thesis investigated dynamic parking recommendation through two complementary optimization frameworks. The first framework focused on dynamic parking-lot recommendation, while the second extended the problem to dynamic parking-spot recommendation by incorporating detailed parking-space characteristics into the optimization process.

The first part of this thesis developed the Dynamic Parking Optimization Framework (DPOF), which addresses the problem of selecting an appropriate parking lot under continuously changing traffic conditions. The framework models the road network as a weighted graph and repeatedly recomputes parking recommendations whenever updated traffic information becomes available. The objective function jointly minimizes total travel time, consisting of driving and walking time, together with parking cost while satisfying user-defined walking-distance and parking-budget constraints. To evaluate the framework comprehensively, an exact optimization model was developed for small-scale instances to provide optimal benchmark solutions, while scalable solution approaches based on a heuristic, Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Ant Colony Optimization (ACO) were implemented for larger transportation networks where exact optimization becomes computationally impractical.

Extensive computational experiments demonstrated that the proposed parking-lot recommendation framework effectively balances travel efficiency and parking cost across a wide range of experimental settings. The exact optimization model consistently produced optimal solutions for small-scale instances and served as a benchmark for evaluating the heuristic and metaheuristic approaches. For larger synthetic networks and SUMO-based road networks, the heuristic and metaheuristic methods maintained high solution quality while providing significantly lower computational times than exact optimization. The experimen-

tal analysis further showed that traffic conditions, walking-distance constraints, parking-budget limits, and network size all influence parking recommendations in predictable ways, confirming the robustness and scalability of the proposed framework for dynamic urban parking scenarios.

The second part of this thesis extended the parking recommendation problem by introducing the Dynamic Parking Spot Assignment Framework (DPSAF). Rather than selecting only a parking lot, the proposed extension performs optimization directly at the individual parking-space level. Each parking space is represented explicitly within the optimization model together with its own availability, parking cost, internal driving time, internal walking time, external walking time, and parent parking facility. Consequently, parking recommendation is formulated as a joint parking-lot and parking-spot optimization problem that captures operational details ignored by conventional parking-lot-based approaches.

To solve this extended problem, the optimization formulation was redesigned so that the decision variable corresponds to individual parking spaces instead of parking lots. The exact optimization model was implemented through exhaustive enumeration for small-scale networks to obtain optimal benchmark solutions, while GA, PSO, and ACO were redesigned to search directly over feasible parking-space candidates using a unified feasibility evaluation and objective-function calculation. The three metaheuristic algorithms differ only in their search strategies, whereas feasibility checking, objective-value computation, and spot evaluation remain identical across all methods. This unified formulation enables a fair comparison of optimization algorithms while ensuring consistent objective evaluation throughout the experimental study.

The computational results demonstrate that incorporating parking-space-level information provides a more realistic representation of urban parking than conventional parking-lot recommendation alone. The extension successfully models internal vehicle movement within parking facilities, pedestrian movement inside parking lots, individual parking-space availability, and spot-specific parking costs while preserving computational scalability. The experimental evaluation showed that the proposed optimization and metaheuristic methods consistently satisfy feasibility constraints while producing high-quality parking recommendations across all tested scenarios. The extension therefore complements the baseline framework by increasing modeling fidelity without fundamentally changing the overall optimization objective.

Taken together, the baseline and extension frameworks developed in this thesis provide a comprehensive optimization framework for dynamic parking recommendation. The baseline framework demonstrates that dynamic parking-lot recommendation can effectively reduce travel time and parking cost through repeated optimization under changing traffic conditions. The extension further improves decision quality by incorporating individual parking-space characteristics into the optimization process, allowing recommendations to account for internal parking-facility movements and parking-space availability. Collectively, the proposed frameworks illustrate that realistic parking recommendation requires in-

tegrating transportation-network conditions with parking-facility characteristics rather than treating them as independent decision problems.

The research presented in this thesis contributes to intelligent transportation systems in several important ways. First, it demonstrates that repeated traffic-aware parking optimization substantially improves parking decisions compared with static recommendation approaches. Second, it introduces a unified objective function that simultaneously considers travel time and parking cost under user-defined operational constraints. Third, it presents a scalable optimization methodology capable of solving both parking-lot and parking-space recommendation problems across synthetic and realistic transportation networks. Finally, the thesis establishes exact optimization benchmarks together with scalable heuristic and metaheuristic algorithms, providing both rigorous evaluation and practical applicability for large-scale deployment.

Future research can extend the proposed frameworks in several directions. Reinforcement Learning, Deep Reinforcement Learning, and hybrid optimization approaches combining learning-based methods with evolutionary optimization may further improve real-time parking recommendation performance. The current single-vehicle formulation can be extended to coordinated multi-vehicle parking assignment where multiple drivers compete simultaneously for available parking spaces. Additional realism may be introduced by incorporating predictive parking availability models, real-time parking reservation systems, connected vehicle communication, dynamic parking pricing, electric-vehicle charging infrastructure, autonomous vehicle parking behavior, and stochastic traffic prediction. Furthermore, future work may investigate digital twin environments that continuously synchronize real parking systems with optimization algorithms to support large-scale intelligent city applications.

Overall, this thesis advances the state of dynamic parking recommendation by developing optimization frameworks that are both computationally efficient and operationally realistic. The proposed baseline framework establishes an effective methodology for dynamic parking-lot recommendation, while the proposed extension further improves realism through parking-space-level optimization that explicitly models parking-space availability, internal driving time, internal walking time, and external pedestrian movement. The results demonstrate that integrating dynamic traffic information with detailed parking-facility modeling provides a practical foundation for next-generation intelligent parking systems capable of supporting efficient, scalable, and real-time urban mobility.

6.1 Limitations of the Work

Although the proposed parking recommendation frameworks provide significant improvements over conventional parking guidance approaches, several limitations should be acknowledged. First, both the baseline and extension frameworks consider a single vehicle making independent parking decisions. While this assumption simplifies the optimization process and allows the proposed algorithms to be

studied in detail, it does not capture interactions among multiple drivers competing simultaneously for the same parking resources. Consequently, the current framework does not model reservation conflicts, cooperative parking assignment, or system-wide parking allocation across multiple vehicles.

Second, the proposed models assume that traffic conditions and parking-space availability are accurately available whenever the optimization process is executed. Although this assumption enables dynamic recommendation under changing conditions, practical deployment would require reliable real-time sensing, communication infrastructure, and data synchronization. Sensor failures, communication delays, inaccurate occupancy detection, and outdated traffic information may influence recommendation quality and should be considered in future implementations.

Third, the extension models parking facilities using predefined internal driving and walking times associated with each parking space. While this representation significantly improves realism compared with traditional parking-lot-based methods, it remains an abstraction of actual parking-facility operations. Factors such as internal vehicle congestion, driver search behavior within parking lots, pedestrian interactions, parking maneuvers, and temporary obstructions are not explicitly modeled.

Fourth, the exact optimization approach is computationally practical only for relatively small problem instances. As network size increases, exhaustive optimization becomes computationally expensive, requiring the use of heuristic and metaheuristic algorithms. Although GA, PSO, and ACO demonstrated strong empirical performance throughout the experiments, they cannot guarantee globally optimal solutions for very large transportation networks. Solution quality therefore depends on algorithm parameters, search behavior, and available computational resources.

Finally, the experimental evaluation was performed using synthetic transportation networks together with SUMO-based simulations that represent realistic urban road conditions. Although these experiments provide extensive validation under controlled environments, additional evaluation using real-world parking datasets collected from multiple cities would further strengthen the practical applicability of the proposed frameworks. Future studies should also investigate user behavior, parking demand uncertainty, seasonal traffic variations, and long-term deployment within operational smart-city parking systems.

Bibliography

- [1] J. Li, H. Qu and L. You, "An Integrated Approach for the Near Real-Time Parking Occupancy Prediction," in *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 4, pp. 3769-3778, April 2023, doi: 10.1109/TITS.2022.3230199.
- [2] J. Li, H. Ma, X. Li and R. Ma, "Semantics-Aware Spatial-Temporal Dynamic Graph Transformer Network for On-Street Parking Occupancy Prediction," in *IEEE Transactions on Intelligent Transportation Systems*, vol. 27, no. 1, pp. 842-856, Jan. 2026, doi: 10.1109/TITS.2025.3629329.
- [3] D. Zhao et al., "MePark: Using Meters as Sensors for Citywide On-Street Parking Availability Prediction," in *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 7, pp. 7244-7257, July 2022, doi: 10.1109/TITS.2021.3067675.
- [4] B. Song, H. Zhou, X. Huang and W. Ma, "Collaborative Parking Vacancy Prediction for Cities With Partial Sensors Missing," in *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 9, pp. 11439-11451, Sept. 2024, doi: 10.1109/TITS.2024.3377734.
- [5] G. O. Boateng et al., "Automated Valet Parking and Charging: A Dynamic Pricing and Reservation-Based Framework Leveraging Multi-Agent Reinforcement Learning," in *IEEE Transactions on Intelligent Vehicles*, vol. 10, no. 2, pp. 1010-1029, Feb. 2025, doi: 10.1109/TIV.2024.3421524.
- [6] S. Saharan, N. Kumar and S. Bawa, "DyPARK: A Dynamic Pricing and Allocation Scheme for Smart On-Street Parking System," in *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 4, pp. 4217-4234, April 2023, doi: 10.1109/TITS.2022.3230851.
- [7] Z. Hu, Y. Cao, X. Li, H. Ai, L. Xu and Z. Liu, "Towards Dynamic Pricing Based Autonomous Valet Parking Management Under Demand Uncertainty," in *IEEE Transactions on Vehicular Technology*, vol. 73, no. 5, pp. 6196-6211, May 2024, doi: 10.1109/TVT.2023.3340629.
- [8] M. Ahmed, M. S. Hamad, A. S. Abdel-Khalik, E. Hamdan, S. Ahmed and N. A. Elmalhy, "Improved Utilization for "Smart Parking Systems" Based on Paging Technique," in *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 5, pp. 3375-3388, May 2024, doi: 10.1109/TITS.2023.3323097.
- [9] V. Hassija, V. Saxena, V. Chamola and F. R. Yu, "A Parking Slot Allocation Framework Based on Virtual Voting and Adaptive Pricing Algorithm," in *IEEE Transactions on Vehicular Technology*, vol. 69, no. 6, pp. 5945-5957, June 2020, doi: 10.1109/TVT.2020.2979637.
- [10] J. Tang, Y. Wang, W. Liu, X. Luo and Z. Zhou, "SCP-Tree: Finding Multiple Nearest Parking Spots With Minimal Group Travel Cost," in *IEEE Access*, vol. 9, pp. 158946-158960, 2021, doi: 10.1109/ACCESS.2021.3131229.

- [11] Tran Thi Kim, N. H. Tran, C. Pham, T. LeAnh, M. T. Thai and C. S. Hong, "Parking Assignment: Minimizing Parking Expenses and Balancing Parking Demand Among Multiple Parking Lots," in *IEEE Transactions on Automation Science and Engineering*, vol. 17, no. 3, pp. 1320-1331, July 2020, doi: 10.1109/TASE.2019.2948200.
- [12] S. Kim, Y. Tak, E. Barmounakis and N. Geroliminis, "Monitoring Outdoor Parking in Urban Areas With Unmanned Aerial Vehicles," in *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 10, pp. 13393-13406, Oct. 2024, doi: 10.1109/TITS.2024.3397588.
- [13] B. Assemi, A. Paz and D. Baker, "On-Street Parking Occupancy Inference Based on Payment Transactions," in *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 8, pp. 10680-10691, Aug. 2022, doi: 10.1109/TITS.2021.3095277.
- [14] Z. Hu, Y. Cao, J. Huang, L. Wu, L. Xu and R. Zhang, "CRPark: A Smart Parking System With Competition and Reservation Services," in *IEEE Transactions on Vehicular Technology*, vol. 74, no. 8, pp. 12981-12996, Aug. 2025, doi: 10.1109/TVT.2025.3555387.
- [15] J. Wang et al., "PRPSV: Parking Efficiency and Reservation Service Optimization Based on Parking Space View," in *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 8, pp. 11696-11711, Aug. 2025, doi: 10.1109/TITS.2025.3574305.
- [16] P. Wang et al., "Optimal Dynamic Supply of Parking Permits Under Uncertainties: A Stochastic Control Approach," in *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 10, pp. 17397-17409, Oct. 2022, doi: 10.1109/TITS.2022.3150347.
- [17] Y. Nie, W. Yang, Z. Chen, N. Lu, L. Huang and H. Huang, "Public Curb Parking Demand Estimation With POI Distribution," in *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 5, pp. 4614-4624, May 2022, doi: 10.1109/TITS.2020.3046841.
- [18] M. A. B. Helal, A. J. Telmoudi, Y. Ouazene and N. Labadie, "A hybrid genetic algorithm for green vehicle routing problem," 2025 *IEEE International Conference on Advanced Systems and Emergent Technologies (ICASET)*, Mammamet-Yasmine, Tunisia, 2025, pp. 1-6, doi: 10.1109/ICASET65966.2025.11232208.
- [19] W. Hu and X. Ma, "Optimization Algorithm of UAVs Task Assignment and Path Planning Based on Dynamic Cluster Particle Swarm Optimization," in *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 10, pp. 18157-18169, Oct. 2025, doi: 10.1109/TITS.2025.3578464.
- [20] H. Wang et al., "Transportation Route Optimization of Municipal Solid Waste Based on Improved Ant Colony Algorithm in Internet of Vehicles," in *IEEE Transactions on Vehicular Technology*, vol. 74, no. 2, pp. 2129-2142, Feb. 2025, doi: 10.1109/TVT.2024.3417976.
- [21] M. Aljohani, S. Olariu, A. Alali and S. Jain, "A Survey of Parking Solutions for Smart Cities," in *IEEE Transactions on Intelligent Trans-*

- portation Systems*, vol. 23, no. 8, pp. 10012-10029, Aug. 2022, doi: 10.1109/TITS.2021.3112825.
- [22] M. R. Alam, S. Saha, M. B. Bostami, M. S. Islam, M. S. Aadeeb and A. K. M. M. Islam, "A Survey on IoT Driven Smart Parking Management System: Approaches, Limitations and Future Research Agenda," in *IEEE Access*, vol. 11, pp. 119523-119543, 2023, doi: 10.1109/ACCESS.2023.3327306.
 - [23] P. Agrawal, P. Mundada, J. Ikhar, N. Rakesh, G. Kaur and L. Pinjarkar, "Maximizing Urban Space A Survey of Smart Parking Techniques and Innovations," 2024 First International Conference on Technological Innovations and Advance Computing (TIACOMP), Bali, Indonesia, 2024, pp. 542-548, doi: 10.1109/TIACOMP64125.2024.00095.
 - [24] H. H. Almalki and M. Ragab, "Smart Parking in Sustainable Environment: Challenges and Future Perspectives," 2025 International Conference on Innovation in Artificial Intelligence and Internet of Things (AIIT), Jeddah, Saudi Arabia, 2025, pp. 1-9, doi: 10.1109/AIIT63112.2025.11082841.
 - [25] M. Abdelmagid, A. Gabour and M. Maalouf, "Dynamic Modeling for the Parking Allocation Problem: A Framework for Real-Time Optimization," 2024 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM), Bangkok, Thailand, 2024, pp. 74-78.
 - [26] J. Kulkarni, S. Tambe, S. Deshmane and P. Vasmatkar, "Smart Automated and Real-Time Parking Management System," 2025 7th International Conference on Inventive Material Science and Applications (ICIMA), Namakkal, India, 2025, pp. 928-932, doi: 10.1109/ICIMA64861.2025.11074057.
 - [27] Y. Amidu, A. F. Gabor and K. ElBassioni, "A Primal-Dual Algorithm for Online Parking Allocation Problems," 2024 IEEE International Conference on Technology Management, Operations and Decisions (ICTMOD), Sharjah, UAE, 2024, pp. 1-5.
 - [28] X. Zheng, R. Zheng and S. Ning, "Parking Space Allocation Model of Intelligent Parking Lot under Peak Demand," 2022 IEEE 25th International Conference on Computer Supported Cooperative Work in Design (CSCWD), Hangzhou, China, 2022, pp. 543-547, doi: 10.1109/CSCWD54268.2022.9776216.
 - [29] T. Lian, "A shared parking spot reservation and allocation model that considers prioritizing commuting demands," ISCTT 2021; 6th International Conference on Information Science, Computer Technology and Transportation, Xishuangbanna, China, 2021, pp. 1-4.
 - [30] R. Hu, L. Zhang and X. Xue, "Parking space allocation based on multi-objective intelligent optimization algorithm," 2020 IEEE 4th Information Technology, Networking, Electronic and Automation Control Conference (ITNEC), Chongqing, China, 2020, pp. 2702-2706, doi: 10.1109/ITNEC48623.2020.9085094.
 - [31] T. Nakazato, Y. Fujimaki and T. Namerikawa, "Parking Lot Allocation Using Rematching and Dynamic Parking Fee Design," *IEEE TCNS*, vol. 9, no. 4, pp. 1692-1703, Dec. 2022.

- [32] J. Wang et al., "PRPSV: Parking Efficiency and Reservation Service Optimization Based on Parking Space View," in *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 8, pp. 11696-11711, Aug. 2025, doi: 10.1109/TITS.2025.3574305.
- [33] S. Saharan, N. Kumar and S. Bawa, "DyPARK: A Dynamic Pricing and Allocation Scheme for Smart On-Street Parking System," in *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 4, pp. 4217-4234, April 2023, doi: 10.1109/TITS.2022.3230851.
- [34] B. Cheng and S. Nian, "Based on genetic algorithm to analyze the advantages and disadvantages of parking lot," 2021 IEEE International Conference on Emergency Science and Information Technology (ICESIT), Chongqing, China, 2021, pp. 534-538, doi: 10.1109/ICESIT53460.2021.9696979.
- [35] Y. -H. Cheng and C. -Y. Kang, "Application of Genetic Algorithm to Path Planning Problem of Automatic Navigation Parking Spaces in Parking Lots," 2023 Sixth International Symposium on Computer, Consumer and Control (IS3C), Taichung, Taiwan, 2023, pp. 118-121, doi: 10.1109/IS3C57901.2023.00040.
- [36] M. M. Abo-Zahhad and M. Abo-Zahhad, "Smart Parking System in Smart Cities Based On the Internet of Things and Machine Learning," 2024 International Telecommunications Conference (ITC-Egypt), Cairo, Egypt, 2024, pp. 190-195, doi: 10.1109/ITC-Egypt61547.2024.10620564.
- [37] U. M. F. Dimlo, S. T. A, O. Lymareva, V. Pandey, J. Saikumar and V. Devi, "An Efficient Parking Management System for Smart Cities Using GA-RBF Model to Optimize Fuel, Time, and Cost," 2024 International Conference on Intelligent Algorithms for Computational Intelligence Systems (IACIS), Hassan, India, 2024, pp. 1-6, doi: 10.1109/IACIS61494.2024.10721975.
- [38] Dong, Na, Fang, Xing, Wu, Ai-guo, A Novel Chaotic Particle Swarm Optimization Algorithm for Parking Space Guidance, *Mathematical Problems in Engineering*, 2016, 5126808, 14 pages, 2016. <https://doi.org/10.1155/2016/51268>
- [39] B. Wang, Y. Liu, X. Hei, L. Wang and Z. Zhang, "A Parking Guidance Method Based on Parallel Particle Swarm Optimization Algorithm," 2014 Tenth International Conference on Computational Intelligence and Security, Kunming, China, 2014, pp. 568-572, doi: 10.1109/CIS.2014.150.
- [40] Adeyiga, Johnson et al. "Comparison of Genetic Algorithm and Particle Swarm Optimization Techniques in Intelligent Parking System." *Journal of Advanced Materials Science and Engineering* (2022): n. pag.
- [41] H. Xie, Z. Zhang and K. Song, "A Self-optimization Algorithm of Multi-style Smart Parking Driven by Experience, Knowledge and Data," 2021 5th CAA International Conference on Vehicular Control and Intelligence (CVCI), Tianjin, China, 2021, pp. 1-7, doi: 10.1109/CVCI54083.2021.9661152.

- [42] Hassoune, Khaoula & Dachry, Wafaa & Moutaouakkil, Fouad & Medromi, Hicham. (2020). Dynamic Parking Guidance Architecture Using Ant Colony Optimization and Multi-agent Systems. *Journal of Advances in Information Technology*. 58-63. 10.12720/jait.11.2.58-63.
- [43] Bhimavarapu, Lekhana & Kommu, Jahnavi & Mandalapu, Manisha & Kalagadda, Kiran & Kusuma, Rambabu. (2025). Towards Smarter Parking: An End-to-End System for Real-Time Space Detection and Path Finding. 1-6. 10.1109/CONIT65521.2025.11167739.
- [44] S. Akandwanaho and I. Govender, ‘Dynamic Street Parking Space Using Memetic Algorithm for Optimization’, *Smart Urban Development*. IntechOpen, Feb. 19, 2020. doi: 10.5772/intechopen.86010.
- [45] A. Baikani, N. Bala and V. Ramalingam, “NeuroPark Guide: Cutting-Edge AI for Parking Solutions,” 2025 3rd International Conference on Communication, Security, and Artificial Intelligence (ICCSAI), Greater Noida, India, 2025, pp. 643-647, doi: 10.1109/ICCSAI64074.2025.11063878.
- [46] K. He, Z. Qing, Z. Chen, Y. Wang, H. Qin and X. Cao, “A Smart Parking Management System Considering Occupancy Rate Prediction and Customer Preferences,” 2025 7th International Conference on Data-driven Optimization of Complex Systems (DOCS), Taiyuan, China, 2025, pp. 277-282, doi: 10.1109/DOCS67533.2025.11200548.
- [47] V. Hassija, V. Saxena, V. Chamola and F. R. Yu, “A Parking Slot Allocation Framework Based on Virtual Voting and Adaptive Pricing Algorithm,” in *IEEE Transactions on Vehicular Technology*, vol. 69, no. 6, pp. 5945-5957, June 2020, doi: 10.1109/TVT.2020.2979637.
- [48] X. Zhang, F. Yuan, Y. Cao and S. Liu, “Reservation Enhanced Autonomous Valet Parking Concerning Practicality Issues,” in *IEEE Systems Journal*, vol. 16, no. 1, pp. 351-361, March 2022, doi: 10.1109/JSYST.2020.3036627.
- [49] Z. Hu, Y. Cao, J. Huang, L. Wu, L. Xu and R. Zhang, “CRPark: A Smart Parking System With Competition and Reservation Services,” in *IEEE Transactions on Vehicular Technology*, vol. 74, no. 8, pp. 12981-12996, Aug. 2025, doi: 10.1109/TVT.2025.3555387.

Glossary

Candidate Parking Lot: A parking lot considered as a feasible recommendation option in the road network. Each candidate lot is evaluated using travel time, walking time, parking cost, and feasibility constraints.

Candidate Parking Spot: An individual parking spot considered as a feasible assignment option inside a parking facility. Each candidate spot is evaluated using availability, internal driving time, internal walking time, external walking time, and parking cost.

Constraint: A limiting condition imposed on the recommendation problem, such as the maximum allowable walking distance or maximum allowable parking cost.

Directed Edge: A road-network connection with a specified direction from one node to another. Directed edges represent directional vehicle movement in the transportation graph.

Dominance Filtering: A candidate-reduction step in which a parking option is removed if another feasible option is no worse in all relevant criteria and strictly better in at least one criterion.

Dynamic Parking Recommendation: A recommendation process in which the selected parking option can be updated as the driver moves and traffic or parking conditions change.

Edge Density: The level of connectivity in the road network, represented by the number of directed edges among nodes.

External Driving Time: The driving time from the current vehicle location to the selected parking lot or to the entrance of the selected parking facility.

External Walking Time: The walking time from the selected parking lot, or from the pedestrian exit of the selected parking facility, to the final destination.

Feasible Parking Option: A parking lot or parking spot that satisfies all imposed constraints, including availability, walking-distance threshold, and parking-cost threshold.

Graph-Based Representation: A mathematical representation of the transportation and parking system in which nodes represent intersections, parking lots, parking spots, entrances, exits, source, and destination, while

edges represent road or walking connections.

Guided Depth-First Search Decoder: A route-construction procedure used by the meta-heuristic comparison methods at the parking-lot level to build a valid route from the source to the selected parking lot.

Internal Driving Time: The vehicle travel time from the entrance of a parking facility to a selected parking spot inside that facility.

Internal Walking Time: The walking time from a selected parking spot to the pedestrian exit inside the parking facility.

Meta-Heuristic Method: A search-based optimization method used to obtain near-optimal solutions for computationally complex problems. In this thesis, GA, PSO, and ACO are used as comparative methods.

Objective Function: A mathematical expression that combines total travel time and parking cost into a single value to be minimized.

Objective Weight: The parameter controlling the trade-off between travel time and parking cost in the objective function.

Parking Facility: A parking lot together with its associated parking spots, entrance, and pedestrian exit.

Parking Spot Assignment: The process of selecting an individual parking spot within a parking facility.

Parking Spot Availability: The current occupancy status of a parking spot indicating whether it can be assigned.

Parking-Cost Threshold: The maximum parking cost acceptable to the user or imposed in the experiment.

Reservation Threshold: The predefined time interval during which an assigned parking spot is reserved for a driver before being released back to the available set.

Road-Network Scalability: The ability of a solution approach to remain computationally practical as network size and problem complexity increase.

Spot-Specific Parking Cost: The parking cost associated with an individual parking spot, which may differ across spots in the same facility.

Synthetic Urban-Like Dataset: An artificially generated dataset designed to mimic urban road-network and parking characteristics while allowing controlled parameter variation.

Walking-Distance Threshold: The maximum allowable walking distance between the selected parking location and the final destination.

Abbreviations

ACO	Ant Colony Optimization
DFS	Depth-First Search
DPLOF	Dynamic Parking Lot Optimization Framework
DPSOF	Dynamic Parking Spot Optimization Framework
GA	Genetic Algorithm
IoT	Internet of Things
ITS	Intelligent Transportation Systems
PSO	Particle Swarm Optimization
SPS	Smart Parking Systems
SUMO	Simulation of Urban MObility

List of Symbols

$G = (N, E)$	Directed graph representing the road network and parking-related traversal structure
N	Set of graph nodes
E	Set of directed edges
i, j	Indices of nodes, parking lots, or parking spots as defined by context
$(i, j) \in E$	Directed edge from node i to node j
d_{ij}	Distance on edge (i, j)
t_{ij}	Travel time on edge (i, j) under current traffic conditions
S	Source node of the driver
A	Current vehicle location during route progression
D	Destination node
P	Set of parking lots or parking facilities
p_i	Parking lot or parking facility i , where $p_i \in P$
m	Number of parking facilities
$L[p]$	Graph node corresponding to parking lot p
a_p	Availability status of parking lot p
C_{p_i}	Parking cost associated with parking lot p_i
$d_{p_i}^{walk}$	Walking distance from parking lot p_i to the destination
$t_{p_i}^{drive}$	External driving time from the current vehicle location to parking lot p_i
$t_{p_i}^{walk}$	External walking time from parking lot p_i to the destination
$t_{p_i}^{total}$	Total travel time associated with parking lot p_i
T_i	Accumulated travel time upon reaching node i
T_D	Total route travel time at the parking-lot level
H_{p_i}	Evaluation score associated with parking lot p_i

F	Set of feasible non-dominated parking lots
p^*	Selected parking lot
S_i	Set of available parking spots in facility p_i at decision time t
s_{ij}	Parking spot j in facility p_i
k_i	Number of available candidate spots in facility p_i
t	Decision time
C_{ij}^s	Parking cost of parking spot s_{ij}
T_i^{road}	External driving time from the current vehicle location to facility p_i
T_{ij}^{veh}	Internal driving time from the facility entrance to parking spot s_{ij}
$T_{ij}^{walk,in}$	Internal walking time from parking spot s_{ij} to the pedestrian exit
$T_i^{walk,out}$	External walking time from facility p_i to the destination
$T_{ij}(t)$	Total travel-time component associated with assigning spot s_{ij}
$q_{ij}(t)$	Availability state of parking spot s_{ij} at time t
τ	Reservation threshold duration
s^*	Selected parking spot
$C_{\max}$	Maximum allowable parking-cost threshold
$W_{\max}$	Maximum allowable walking-distance threshold
α	Objective-weight parameter balancing travel time and parking cost
v_{walk}	Walking speed of the user
B	Large positive constant used in constraint formulations
Z	Objective function value
x_{ij}	Binary decision variable indicating whether edge (i, j) is selected
$x_{L[p],D}$	Binary decision variable indicating whether the walking link from parking lot p to D is selected
$Cost$	Parking cost of the selected parking option
z_{ij}	Binary decision variable indicating whether parking spot s_{ij} is assigned